The Exact Mean-Square Stability Boundary of Heavy-Ball SGD: a Lyapunov Equation and a Secular Equation

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Abstract

We study the linear mean-square stability of heavy-ball SGD on a quadratic loss, $w_{t+1} = ((1+\beta)I - \eta H_{B_t})w_t - \beta w_{t-1} - \eta\xi_t$, with minibatch Hessians that need not commute with each other or with their mean H. We derive the exact stationary second-moment equation (Theorem 2), relate it exactly to the small-step Schur-complement expansion of Andreyev et al. [2026], and show that beyond the threshold, $\rho(\mathcal{T}) \geq 1$ for the second-moment operator, the second moment grows at least linearly (Corollary 4), the mean-square part of the instability criterion they list as open for SGD with momentum. For random-feature minibatches with independent coordinates of any kurtosis the boundary is the root of a rank-one secular equation (Theorem 5); at $\beta = 0$, $B = 1$ and Gaussian features this is the mean-square condition of the LMS filter. For arbitrary per-sample Hessians it is the reciprocal of the top eigenvalue of a symmetrisable operator (Theorem 9), the criterion that Cai et al. [2026, Theorem F.1] obtained concurrently and independently. At the threshold an identity places a modal curvature between the two batch-sharpness plateaus $2(1-\beta)/\eta$ and $2(1+\beta)/\eta$ reported by Andreyev et al. [2026], in the random-feature model a stationary batch sharpness reaches them in the two batch limits, and a variational test gives sufficient instability conditions through batch sharpness (Theorem 13); in a quasi-stationary slow-fast model progressive sharpening saturates just below the threshold (Theorem 14).

The secular root matches sample paths with median error 0.19–3.82 % in 29 of 30 cells and the spectral radius of the exact operator within 6.11×10^{-10} in 36 cells. On tethered Hessians of networks with $P = 33$ and 3 610 parameters the threshold of Theorem 9 is where that radius crosses 1. At the iterates of a 33-parameter network trained with weight decay, runs train up to 5.1× above an $|H|$ reading of the threshold without diverging; six predictions written from a first run held on three new seeds. A two-sided bracket of the threshold with bounded rounding errors places training at 0.81–1.10 times the threshold of the Gauss–Newton surrogate on networks with 3 610 and 12 826 parameters, in an exploratory measurement. Code and raw files reproduce every number; failed predictions are reported with the rest.

1 Introduction

With minibatch noise, the quantity that hovers at $2/\eta$ during training is the batch sharpness, the edge of stochastic stability [Andreyev and Beneventano, 2024], and the linear mean-square

*This paper is the second-moment part of a companion manuscript on identifying the edge of stability by extrapolating the step variance, as Southwell [1932] located the buckling load of a column from sub-critical deflections. The estimator, its minimax analysis and its measurements on networks are in that manuscript and are not repeated here; the network checkpoints of Section 4.2 come from it.

stability threshold lies below the deterministic $2(1+\beta)/\lambda_1$ [Wu et al., 2018, Andreyev et al., 2026]. Without momentum, Mulayoff and Michaeli [2024] give the exact mean-square threshold of SGD on a quadratic in any dimension. With heavy-ball momentum, Andreyev et al. [2026] obtain the threshold to leading order, in the noise-dominated regime and for small η, and state that the precise instability criterion of SGD with momentum is an open question. In high-dimensional limits, Paquette and Paquette [2021] and Lee et al. [2022] describe mini-batch SGD with momentum through Volterra equations; in the proof of their Proposition 2, Lee et al. [2022] compute the norm of the Volterra kernel and state two sufficient conditions for it to be below one.

This paper gives the exact mean-square boundary of heavy-ball SGD on the quadratic model at every $\eta > 0$ and $\beta \in [0, 1)$, and identifies it with the conditions above where they apply:

- The exact stationary second-moment equation with a matrix curvature noise that need not commute with H (Theorem 2), its exact relation $I - T_{\text{slow}} = \mathcal{S} \circ \mathcal{L}$ to the Schur complement of Andreyev et al. [2026, Appendix D] and their remainder to leading order (Lemma 15), the reduction of the spectral radius to the symmetric matrices (Lemma 3) and that for $\rho(\mathcal{T}) \geq 1$ the second moment grows at least linearly, which also meets Andreyev et al. [2026, Definition A.1], the item they list as open for SGD with momentum (Corollary 4; additive noise of unbounded support meets that definition at every η, so the moment bound is the content); its relation to batch sharpness at the threshold (Theorem 13) and saturation in a slow-fast model (Theorem 14).

- For random-feature minibatches, the boundary as the root of a rank-one secular equation with bounds by λ_1 and $\operatorname{tr} H$ (Theorem 5), the two plateaus and the Rademacher separation (Corollary 6), and the gap between mean-square and almost-sure stability along the top mode (Proposition 7).

- A curvature-noise covariance $\Sigma_z \neq H$ (Theorem 8), where a mode with small curvature and non-zero curvature noise drives the ceiling to zero, and arbitrary non-commuting per-sample Hessians (Theorem 9), where the boundary is the reciprocal of the top eigenvalue of a symmetrisable operator; at $\beta = 0$ this is Theorem 5 of Mulayoff and Michaeli [2024]. Appendix B bounds the gap between the threshold and its diagonal part, which closes as $\beta \to 1$, and gives the burn-in of the second moment in Loewner order.

- Measurements of the secular root against sample paths and against the exact second-moment operator, of the linearised minibatch iteration on two network checkpoints, of the operator of Theorem 9 on tethered network Hessians, and of the threshold at the iterates of training runs of a small network, replicated on new seeds, with predictions written before each run; and a certified two-sided bracket of the threshold at training checkpoints of networks with thousands of parameters (Section 4).

2 Setting and the one-dimensional ceiling

Let $\mathcal{L}(\Theta; B)$ be the loss of a minibatch B at parameters $\Theta \in \mathbb{R}^P$, $\mathcal{L}(\Theta)$ its mean, and heavy-ball SGD $\Theta_{t+1} = \Theta_t + \beta(\Theta_t - \Theta_{t-1}) - \eta \nabla \mathcal{L}(\Theta_t; B_t)$ with i.i.d. minibatches. Near a checkpoint Θ_0 the quadratic model replaces $\nabla \mathcal{L}(\Theta; B_t)$ by $\nabla \mathcal{L}(\Theta_0; B_t) + H_{B_t}(\Theta - \Theta_0)$ with $H_{B_t} = \nabla^2 \mathcal{L}(\Theta_0; B_t)$ and mean $H = \nabla^2 \mathcal{L}(\Theta_0)$; with $w_t = \Theta_t - \Theta_0$, $\xi_t = \nabla \mathcal{L}(\Theta_0; B_t) - \nabla \mathcal{L}(\Theta_0)$ and $\nabla \mathcal{L}(\Theta_0) = 0$ this is the recursion (2) of Section 3. We write $\Delta w_t = w_{t+1} - w_t$ for the step, $x_t = \langle w_t, u_1 \rangle$ and $\Delta x_t = x_{t+1} - x_t$ for their components along the top eigenvector. Let $H = \sum_k \lambda_k u_k u_k^\top$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_P$ (a gap $\lambda_1 > \lambda_2$ is assumed where u_1 must be unique, in Propositions 1 and 7 and in Corollary 6), and

allow a proximal tether $\frac{\mu}{2}\|\Theta-\Theta_0\|^2$ with $\mu \geq 0$ and $\lambda_P + \mu > 0$, which shifts every eigenvalue by μ and keeps the eigenvectors. Write $\kappa_\beta = \frac{1+\beta}{1-\beta}$ and $\eta^{(\mu)}_{\mathrm{crit}}(\beta) = 2(1+\beta)/(\lambda_1+\mu)$, the deterministic threshold of heavy-ball gradient descent.

Proposition 1 (One-dimensional mean-square ceiling; Wu et al. 2018, Andreyev et al. 2026)**.** *On* $\mathrm{span}(u_1)$*, let the restoring force be evaluated on a fresh minibatch* B_t *at each step, with directional curvature variance* $\nu_B^2 = \mathrm{Var}(u_1^\top \nabla^2 \mathcal{L}(\Theta_0; B_t) u_1)$*, and let additive noise of variance* $\sigma_1^2 > 0$ *drive the mode. The stationary second moment is finite if and only if* $\eta < \eta^{(\mu),\mathrm{EoSS}}_{\mathrm{crit}}(B,\beta)$*, where*

$$\eta^{(\mu),\mathrm{EoSS}}_{\mathrm{crit}}(B,\beta) = \frac{\eta^{(\mu)}_{\mathrm{crit}}(\beta)}{1+\kappa_B(\beta)}, \qquad \kappa_B(\beta) = \kappa_\beta \frac{\nu_B^2}{(\lambda_1+\mu)^2}. \tag{1}$$

This is the exact one-dimensional mean-square threshold of heavy-ball SGD of Andreyev et al. [2026, Section 4.1 and Appendix B], $\eta_{\max} = 2a(1+\beta)/\big(a^2 + \kappa_\beta \sigma_b^2\big)$ with $a = \lambda_1 + \mu$ and $\sigma_b = \nu_B$, obtained from the linear recursion of $(\mathbb{E}[x_t^2], \mathbb{E}[x_t x_{t-1}], \mathbb{E}[x_{t-1}^2])$; for $\beta = 0$ it is the linear-stability condition $\eta < 2\lambda_1/(\lambda_1^2 + \nu_B^2)$ of Wu et al. [2018], and it is the necessary condition (17) of Mulayoff and Michaeli [2024] along u_1. Andreyev et al. [2026] use it to explain the two plateaus $2(1+\beta)/\eta$ and $2(1-\beta)/\eta$ of batch sharpness. Without curvature noise, the stationary normalised step variance of the mode is $\mathbb{E}[\Delta x_t^2]/\eta^2 = \sigma_1^2/\big((1-\beta^2)(1-\eta(\lambda_1+\mu)/(2(1+\beta)))\big)$, by the Yule–Walker equations of the second-order autoregression. The rest of this paper is about $\mathbb{R}^P$.

3 The exact second-moment equation in $\mathbb{R}^P$ and the stochastic ceiling

Proposition 1 is one-dimensional. The next two statements give the P-dimensional second-moment equation of heavy-ball SGD with a matrix curvature noise that need not commute with H, and the exact mean-square ceiling on a quadratic with random-feature minibatches. Let $H = \sum_k \lambda_k u_k u_k^\top \succ 0$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_P > 0$, and

$$w_{t+1} = \big((1+\beta)I - \eta H_{B_t}\big) w_t - \beta w_{t-1} - \eta \xi_t, \tag{2}$$

with (H_{B_t}, ξ_t) i.i.d. and independent of the past, $\mathbb{E}[H_{B_t}] = H$, $\mathbb{E}[\xi_t] = 0$, $\mathbb{E}[\xi_t \xi_t^\top] = \Sigma_\xi$, and $\mathcal{V}(X) \triangleq \mathbb{E}[(H_{B_t} - H) X (H_{B_t} - H)]$. Write $\Sigma_0 = \mathbb{E}[w_t w_t^\top]$, $D_\Theta = \mathbb{E}[\Delta w_t \Delta w_t^\top]$ and $[X, Y] = XY - YX$.

Theorem 2 (Exact Lyapunov equation and virial identity)**.** *In any covariance-stationary regime of* (2) *with* $\eta > 0$ *and* $\mathbb{E}[w_t] = 0$ *(automatic when* $H \succ 0$*, see the proof of Theorem 5), without any commutation assumption: (1)* $D_\Theta = \frac{\eta}{1+\beta}(H\Sigma_0 + \Sigma_0 H)$*; in the eigenbasis of* H*,* $\frac{1+\beta}{2\eta}\,\mathbb{E}[\langle \Delta w_t, u_k\rangle^2]/\mathbb{E}[\langle w_t, u_k\rangle^2] = \lambda_k$ *for every mode with* $\mathbb{E}[\langle w_t, u_k\rangle^2] > 0$*, identically in* β*,* $\mathcal{V}$ *and* Σ_ξ*. (2)* $(H\Sigma_0 + \Sigma_0 H) - \frac{\eta}{1+\beta} H\Sigma_0 H + \frac{\eta\beta}{(1-\beta)(1-\beta^2)}[H,[H,\Sigma_0]] - \frac{\eta}{1-\beta}\mathcal{V}(\Sigma_0) = \frac{\eta}{1-\beta}\Sigma_\xi$*. (3)* $u_j^\top [H,[H,\Sigma_0]] u_k = (\lambda_j - \lambda_k)^2 (\Sigma_0)_{jk}$ *vanishes on the diagonal; for* $\mathcal{V} = 0$ *and any* Σ_ξ *the normalized step cross-covariance* $C_{jk} = \eta^{-2}(D_\Theta)_{jk}$ *equals*

$$\frac{u_j^\top \Sigma_\xi u_k}{(1-\beta^2)\Big[1 - \frac{\eta\lambda_j\lambda_k}{(1+\beta)(\lambda_j+\lambda_k)} + \frac{\eta\beta(\lambda_j-\lambda_k)^2}{(1-\beta)(1-\beta^2)(\lambda_j+\lambda_k)}\Big]},$$

whose diagonal is the one-dimensional stationary step variance of Section 2 mode by mode.

At $\beta = 0$, part (2) has the form of the stationary covariance equation $(2C - \eta D)\,\mathrm{vec}\,\Sigma_0 = \eta p\,\mathrm{vec}\,\Sigma_g$ of Mulayoff and Michaeli [2024, Theorem 12], which they state at a regular minimum

for sampling without replacement, $p = (n - B)/(B(n - 1))$; sampling with replacement gives $p = 1/B$, their large-n value, and with that p the stability operator is theirs. Theorem 2 extends that exact equation to heavy-ball momentum with a curvature noise that need not commute with H; what $\beta > 0$ adds is the term $\frac{\eta\beta}{(1-\beta)(1-\beta^2)}[H,[H,\Sigma_0]]$, which vanishes on the diagonal of the eigenbasis and is stabilizing off it, and the factors $1/(1 \pm \beta)$. Andreyev et al. [2026, Appendix D] reach the stability boundary of the same system through the Schur complement $T_{\text{slow}}(\eta, \beta)$ of the second-moment operator, which they expand for small η with a remainder they state to be of order $\eta^2\|H\|^2/(1-\beta)^3$ (their Appendix H.1.1 evaluates this scale to 16.28 on their network). Theorem 2(2) gives that boundary without the expansion, and the two are related exactly: $I - T_{\text{slow}} = \mathcal{S} \circ \mathcal{L}$, where $\mathcal{L}(\Sigma)$ is $\eta/(1-\beta)$ times the left side of the equation of Theorem 2(2), curvature-noise term included, and $\mathcal{S}^{-1}(X) = X + \frac{\beta\eta}{(1-\beta)^2(1+\beta)}(HX + XH)$ does not depend on the curvature noise (Appendix A). For $H \succeq 0$, $\mathcal{S}$ is positive definite at every $\eta \geq 0$, so T_{slow} exists at every η, not only small ones, and the two equations have the same solutions for every Σ_ξ and are singular at the same η; $\mathcal{S} = I$ only at $\beta = 0$. Their remainder follows exactly (Lemma 15): $R(\Sigma) = \frac{2\beta\eta^2}{(1-\beta)^3}H\Sigma H + R_3(\Sigma)$ with an explicit $R_3 = O(\eta^3)$, so the scale $\eta^2\|H\|^2/(1-\beta)^3$ of their bound is attained, with constant 2β. It is carried by $H\Sigma H$: the $H^2\Sigma$ and ΣH^2 parts of our double commutator are absorbed by the left factor $\mathcal{S}$. The two readings differ off the diagonal of the eigenbasis, where $(H\Sigma H)_{jk} = \lambda_j\lambda_k\Sigma_{jk}$ and $[H,[H,\Sigma]]_{jk} = (\lambda_j - \lambda_k)^2\Sigma_{jk}$, and an entry with $\lambda_k = 0$ and no curvature noise on it decides between them: there the remainder vanishes identically, while the double-commutator entry does not. The expansion is also not uniform in β: the next term is not small against the leading one once $\beta\eta\|H\| \gtrsim (1-\beta)^2$ (Lemma 15 gives the full condition). On the run of their Appendix H.1.1 ($\eta = 0.01$, $\beta = 0.9$, $\lambda_{\max} = 12.76$, where they evaluate the scale to 16.28) the ratio $\beta\eta\lambda_{\max}/(1-\beta)^2$ is 11.5 and the multiplier of $\mathcal{S}$ on the top mode is 0.076, so neither their expansion nor the leading order of the remainder describes the operator there; equation (2) gives the boundary at any η. Part (1) is a matrix form of the kind of the first fluctuation–dissipation relation of Yaida [2019]; the co-diagonal additive-noise case of the stationary covariance is in Gitman et al. [2019]. Proofs in Appendix A.

Lemma 3 (The second-moment operator is decided on the symmetric matrices)**.** *Let* $\mathcal{T}(Q) = \mathbb{E}[F_tQF_t^\top]$ *act on all real* $2P \times 2P$ *matrices, with* $F_t = \begin{pmatrix} (1+\beta)I - \eta H_{B_t} & -\beta I \\ I & 0 \end{pmatrix}$ *the transition of* (2)*. Then* $\mathcal{T}$ *commutes with transposition and is positive,* $\|\mathcal{T}^k\| = \|\mathcal{T}^k(I)\|$ *for every* k *in the norm induced by the spectral norm, and*

$$\rho(\mathcal{T}) = \lim_{k\to\infty} \|\mathcal{T}^k(I)\|^{1/k} = \rho\big(\mathcal{T}|_{\mathbb{S}^{2P}}\big), \qquad \|\mathcal{T}^k(Q)\| \leq \|Q\|\,\|\mathcal{T}^k(I)\| \text{ for every } Q.$$

Proof. $\mathcal{T}(Q)^\top = \mathbb{E}[F_tQ^\top F_t^\top] = \mathcal{T}(Q^\top)$, so $\mathcal{T}$ preserves the symmetric and the antisymmetric matrices. $\mathcal{T}$ is in Kraus form, hence positive (even completely positive) on the complexified matrix algebra, and so is every $\mathcal{T}^k$; the Russo–Dye theorem gives $\|\mathcal{T}^k\| = \|\mathcal{T}^k(I)\|$. By Gelfand's formula, which does not depend on the norm, $\rho(\mathcal{T}) = \lim_k \|\mathcal{T}^k\|^{1/k} = \lim_k \|\mathcal{T}^k(I)\|^{1/k} \leq \rho(\mathcal{T}|_{\mathbb{S}^{2P}})$, because $\mathcal{T}^k(I) \in \mathbb{S}^{2P}$; the reverse inequality is trivial. The last bound is $\|\mathcal{T}^k(Q)\| \leq \|\mathcal{T}^k\|\,\|Q\|$. □

So the stability of the whole recursion is decided by the orbit of the identity, the antisymmetric part never dominates, and Theorem 5 can be stated on $\mathbb{S}^{2P}$ without loss. The orbit also gives a matrix-free way to compute the ceiling: iterate $Q \mapsto \mathcal{T}(Q)/\|\mathcal{T}(Q)\|$ from $Q_0 = I$, which is a power iteration on the positive semidefinite cone at the cost of one application of $\mathcal{T}$ per step.

Corollary 4 (Mean-square instability criterion)**.** *In the setting of* (2) *with* $\Sigma_\xi \succ 0$*, let* $\mathcal{T}$ *be as in Lemma 3, and assume either that* ξ_t *is independent of* H_{B_t} *or that* $\mathbb{E}[w_0] = \mathbb{E}[w_{-1}] = 0$*. Then*

$\sup_t \mathbb{E}\|w_t\|^2 < \infty$ *if and only if* $\rho(\mathcal{T}) < 1$. *If* $\rho(\mathcal{T}) \geq 1$, *then* $\mathbb{E}\|w_t\|^2 + \mathbb{E}\|w_{t-1}\|^2 \geq c\,t$ *for every* $t \geq 1$ *with a constant* $c > 0$, *so* $\sup_t \mathbb{E}\|w_t\|^2 = \infty$, *and for every compact* $K \subset \mathbb{R}^P$ *there is a finite* T *with* $\Pr(w_T \notin K) > 0$. *For the random-feature minibatches of Theorem 5 below,* $\rho(\mathcal{T}) \geq 1$ *if and only if* $\eta \geq \eta_{\mathrm{crit}}^{\mathrm{EoSS},P}(\beta, B)$.

That bounded second moments of a linear recursion with i.i.d. coefficients are equivalent to $\rho(\mathbb{E}[F_t \otimes F_t]) < 1$ is standard [Costa et al., 2005, Chapter 3], and it is the criterion used for SGD without momentum by Ma and Ying [2021] and Mulayoff and Michaeli [2024]. The corollary records it for (2), with the divergence statement used below. Its last conclusion has the form of the escape condition of Andreyev et al. [2026, Definition A.1], but on this model that condition does not single out $\rho(\mathcal{T})$: when ξ_t has unbounded support (for instance Gaussian), $\Pr(w_1 \notin K) > 0$ for every compact K at every η, also where $\rho(\mathcal{T}) < 1$; with Gaussian features one minibatch has unbounded curvature, so the same holds from every start $w_0 \neq 0$ without additive noise; and $\rho(\mathcal{T})$ does not depend on the initial parameter, so as a function $f(\theta_0)$ it is constant and the definition then asks for escape from every start; read as escape in probability, $\Pr(w_t \notin K) \to 1$ for every compact K, it does not follow from $\rho(\mathcal{T}) \geq 1$, since between the mean-square and the almost-sure thresholds of Proposition 7 typical trajectories decay. The content of the corollary is the moment bound. Without additive noise, $\Sigma_\xi = 0$, the same eigenmatrix gives $\langle W, Q_t\rangle = \rho(\mathcal{T})^t \langle W, Q_0\rangle$, so $\mathbb{E}\|w_t\|^2 + \mathbb{E}\|w_{t-1}\|^2 \geq \rho(\mathcal{T})^t \langle W, Q_0\rangle / \operatorname{tr} W$ for every start with $\langle W, Q_0\rangle > 0$, for instance a deterministic $w_0 \notin \ker W_{11}$ with $w_{-1} = 0$: at $\rho(\mathcal{T}) \geq 1$ the second moment from such a start does not decay, and for $\rho(\mathcal{T}) > 1$ it grows geometrically. It needs $\Sigma_\xi \succ 0$: for the homogeneous recursion, the one of their Appendix D, $\rho(\mathcal{T}) \geq 1$ does not force escape (the start $w_0 = w_{-1} = 0$ stays at 0), and the bound $\langle W, Q_t\rangle \geq tg$ is what the additive noise supplies.

Proof. Let $s_t = (w_t, w_{t-1})$, $Q_t = \mathbb{E}[s_t s_t^\top]$ and $G = \operatorname{blkdiag}(\eta^2 \Sigma_\xi, 0)$. The cross terms $\mathbb{E}[F_t s_t (\xi_t^\top, 0)]$ vanish: s_t is independent of (F_t, ξ_t), and either F_t is independent of ξ_t, which has mean 0, or $\mathbb{E}[s_t] = 0$ for every t, because $\mathbb{E}[w_t]$ follows the deterministic heavy-ball recursion with H from 0. So $Q_{t+1} = \mathcal{T}(Q_t) + G$. If $\rho(\mathcal{T}) < 1$, $Q_t = \mathcal{T}^t(Q_0) + \sum_{s<t} \mathcal{T}^s(G)$ is bounded. The adjoint $\mathcal{T}^*(Y) = \mathbb{E}[F_t^\top Y F_t]$ maps the positive semidefinite cone of $\mathbb{S}^{2P}$ into itself, so by the Krein–Rutman theorem in finite dimension [Berman and Plemmons, 1994, Thm. 1.3.2] it has an eigenvector $W \succeq 0$, $W \neq 0$, with eigenvalue $\rho(\mathcal{T}^*) = \rho(\mathcal{T})$. Its upper-left block W_{11} is not zero: if it were, $W = \operatorname{blkdiag}(0, W_{22})$, and $F_t^\top W F_t = \operatorname{blkdiag}(W_{22}, 0)$ for every F_t, so $\rho W = \mathcal{T}^*(W)$ forces $W_{22} = 0$. Hence $g = \langle W, G\rangle = \eta^2 \operatorname{tr}(W_{11}\Sigma_\xi) > 0$ and $\langle W, Q_{t+1}\rangle = \rho(\mathcal{T})\langle W, Q_t\rangle + g$. For $\rho(\mathcal{T}) \geq 1$ this gives $\langle W, Q_t\rangle \geq tg$, and $\langle W, Q_t\rangle \leq \operatorname{tr} W \operatorname{tr} Q_t = \operatorname{tr} W\,(\mathbb{E}\|w_t\|^2 + \mathbb{E}\|w_{t-1}\|^2)$, so the bound holds with $c = g/\operatorname{tr} W$. If $w_T \in K$ almost surely for every T, then $\mathbb{E}\|w_T\|^2 \leq \max_K \|w\|^2$ for every T, a contradiction. The last statement is Theorem 5, whose proof shows $\rho(\mathcal{T}) < 1$ for $\eta < \eta_{\mathrm{crit}}^{\mathrm{EoSS},P}$ and the absence of a finite stationary second moment otherwise; the latter excludes $\rho(\mathcal{T}) < 1$, since then $\sum_s \mathcal{T}^s(G)$ would be one. □

With three equiprobable non-commuting 2×2 minibatch Hessians and $\beta = 0.9$, the exact recursion stays bounded at $0.9\,\eta^\star$ ($\rho(\mathcal{T}) = 0.9886$), and $\langle W, Q_t\rangle \geq tg$ holds for $t \leq 3000$ at $\eta^\star$ and $1.02\,\eta^\star$ (`paper/checks/ms-instability-criterion.mjs`, which also matches the recursion to a Monte Carlo run).

The open question of Andreyev et al. [2026]. For SGD without momentum, Andreyev and Beneventano [2024] prove that on the quadratic approximation batch sharpness above $2/\eta$ forces divergence. For momentum, Andreyev et al. [2026, Section 3] write that "for SGDM and SGDN, we do not have an analogous theoretical result" and, in a later paragraph, that "developing such a theory for momentum methods, including identifying the precise instability criterion and proving

that it is saturated under progressive sharpening, remains an important open question"; the first item of their future work (Section 6.2) is "finding a valid instability criterion for SGDM and SGDN". Their Definition A.1 fixes an open set U of parameters, where a local model of the loss is trusted, and calls $f : U \to \mathbb{R}$ a valid instability criterion with threshold c on U if $f(\theta_0) > c$ implies that the iterates leave every compact subset of U, equivalently that for every compact $K \subset U$ containing θ_0 there is a finite T with $\Pr(\theta_T \notin K \mid \theta_0) > 0$; on the quadratic model of this paper $U = \mathbb{R}^P$. Their Appendix D writes the homogeneous second-moment recursion of linearized SGDM, states that its mean-square stability is $\rho(T_{\mathrm{HB}}) < 1$, and reduces this to $\rho(T_{\mathrm{slow}}(\eta, \beta)) < 1$ for sufficiently small η (their eq. (31)); it does not show that $\rho \geq 1$ forces escape in the sense of their Definition A.1, and for the homogeneous recursion it does not (the start $w_0 = w_{-1} = 0$ stays at 0). Their Theorem 4.1 keeps the leading order of T_{slow}, "in the noise-dominated regime and for sufficiently small η": it is the SGD condition of Ma and Ying [2021] at $\eta_{\mathrm{eff}} = \eta/(1-\beta)$, the step-size rescaling that Yuan et al. [2016] establish for momentum methods with a small constant step, $\rho(I - \eta_{\mathrm{eff}} K_H + \eta_{\mathrm{eff}}^2 G) < 1$ in the notation of Lemma 15, and the dropped remainder is bounded only by $\|R\| \lesssim \eta^2 \|H\|^2/(1-\beta)^3$. For heavy-ball SGD on the quadratic model this paper adds, to the linear mean-square part of their question: (i) a closed stationary equation that decides the boundary at every $\eta > 0$ and $\beta \in [0, 1)$, Theorem 2(2), with no small-η and no noise-dominated assumption, and its exact relation $I - T_{\mathrm{slow}} = \mathcal{S} \circ \mathcal{L}$ to their Schur complement; (ii) the generality of their Appendix D in the minibatch Hessians, which need not commute with each other or with H, without the small-η expansion; (iii) that for $\rho(\mathcal{T}) \geq 1$ the second moment grows at least linearly, $\mathbb{E}\|w_t\|^2 + \mathbb{E}\|w_{t-1}\|^2 \geq ct$ (Corollary 4), whose probabilistic conclusion has the form of their Definition A.1 but is also met below $\rho(\mathcal{T}) = 1$ (additive noise of unbounded support meets it at every η), so it does not settle the first item of their Section 6.2; the content is the moment bound, the heavy-ball analogue of the divergence theorem of Andreyev and Beneventano [2024] for batch sharpness at $\beta = 0$, stated for $\rho(\mathcal{T})$ and not for batch sharpness; the argument is the standard one for linear systems with i.i.d. coefficients [Costa et al., 2005] and needs the additive noise $\Sigma_\xi \succ 0$, which their model, an interpolating minimum, does not have (their footnote 5 states that relaxing interpolation changes the steady-state variance but not the stability criteria); (iv) their remainder exactly, $R = \frac{2\beta\eta^2}{(1-\beta)^3} H\Sigma H + R_3$, which attains the scale of their bound and is not small once $\beta\eta\|H\| \gtrsim (1-\beta)^2$ (Lemma 15), a ratio equal to 11.5 on the run where they evaluate it; (v) for random-feature minibatches, the boundary as the root of a secular equation (Theorem 5), whose two limits take the values of their plateaus $2(1+\beta)/\eta$ and $2(1-\beta)/\eta$, through λ_1 and $\operatorname{tr} H/B$ (Corollary 6). For SGD with Nesterov momentum, the exact mean-square criterion on random quadratics in any dimension is the outer-momentum criterion of Sharma et al. [2026, Theorem 1] with one local step. Concurrently and independently, Cai et al. [2026, Theorem F.1] give the exact mean-square criterion of the homogeneous heavy-ball recursion with i.i.d. curvature noise that need not commute with the mean; it is the threshold of Theorem 9, and their comparison of two momentum values (their eq. (67)) gives the monotonicity of Corollary 10 without strictness. Whether training saturates the criterion is measured, on one 33-parameter network and with the quasi-static diagnostic of Section 4.4, in Phases 5 and 5b: the runs cross the $|H|$ reading between $\eta = 0.1$ and 0.15 and train up to 5.1× above it without diverging, at checkpoints where H is indefinite and the linearised iteration has no mean-square stable step. On the networks with $P = 3\,610$ and $12\,826$ parameters, Phase 6 brackets the threshold of the Gauss–Newton surrogate from both sides and finds training at it, within 0.81–$1.10\,\eta$ over the window, for $\eta \geq 0.02$ (Section 4.5). Theorem 13 relates the criterion to batch sharpness: at the threshold an identity puts a modal curvature between the two SGDM plateau values $2(1-\beta)/\eta^\star$ and $2(1+\beta)/\eta^\star$, a stationary batch sharpness lies above it for positive semidefinite minibatch Hessians and, in the random-feature model, reaches the plateaus in the two batch limits, and a

variational test gives sufficient instability conditions through a fixed-checkpoint batch sharpness and the noise share; Theorem 14 proves saturation in a quasi-stationary slow-fast model, with an equilibrium at distance $\Theta(\sigma_\xi^2/v_{\mathrm{PS}})$ below the threshold, and a secant equilibrium above the tangent threshold. Not answered here: the dynamics of training beyond the quadratic and slow-fast models, which is what bounds those runs, and a threshold on the batch sharpness of Andreyev et al. [2026, Definition 3.1] alone. The criterion is also a mean-square one: between the mean-square and the almost-sure thresholds of Proposition 7 the second moment diverges while typical trajectories decay (the failed cell of Phase 1, Section 4.1).

Theorem 5 (Secular equation of the stochastic ceiling)**.** *Let $H_{B_t} = \frac{1}{B}\sum_{b=1}^{B} z_{i_b} z_{i_b}^\top$ with $z_i = H^{1/2}\omega_i$, where the coordinates $\omega_{i,k}$ (in the eigenbasis of H) are independent with mean 0, variance 1 and $\mathbb{E}[\omega_{i,k}^4] = 3 + \kappa_{\mathrm{feat}}$, $\kappa_{\mathrm{feat}} \geq -2$ (0 for Gaussian features, -2 for Rademacher features, for which $\omega^4 \equiv 1$), the B draws i.i.d.; let $\sigma_{\xi,k}^2 = u_k^\top \Sigma_\xi u_k > 0$, $\kappa_\beta = \frac{1+\beta}{1-\beta}$, $\alpha_{\mathrm{diag}} = 1 + \kappa_\beta \frac{1+\kappa_{\mathrm{feat}}}{B}$, $\Lambda = \mathrm{diag}(\lambda_k)$ and $\lambda = (\lambda_1, \dots, \lambda_P)^\top$. Then $\mathrm{Var}(u_k^\top H_{B_t} u_j) = \frac{\lambda_k \lambda_j}{B}\big(1 + (1+\kappa_{\mathrm{feat}})\delta_{kj}\big)$, the off-diagonal entries of Σ_0 decouple from each other and from the diagonal, the diagonal normalized step variances $c = (C_{kk})_k$ solve*

$$\Big[I - \tfrac{\eta}{2(1+\beta)}\big(\alpha_{\mathrm{diag}}\Lambda + \tfrac{\kappa_\beta}{B}\lambda \mathbf{1}^\top\big)\Big] c = \frac{\sigma_\xi^2}{1-\beta^2}, \tag{3}$$

and a finite stationary second moment of (w_t, w_{t-1}) exists if and only if $\eta < \eta_{\mathrm{crit}}^{\mathrm{EoSS},P}(\beta, B)$, the unique root in $\big(0, \frac{2(1+\beta)}{\lambda_1 \max(\alpha_{\mathrm{diag}}, 1)}\big]$ (the right end is attained only without curvature noise on the top mode, as for $P = 1$ and Rademacher features) of

$$\frac{\eta}{2(1-\beta)B} \sum_{k=1}^{P} \frac{\lambda_k}{1 - \frac{\eta\lambda_k}{2(1+\beta)}\alpha_{\mathrm{diag}}} = 1. \tag{4}$$

With $\mu^\star = 2(1+\beta)/\eta_{\mathrm{crit}}^{\mathrm{EoSS},P}(\beta, B)$,

$$\max(\alpha_{\mathrm{diag}}, 0)\lambda_1 + \kappa_\beta \frac{\mathrm{tr}\, H}{B} \;\geq\; \mu^\star \;\geq\; \max\Big(\alpha_{\mathrm{diag}}\lambda_1 + \kappa_\beta \frac{\lambda_1}{B},\; \min(\alpha_{\mathrm{diag}}, 0)\lambda_1 + \kappa_\beta \frac{\mathrm{tr}\, H}{B}\Big), \tag{5}$$

with $\alpha_{\mathrm{diag}}\lambda_1 + \kappa_\beta\lambda_1/B = \lambda_1(1 + \kappa_\beta(2+\kappa_{\mathrm{feat}})/B) \geq \lambda_1$. For $P = 1$, (4) is the ceiling of Proposition 1 with $\nu_B^2 = \lambda_1^2(2 + \kappa_{\mathrm{feat}})/B$.

Corollary 6 (Two plateaus; the Rademacher separation)**.** *Write $\varepsilon_B = \kappa_\beta/B$ and $T(\mu) = \sum_{k\geq 2} \lambda_k/(\mu - \alpha_{\mathrm{diag}}\lambda_k)$. At the ceiling, $\mu^\star = 2(1+\beta)/\eta_{\mathrm{crit}}^{\mathrm{EoSS},P}$ satisfies exactly $\mu^\star = \alpha_{\mathrm{diag}}\lambda_1 + \varepsilon_B\lambda_1/\big(1 - \varepsilon_B T(\mu^\star)\big)$. If $\alpha_{\mathrm{diag}} > 0$, $\lambda_1 > \lambda_2$ and, with $S_\perp = \sum_{k\geq 2} \lambda_k/(\lambda_1 - \lambda_k)$, $\varepsilon_B S_\perp/\alpha_{\mathrm{diag}} \leq \frac{1}{2}$ (large batch with a spectral gap), then $\lambda_1 = \frac{2(1+\beta)}{\eta}\big(1 - \varepsilon_B(2 + \kappa_{\mathrm{feat}}) + \varrho\big)$ at $\eta = \eta_{\mathrm{crit}}^{\mathrm{EoSS},P}$, with $-2\varepsilon_B^2 S_\perp/\alpha_{\mathrm{diag}} \leq \varrho \leq \varepsilon_B^2\big(2 + \kappa_{\mathrm{feat}} + 2\varepsilon_B S_\perp/\alpha_{\mathrm{diag}}\big)^2$. The gap is needed, not an artifact of the proof: with $\lambda_1 = \lambda_2$ the top eigenvalue on those two modes is $\lambda_1(1 + \varepsilon_B(3 + \kappa_{\mathrm{feat}}))$, so the first-order coefficient itself changes. As $\kappa_\beta \,\mathrm{tr}\, H/(B\lambda_1) \to \infty$ with $\lambda_1/\,\mathrm{tr}\, H \to 0$ (small batch, high intrinsic rank), $\mathrm{tr}\, H/B = \frac{2(1-\beta)}{\eta}(1 + o(1))$ at the ceiling. The two empirical plateaus of Andreyev et al. [2026, eqs. 1–2], $2(1+\beta)/\eta$ and $2(1-\beta)/\eta$, are the values of the two limits of (4), taken by λ_1 and by the batch trace-sharpness $\mathrm{tr}\, H/B$ respectively, not by batch sharpness itself; Theorem 13(2) gives the two limits of a stationary batch sharpness, and the large-batch plateau of SGD with Nesterov momentum in Andreyev et al. [2026], $2(1+\beta)/(\eta(1+2\beta))$, is not covered. For Rademacher features the directional curvature along u_1 is deterministic, $\mathrm{Var}(u_1^\top H_{B_t} u_1) = 0$, so the one-dimensional ceiling of Proposition 1 is $2(1+\beta)/\lambda_1$ for every B, while by (5), with $\mathrm{tr}\, H_\perp = \mathrm{tr}\, H - \lambda_1$, $\eta_{\mathrm{crit}}^{\mathrm{EoSS},P} \leq 2(1+\beta)/(\lambda_1 + \kappa_\beta \,\mathrm{tr}\, H_\perp/B)$*

for $B \le \kappa_\beta$ and $\eta_{\rm crit}^{{\rm EoSS},P} \le 2(1+\beta)/\max(\lambda_1, \kappa_\beta \operatorname{tr} H/B)$ for $B > \kappa_\beta$: the cross-mode couplings $\lambda_1\lambda_k/B$ lower the ceiling, and a one-dimensional probe along u_1 cannot see them.

The same structure, a cone-preserving second-moment operator with a rank-one global coupling, the Krein–Rutman theorem, a secular equation and bounds by $\operatorname{tr} H$ and $\lambda_{\max}$ with the factor $2(1-\beta)$, appears in the mean-square stability of full-batch zeroth-order methods with Gaussian two-point directions [Song et al., 2026]: ZO-GD, ZO-GDM, and ZO-Adam under the assumption that the preconditioner commutes with H; mini-batch zeroth-order methods are left there as future work. There the randomness is the search direction and the Hessian is the full-batch one. Here it is the minibatch: the sampled Hessians need not commute with each other or with H (Theorem 2, Lemma 3), the coupling carries the feature kurtosis through $\alpha_{\rm diag}$, and the curvature-noise covariance need not be H (Theorem 8). A small-batch edge of stability governed by a trace rather than by the top Hessian eigenvalue also appears in the high-dimensional model of Agarwala and Pennington [2024], through the trace of the neural tangent kernel.

Proposition 7 (Mean-square against almost-sure stability for all $\beta \in [0,1)$; a distribution-free trace identity)**.** *(1) For the homogeneous Heavy-Ball recursion $x_{t+1} = (1+\beta-\eta(\lambda_1+\zeta_t))x_t - \beta x_{t-1}$ along u_1, with $\beta \in [0,1)$, $\mathbb{E}\zeta_t = 0$ and $\operatorname{Var}\zeta_t = \nu_{11,B}^2 < \infty$, the mean-square threshold is exactly $\frac{2(1+\beta)}{\lambda_1}(1+\kappa_\beta\nu_{11,B}^2/\lambda_1^2)^{-1} = \frac{2(1+\beta)}{\lambda_1}\big(1-\kappa_\beta\nu_{11,B}^2/\lambda_1^2+\varrho\big)$, where $0 \le \varrho \le \big(\kappa_\beta\nu_{11,B}^2/\lambda_1^2\big)^2$. Let $u_t = 2\kappa_\beta\delta + 2\kappa_\beta(1+\delta)\zeta_t/\lambda_1$ at $\eta = \frac{2(1+\beta)}{\lambda_1}(1+\delta)$, and write $m_{3,B} = \mathbb{E}|\zeta_t|^3$. Set $\varepsilon_0(0) = \frac12$ and $\varepsilon_0(\beta) = \frac{1-\beta}{8(1+\beta)}$ for $\beta \in (0,1)$. If $|u_t| \le \varepsilon_0(\beta)$ almost surely, $m_{3,B} < \infty$ and $|\delta| \le \frac12$, then the top Lyapunov exponent $\gamma_{\rm Lyap}(\eta,\beta) = \lim_{t\to\infty}\frac1t\log\|M_t\cdots M_1\|$ of the companion matrix M_t [Furstenberg and Kesten, 1960, Caughey and Gray, 1965, Khasminskii, 1967, Kozin, 1969, Ariaratnam and Tam, 1979, Arnold, 1984] satisfies $\gamma_{\rm Lyap}(\eta,\beta) = 2\kappa_\beta\delta - 2\kappa_\beta^2\nu_{11,B}^2/\lambda_1^2 + O\big(\delta^2 + m_{3,B}/\lambda_1^3 + \nu_{11,B}^4/\lambda_1^4\big)$, with an implied constant depending only on β. If in addition $|\zeta_t| \le c_0(\beta)\lambda_1$ almost surely for a positive $c_0(\beta)$ depending only on β, then on the range $|\delta| \le c_1(\beta)$ the exponent is negative below and positive above a window of width $O(m_{3,B}/\lambda_1^3 + \nu_{11,B}^4/\lambda_1^4)$ around $\delta = \kappa_\beta\nu_{11,B}^2/\lambda_1^2$, so the almost-sure threshold there is $\eta_{\rm crit}^{\rm a.s.}(\beta) = \frac{2(1+\beta)}{\lambda_1}\big(1+\kappa_\beta\nu_{11,B}^2/\lambda_1^2 + O(m_{3,B}/\lambda_1^3 + \nu_{11,B}^4/\lambda_1^4)\big)$. At $\beta = 0$, for each fixed $p > 0$ there is a $c_p > 0$ such that $|\zeta_t| \le c_p\lambda_1$ almost surely implies that the threshold of $\mathbb{E}|1-\eta(\lambda_1+\zeta_t)|^p = 1$ is $\frac{2}{\lambda_1}\big(1-(p-1)\nu_{11,B}^2/\lambda_1^2 + O_p(m_{3,B}/\lambda_1^3 + \nu_{11,B}^4/\lambda_1^4)\big)$. At $\beta = 0$ the support condition can be replaced by a density bound near the singular multiplier: if $|\delta| \le \frac18$, $m_{3,B} < \infty$ and the law of $1+u_t$ has a density bounded by $M_{\rm sing}$ on $[-\frac12,\frac12]$, then $\big|\gamma_{\rm Lyap}(\eta,0) - (\mathbb{E}u_t - \frac12\mathbb{E}u_t^2)\big| \le 11\,\mathbb{E}|u_t|^3 + (1+\log 2)M_{\rm sing}$, and the root at $\beta = 0$ is $\frac{2}{\lambda_1}\big(1+\nu_{11,B}^2/\lambda_1^2 + O(m_{3,B}/\lambda_1^3 + \nu_{11,B}^4/\lambda_1^4 + M_{\rm sing})\big)$. With $\varepsilon = \nu_{11,B}^2/\lambda_1^2$, Gaussian ζ_t has $M_{\rm sing} \le (2\pi\frac{49}{16}\varepsilon)^{-1/2}e^{-1/(162\varepsilon)}$, and $\zeta_t = \lambda_1(\chi_B^2/B - 1)$ has $M_{\rm sing} \le \frac{18}{7}\sqrt{B/(2\pi)}\,e^{-0.0056B}$; both are smaller than every power of $1/B$ as $B \to \infty$, but not small at moderate B. A finite third moment alone does not give these remainders: if the law of u_t charges every neighbourhood of -1 with an unbounded density, then $\mathbb{E}\big[|\log|1+u_t|\,|\,\mathbf{1}_{|1+u_t|\le 1/2}\big]/m_{3,B}$ can be arbitrarily large. For $\beta \in (0,1)$ with unbounded ζ_t the remainder is left open. If $\zeta_t = 0$ almost surely, both thresholds equal $2(1+\beta)/\lambda_1$, which is the Rademacher case $\nu_{11,B}^2 = 0$. Rosenthal's inequality gives $\nu_{11,B}^2 = O(1/B)$ and $m_{3,B} = O(B^{-3/2})$ for an average of B i.i.d. summands with a finite third moment; those rates enter the remainders only when $|u_t| \le \varepsilon_0(\beta)$ almost surely. (2) In the setting of Theorem 2 with $H_{B_t} = \frac1B\sum_b H_{i_b}$, $H_i \succeq 0$ i.i.d., $\mathbb{E}H_i = H$ and no further structure, every stationary Σ_0 satisfies $\operatorname{tr}\big([2H - \frac{\eta}{1+\beta}H^2 - \frac{\eta}{(1-\beta)B}(\mathbb{E}H_i^2 - H^2)]\Sigma_0\big) = \frac{\eta}{1-\beta}\operatorname{tr}\Sigma_\xi$, since $\operatorname{tr}[H,[H,\Sigma_0]] = 0$; multiplying by H^{-1} before the trace gives $\operatorname{tr}\big([2I - \frac{\eta}{1+\beta}(H + \frac{\kappa_\beta}{B}\mathbb{E}[(H_i-H)H^{-1}(H_i-H)])]\Sigma_0\big) = \frac{\eta}{1-\beta}\operatorname{tr}(H^{-1}\Sigma_\xi) > 0$ with a positive semidefinite correction, hence the distribution-free necessary condition $\eta_{\rm crit}^{{\rm EoSS},P} \le 2(1+\beta)/\lambda_{\min}\big(H + \frac{\kappa_\beta}{B}\mathbb{E}[(H_i-H)H^{-1}(H_i-H)]\big)$ for every $B \ge 1$; and if $\mathbb{E}[H_iXH_i] \succeq c_0\operatorname{tr}(HX)H$*

for $X \succeq 0$ and $B \geq \kappa_\beta$, $\eta_{\text{crit}}^{\text{EoSS},P} \leq 2(1-\beta)B/(c_0 \operatorname{tr} H)$ (equality is possible: $H_i \in \{0,2\}$, $\beta = 0$, $B = 1$, $c_0 = 2$). Proof in Appendix A.

Numerically, $M_{\text{sing}} \approx 0.4$ at $B = 64$ and below 10^{-6} from $B = 1024$ (`paper/checks/g3-singular-density.mjs`). For $\lambda_1 = 1$, $B = 64$, $\beta = 0$ and $\zeta_t = \chi_B^2/B - 1$, the mean-square formula gives 1.9394; the root of $\mathbb{E} \log|1 - \eta\chi_B^2/B| = 0$ is 2.0688 by quadrature, against the first-order values 1.9375 and 2.0625, and a Gaussian ζ_t of the same variance has quadrature root 2.078 (`paper/checks/g3-lyapunov-remainder.mjs`); these two roots are values of the integral. Part (1) is the mechanism behind the one failed cell of Phase 1 (Section 4.1): a finite sample-path harness follows typical trajectories, so its threshold can lie above the mean-square one, toward the almost-sure one, and Theorem 5 is about the mean-square one. Jensen's inequality puts the almost-sure threshold at or above the mean-square one; under the support bound they differ by $\pm\kappa_\beta \nu_{11,B}^2/\lambda_1^2$ to first order.

Theorem 8 (Secular equation for a curvature-noise covariance that is not the Hessian; flat-mode collapse)**.** *In the setting of Theorem 2 let $H = \sum_k \lambda_k u_k u_k^\top \succ 0$ and $H_{B_t} = H + \frac{1}{B}\sum_b (z_{i_b} z_{i_b}^\top - \Sigma_z)$ with $z_i = \Sigma_z^{1/2}\omega_i$, $\Sigma_z = \operatorname{diag}(s_1, \ldots, s_P) \succeq 0$ in the eigenbasis of H (for instance a Gauss–Newton covariance $\Sigma_z = H_{\text{GN}} \neq H$ that commutes with H), and $\omega_{i,k}$ independent, standardised, with $\mathbb{E}\omega_{i,k}^4 = 3 + \kappa_{\text{feat}}$, $\kappa_{\text{feat}} \geq -2$. Write $\rho_k = s_k^2/\lambda_k$, $d_k = \lambda_k + \kappa_\beta \frac{1+\kappa_{\text{feat}}}{B}\rho_k$ and $\mathcal{T}_{\text{flat}} = \sum_k \rho_k = \operatorname{tr}(H^{-1}\Sigma_z^2)$ ($\Sigma_z = H$ gives $\rho_k = \lambda_k$, $d_k = \alpha_{\text{diag}}\lambda_k$ and Theorem 5). (1) In the eigenbasis, $(\mathcal{V}(X))_{kk} = \frac{s_k}{B}\sum_j s_j X_{jj} + \frac{1+\kappa_{\text{feat}}}{B} s_k^2 X_{kk}$ and $(\mathcal{V}(X))_{jk} = \frac{s_j s_k}{B} X_{kj}$ for $j \neq k$ ($= \frac{s_j s_k}{B} X_{jk}$ for symmetric X), so the diagonal of the stationary equation closes on $\gamma_{0,k} = (\Sigma_0)_{kk}$ and reads $2\lambda_k\gamma_{0,k} - \frac{\eta}{1+\beta}\lambda_k^2\gamma_{0,k} - \frac{\eta}{1-\beta}\big(\frac{s_k}{B}\sum_j s_j\gamma_{0,j} + \frac{1+\kappa_{\text{feat}}}{B}s_k^2\gamma_{0,k}\big) = \frac{\eta}{1-\beta}u_k^\top\Sigma_\xi u_k$. (2) For $\Sigma_\xi \succ 0$, a finite stationary second moment exists if and only if $\eta < \eta^\star \triangleq 2(1+\beta)/\lambda_{\max}(\widetilde{\mathcal{M}})$, $\widetilde{\mathcal{M}} = \operatorname{diag}(d_k) + \frac{\kappa_\beta}{B} v v^\top$, $v_k = \sqrt{\rho_k}$. With $\mathcal{A} = \{k : s_k > 0\}$, a mode outside $\mathcal{A}$ is decoupled ($d_k = \lambda_k$, $v_k = 0$), so $\eta^\star$ is the smaller of $\min_{k\notin\mathcal{A}} 2(1+\beta)/\lambda_k$ and, when $\mathcal{A} \neq \emptyset$, the unique root in $(0, \min_{k\in\mathcal{A}} 2(1+\beta)/\max(d_k, \lambda_k)]$ of the secular equation (whose terms with $k \notin \mathcal{A}$ vanish)*

$$\frac{\eta}{2(1-\beta)B}\sum_{k=1}^{P}\frac{s_k^2/\lambda_k}{1 - \frac{\eta\lambda_k}{2(1+\beta)} - \frac{\eta(1+\kappa_{\text{feat}})}{2(1-\beta)B}\frac{s_k^2}{\lambda_k}} = 1.$$

(3) With $\mu^\star = 2(1+\beta)/\eta^\star$, $\max_k d_k + \frac{\kappa_\beta}{B}\mathcal{T}_{\text{flat}} \geq \mu^\star \geq \max\big\{\max_k(\lambda_k + \kappa_\beta\frac{2+\kappa_{\text{feat}}}{B}\rho_k),\ \min_k d_k + \frac{\kappa_\beta}{B}\mathcal{T}_{\text{flat}}\big\}$; in particular $\eta^\star \leq 2(1+\beta)/\lambda_1$ for every $\kappa_{\text{feat}} \geq -2$, and $\eta^\star \leq 2(1-\beta)B/\mathcal{T}_{\text{flat}}$ whenever $\min_k d_k \geq 0$ (for instance $\kappa_{\text{feat}} \geq -1$); without that condition the second bound can fail. For every $\kappa_{\text{feat}} \geq -2$ and $P \geq 2$, $\eta^\star \leq \min_{j\neq k}\frac{2(1-\beta)B\sqrt{\lambda_j\lambda_k}}{s_j s_k}$, so a mode with $\lambda_k \to 0$ at fixed $s_k > 0$, with some $s_j > 0$, $j \neq k$, drives $\eta^\star$ to zero as $O(\sqrt{\lambda_k})$, and with $P = 1$ and Rademacher features there is no collapse: curvature noise in a flat direction has no restoring force to balance it, a mechanism excluded by $\Sigma_z = H$. A tether $\frac{\mu}{2}\|w - \Theta_0\|^2$ leaves s_k unchanged and replaces λ_k by $\lambda_k + \mu$, so $\rho_k(\mu) \leq s_k^2/\mu$ and $\mathcal{T}_{\text{flat}}(\mu) \leq \operatorname{tr}(\Sigma_z^2)/\mu$. Proof in Appendix A.

Examples for part (3): with Rademacher features, $\Sigma_z = H = I_2$, $B = 1$ and $\beta = \frac{1}{2}$, $\eta^\star = 0.75$ exceeds the bound $2(1-\beta)B/\mathcal{T}_{\text{flat}} = 0.5$, which therefore needs $\min_k d_k \geq 0$; with $\lambda = (1, 10^{-4})$, $s = (1,1)$, $\beta = \frac{1}{2}$, $B = 1$ and Rademacher features, $\eta^\star = 0.00998$ against the bound 0.01. On the network checkpoints of Phase 2 the negative directions of H dominate instead of flat modes (Section 4.3).

Theorem 9 (Exact mean-square threshold in $\mathbb{R}^P$ for arbitrary non-commuting per-sample Hessians)**.** *In the setting of Theorem 2, let $H = \frac{1}{N}\sum_{i=1}^N H_i \succ 0$ (or $H + \mu I \succ 0$ under a tether*

$\mu > 0$) *have eigenvalues* $\lambda_1 \geq \cdots \geq \lambda_P > 0$ *and top eigenspace* $\mathcal{U}_1 = \ker(H - \lambda_1 I)$ *with orthogonal projector* P_1. *Let* $H_{B_t} = \frac{1}{B}\sum_{b=1}^{B} H_{i_b}$ *be sampled with replacement of batch size* B *from arbitrary symmetric per-sample Hessians* $H_i \in \mathbb{S}^P$ *(without any feature or commutation assumption), write* $\Delta H_i \triangleq H_i - H$, *and define* $\mathcal{V}(X) \triangleq \mathbb{E}[(H_{B_t} - H)X(H_{B_t} - H)] = \frac{1}{B}\mathbb{E}[\Delta H_i X \Delta H_i] \neq 0$. *Let* $\mathcal{L}_{\eta,\beta}(X) = (HX + XH) - \frac{\eta}{1+\beta}HXH + \frac{\eta\beta}{(1-\beta)(1-\beta^2)}[H,[H,X]]$, *which multiplies entry* (j,k) *in the eigenbasis of* H *by* $\ell_{jk}(\eta,\beta) = (\lambda_j + \lambda_k) + \eta\, b_{jk}(\beta)$, *positive for* $\eta < \eta_{\mathrm{det}}$ *(part (1)), where* $b_{jk}(\beta) \triangleq -\frac{\lambda_j\lambda_k}{1+\beta} + \frac{\beta(\lambda_j-\lambda_k)^2}{(1-\beta)(1-\beta^2)}$, *and for* $\eta \in (0, \eta_{\mathrm{det}})$ *with* $\eta_{\mathrm{det}} \triangleq \frac{2(1+\beta)}{\lambda_1}$, *set* $\mathcal{N}_{\eta,\beta} \triangleq \eta\mathcal{L}_{\eta,\beta}^{-1}$ *and* $\mathcal{K}_{\eta,\beta}(X) \triangleq \frac{1}{1-\beta}\mathcal{N}_{\eta,\beta}(\mathcal{V}(X)) = \frac{\eta}{1-\beta}\mathcal{L}_{\eta,\beta}^{-1}(\mathcal{V}(X))$ *on* $\mathbb{S}^P$. *(1)* (Invertibility, cone preservation, real spectrum with the inertia of $\mathcal{V}$, and regular splitting.) *For every* $\beta \in [0,1)$ *and* $\eta \in (0, \eta_{\mathrm{det}})$, $\mathcal{L}_{\eta,\beta}$ *is invertible,* $[1/\ell_{jk}(\eta,\beta)]_{j,k=1}^{P} \succeq 0$ *is positive semidefinite (singular when two eigenvalues of* H *coincide), and* $\mathcal{L}_{\eta,\beta}^{-1}$ *and* $\mathcal{K}_{\eta,\beta}$ *preserve the cone* $\mathbb{S}_+^P$. *On* $\mathcal{H} = (\mathbb{S}^P, \langle X, Y\rangle_F = \mathrm{tr}(XY))$, $\mathcal{N}_{\eta,\beta} \succ_{\mathcal{H}} 0$ *is self-adjoint and positive definite, whereas* $\mathcal{V}$ *is self-adjoint on* $\mathcal{H}$ *but in general indefinite (for* $\Delta H = \mathrm{diag}(1,-1)$, $\mathcal{V}(E_{12}+E_{21}) = -(E_{12}+E_{21})$*); hence* $\mathcal{K}_{\eta,\beta}$ *is similar to the symmetric operator* $\frac{1}{1-\beta}\mathcal{N}_{\eta,\beta}^{1/2}\mathcal{V}\mathcal{N}_{\eta,\beta}^{1/2}$, *has a real spectrum with the same inertia as* $\mathcal{V}$ *(in general including negative eigenvalues), and by the Krein–Rutman theorem [Krein and Rutman, 1948] its spectral radius* $\rho(\mathcal{K}_{\eta,\beta}) = \lambda_{\max}(\mathcal{K}_{\eta,\beta})$ *is its largest eigenvalue, with a PSD eigenmatrix* $X_\eta \in \mathbb{S}_+^P \setminus \{0\}$. *Splitting* $\mathcal{T}(\eta,\beta) = \mathcal{T}_0 + \Delta$ *on* $\mathbb{S}^{2P}$, $\rho(\mathcal{T}(\eta,\beta)) < 1 \iff \rho((I-\mathcal{T}_0)^{-1}\Delta) < 1 \iff \rho(\mathcal{K}_{\eta,\beta}) < 1$ *and* $\rho(\mathcal{T}(\eta,\beta)) = 1 \iff \rho(\mathcal{K}_{\eta,\beta}) = 1$, *because* $\rho((I-\mathcal{T}_0)^{-1}\Delta) = \rho(\mathcal{K}_{\eta,\beta})$ *by the factorization of Theorem 24(1) in Section B.2. (2)* (Variational Rayleigh quotient and strict monotonicity for all $\beta \in [0,1)$.) *Since* $\mathcal{K}_{\eta,\beta}$ *is self-adjoint in the inner product weighted by* $\mathcal{N}_{\eta,\beta}^{-1}$ *and* $\rho(\mathcal{K}_{\eta,\beta}) = \lambda_{\max}(\mathcal{K}_{\eta,\beta})$,

$$\rho(\mathcal{K}_{\eta,\beta}) = \frac{1}{1-\beta}\max_{X\in\mathbb{S}^P\setminus\{0\}} \frac{\langle X, \mathcal{V}(X)\rangle_F}{\sum_{j,k=1}^{P}\big(\ell_{jk}(\eta,\beta)/\eta\big)\widetilde{X}_{jk}^2}.$$

where $\widetilde{X} = U^\top X U$ *is* X *in the eigenbasis* $U = [u_1,\ldots,u_P]$ *of* H *and the sum runs over all* P^2 *entries. The numerator does not depend on* η *and its maximum is strictly positive because* $X = I_P$ *gives* $\langle I_P, \mathcal{V}(I_P)\rangle_F = \frac{1}{B}\mathbb{E}\|\Delta H_i\|_F^2 > 0$. *Each denominator weight* $\ell_{jk}(\eta,\beta)/\eta = (\lambda_j+\lambda_k)/\eta + b_{jk}(\beta)$ *is strictly decreasing in* η *since* $\lambda_j+\lambda_k > 0$ *and tends to* $+\infty$ *as* $\eta \downarrow 0$; *hence* $\eta \mapsto \rho(\mathcal{K}_{\eta,\beta})$ *is continuous, strictly increasing on* $(0,\eta_{\mathrm{det}})$ *for every* $\beta \in [0,1)$, *and satisfies* $\lim_{\eta\downarrow 0}\rho(\mathcal{K}_{\eta,\beta}) = 0$ *(at* $\beta = 0$, $\eta \mapsto \mathcal{N}_{\eta,0}$ *even grows in the order of completely positive maps on* $\mathbb{S}_+^P$, *since with* $\mu_j = \lambda_j^{-1}$ *and* $x_j = \mu_j - \eta/2 > 0$, $[\frac{\lambda_j+\lambda_k}{\ell_{jk}(\eta,0)^2}]$ *is congruent via* $\mathrm{diag}(\mu)$ *to* $[\frac{1}{x_j+x_k} + \frac{\eta}{(x_j+x_k)^2}] = \int_0^\infty (1+\eta s)e^{-s(x_j+x_k)}\,ds \succeq 0$, *positive definite when the* λ_j *are distinct). (3)* (Necessary and sufficient blow-up, the two rates, and uniqueness of the exact ceiling.) *Near* η_{det}, *only the top-block weights* ($\lambda_j = \lambda_k = \lambda_1$) *tend to zero, as* $\ell_{11}(\eta,\beta)/\eta \asymp \eta_{\mathrm{det}} - \eta$, *while all other weights stay bounded away from zero. By the Rayleigh quotient in (2): if* $P_1\Delta H_i P_1 \neq 0$ *for some sample* i *(for simple* λ_1, $\nu_{11,B}^2 > 0$*),* $X = P_1$ *gives* $\rho(\mathcal{K}_{\eta,\beta}) = \Theta((\eta_{\mathrm{det}} - \eta)^{-1})$; *if* $P_1\Delta H_i P_1 = 0$ *for all* i *but* $\Delta H_i P_1 \neq 0$ *for some* i *(pure cross-mode coupling, as in the Rademacher case of Corollary 6),* $X = sP_1 + \mathcal{V}(P_1)$ *(*$\mathcal{V}(P_1) \neq 0$*) gives* $\rho(\mathcal{K}_{\eta,\beta}) = \Theta((\eta_{\mathrm{det}} - \eta)^{-1/2})$; *if* $\Delta H_i P_1 = 0$ *for all* i, $\mathcal{V}$ *neither reads from nor writes to the top block,* $\mathcal{K}_{\eta,\beta}$ *reduces to the complement where all weights stay bounded away from zero, and* $\rho(\mathcal{K}_{\eta,\beta}) = O(1)$ *remains bounded. Thus* $\lim_{\eta\uparrow\eta_{\mathrm{det}}}\rho(\mathcal{K}_{\eta,\beta}) = +\infty \iff \Delta H_i P_1 \not\equiv 0$, *in which case there is a unique ceiling* $\eta_{\mathrm{crit}}^{\mathrm{EoSS},P}(\beta,B) \in (0,\eta_{\mathrm{det}})$ *with* $\rho(\mathcal{K}_{\eta,\beta}) = 1$, *its Krein–Rutman eigenmatrix* $\Sigma_0^\star \in \mathbb{S}_+^P \setminus \{0\}$ *is the second moment that loses stability first, and with* $\mathcal{L}_0(X) = HX + XH$ *and* $\mathcal{B}_\beta(X) = \frac{1}{1+\beta}HXH - \frac{\beta}{(1-\beta)(1-\beta^2)}[H,[H,X]] + \frac{1}{1-\beta}\mathcal{V}(X)$, $\eta_{\mathrm{crit}}^{\mathrm{EoSS},P}(\beta,B) = 1/\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2})$. *In every case the mean-square ceiling is* $1/\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}) \leq \eta_{\mathrm{det}}$; *when* $\Delta H_i P_1 \equiv 0$ *it can equal* η_{det}. *Proof in Appendix A.*

At $\beta = 0$, $\mathcal{B}_0(X) = HXH + \mathcal{V}(X) = (1 - \frac{1}{B})HXH + \frac{1}{B}\mathbb{E}[H_i X H_i]$ and $\mathcal{L}_0 = 2C$ with $C =$

$\frac{1}{2}H \oplus H$, so the last formula has the form $\eta^\star = 2/\lambda_{\max}(C^{-1}D)$ of Mulayoff and Michaeli [2024, Theorem 5], which they prove at interpolating minima for sampling without replacement (with $C^\dagger$), here with $p = 1/B$ for sampling with replacement and $H \succ 0$; Theorem 9 extends it to heavy-ball momentum, where the double commutator and the factors $1/(1 \pm \beta)$ enter $\mathcal{B}_\beta$. Against the full second-moment operator, in seven models with non-commuting per-sample Hessians ($P \in \{3, 4, 5\}$, $N \in \{4, 5, 6, 8\}$ samples, $B \in \{1, 2\}$, two of them with repeated eigenvalues of H) at $\beta \in \{0.5, 0.9\}$, $\rho(\mathcal{T})$ is below 1 at $0.999\,\eta^\star$ and above 1 at $1.001\,\eta^\star$ in all 14 cases, with $|\rho(\mathcal{T}(\eta^\star)) - 1| \le 1.1 \times 10^{-13}$ (`paper/checks/diagonalization-burnin.py`). Theorems 2 and 9 use the minibatch law only through $\mathcal{V}(X) = \mathbb{E}[(H_{B_t} - H)X(H_{B_t} - H)]$ and the independence of (H_{B_t}, ξ_t) across steps. For B of N fixed matrices $H_1, \dots, H_N$ with mean H drawn without replacement within each step, independently across steps, the covariances of the inclusion indicators and $\sum_i \Delta H_i = 0$ give $\mathcal{V}(X) = p\,\frac{1}{N}\sum_i \Delta H_i X \Delta H_i$ with $p = \frac{N-B}{B(N-1)}$, in place of $p = \frac{1}{B}$ for draws with replacement (`paper/checks/second-moment-renewal.py`); Theorem 9 is exact for both, and at $\beta = 0$ it is Theorem 5 of Mulayoff and Michaeli [2024] with their p. In Theorem 5 the factor $\frac{1}{B}$ enters only through $\mathcal{V} = p\mathcal{V}_1$ with $\mathcal{V}_1(X) = \mathbb{E}[(zz^\top - H)X(zz^\top - H)]$, so a law with $\mathcal{V} = p\mathcal{V}_1$ replaces $\frac{1}{B}$ by p in (4) and in $\alpha_{\text{diag}} = 1 + p\kappa_\beta(1 + \kappa_{\text{feat}})$. A pool of n i.i.d. features drawn once and then sampled without replacement does not have that form: centred at its own mean, it gives $\mathbb{E}_{\text{pool}}[(H_i - \hat{H})X(H_i - \hat{H})] = \frac{n-1}{n}\mathcal{V}_1(X)$ (with $\tilde{H}_j = H_j - H$ independent and centred, $H_i - \hat{H} = \frac{n-1}{n}\tilde{H}_i - \frac{1}{n}\sum_{j \ne i}\tilde{H}_j$, the cross terms vanish, and the coefficient is $\frac{(n-1)^2}{n^2} + \frac{n-1}{n^2} = \frac{n-1}{n}$), so its noise operator averaged over pool draws is $\frac{n-B}{B(n-1)} \cdot \frac{n-1}{n}\mathcal{V}_1 = \frac{1-\zeta}{B}\mathcal{V}_1$, $\zeta = B/n$. The secular root with $p = (1 - \zeta)/B$ uses that average and approximates the threshold of one realised pool, which Theorem 9 gives exactly. A pool that is the whole support of a law with independent coordinates, such as the 2^P sign vectors of Rademacher features, has $\hat{H} = H$ and $\mathcal{V} = \frac{n-B}{B(n-1)}\mathcal{V}_1$ exactly, so for it (4) with $p = \frac{n-B}{B(n-1)}$ is exact.

Two further cases reduce to the theorems. (a) Curvature outside $H \succ 0$: if $\lambda_{\min}(H) < 0$, the deterministic iteration has a root above 1 on the mode with $\lambda < 0$, and $\mathcal{T} \succeq \mathcal{T}_0$ gives $\rho(\mathcal{T}) \ge \rho(\bar{F})^2 > 1$ at every $\eta > 0$, so there is no mean-square stable rate. If $H \succeq 0$ is singular and every $H_i \succeq 0$, as at an interpolating minimum, then $\ker H = \bigcap_i \ker H_i$ and every H_i maps into range H; the component of w_t in $\ker H$ follows $w_{t+1} = (1 + \beta)w_t - \beta w_{t-1}$ plus the noise there, independently of the rest. With Σ_ξ supported on range H it stays bounded, and the second moment is bounded if and only if η is below the threshold of Theorem 9 on range H, with $\mathcal{L}_0$ inverted there (at $\beta = 0$ the pseudo-inverse $C^\dagger$ of Mulayoff and Michaeli [2024]); with noise in $\ker H$ it grows linearly at every η. Without $H_i \succeq 0$, curvature noise on a direction of zero curvature drives the threshold to 0, the limit $\lambda_k \to 0$ of Theorem 8(3). (b) A fixed preconditioner $P_{\text{pre}} \succ 0$, $w_{t+1} = w_t + \beta(w_t - w_{t-1}) - \eta P_{\text{pre}}(H_{B_t} w_t + \xi_t)$, need not commute with H or the H_i: $\tilde{w}_t = P_{\text{pre}}^{-1/2} w_t$ follows (2) with $\tilde{H}_i = P_{\text{pre}}^{1/2} H_i P_{\text{pre}}^{1/2}$ and $\tilde{\xi}_t = P_{\text{pre}}^{1/2}\xi_t$, so Theorems 2 and 9 apply with $\tilde{H} = P_{\text{pre}}^{1/2} H P_{\text{pre}}^{1/2}$, whose eigenvalues are those of $P_{\text{pre}}H$. For Gaussian features $z_i \sim \mathcal{N}(0, H)$, $P_{\text{pre}}^{1/2} z_i \sim \mathcal{N}(0, \tilde{H})$ again has independent coordinates in the eigenbasis of $\tilde{H}$, so Theorem 5 holds with the eigenvalues of $P_{\text{pre}}H$; for other feature laws that independence can be lost. (c) Random reshuffling, with independent permutations between epochs of $m = N/B$ disjoint batches, is not i.i.d. across steps, but with ξ_t of mean zero, independent across steps and independent of the permutation and of the minibatch Hessians (an assumption that excludes noise drawn from the same batch as H_{B_t}), the second moment of (w_{em}, w_{em-1}) at epoch boundaries obeys $Q_{e+1} = \mathcal{T}_{\text{RR}}(Q_e) + G_{\text{RR}}$, with $\mathcal{T}_{\text{RR}}(Q) = \mathbb{E}_\pi[F_{\text{ep}} Q F_{\text{ep}}^\top]$ for the epoch product $F_{\text{ep}} = F_{e,m} \cdots F_{e,1}$, i.i.d. across epochs, and $\sum_{r=1}^m (H_{B_{e,r}} - H) = 0$ in every epoch. So $\rho(\mathcal{T}_{\text{RR}}) < 1$ bounds the second moment, and for $G_{\text{RR}} \succ 0$ the argument of Corollary 4 makes it also necessary; a closed form of $\rho(\mathcal{T}_{\text{RR}})$ like (4) is not known. Within an epoch the batch errors $\Delta_r = H_{B_{e,r}} - H$ are exchangeable and sum to 0, so $\mathbb{E}_\pi[\Delta_r X \Delta_s] = \frac{m\delta_{rs} - 1}{m - 1}\mathcal{V}(X)$ with $\mathcal{V}$ the without-replacement

operator above; for $m = 2$ and $\beta = 0$, $\Delta_2 = -\Delta_1$ and $F_{\mathrm{ep}} = F_2F_1 = (I - \eta H)^2 + \eta^2[H, \Delta_1] - \eta^2\Delta_1^2$ exactly. Along u_1 at $\beta = 0$, with batch errors ζ_r, $\sum_r \zeta_r = 0$ gives $\sum_{s<r} \zeta_s\zeta_r = -\frac{1}{2}\sum_r \zeta_r^2$, hence $\mathbb{E}_\pi[F_{\mathrm{ep}}^2] = a^{2m} - m\, a^{2m-2}\eta^2\nu_{11,B}^2$ with $a = 1 - \eta\lambda_1$, up to terms of higher order in the batch errors, against $a^{2m} + m\, a^{2m-2}\eta^2\nu_{11,B}^2$ to the same order for m independent steps: the first-order curvature-noise correction changes sign, and the one-dimensional mean-square threshold of reshuffling is $\frac{2}{\lambda_1}(1 + \nu_{11,B}^2/\lambda_1^2)$ to first order, the first-order almost-sure threshold of Proposition 7(1), instead of $\frac{2}{\lambda_1}(1 - \nu_{11,B}^2/\lambda_1^2)$. On a scalar pool of 8 values with $B = 2$ and $\nu_{11,B}^2 = 0.00424$, exact enumeration over the permutations gives 2.00854 against 2.00849, and 1.99155 against 1.99151 for independent steps (`paper/checks/second-moment-renewal.py`).

Corollary 10 (The effective step at the threshold increases with β)**.** *Let $H \succ 0$ and let the law of the per-sample Hessians, hence $\mathcal{V}$, not depend on β, and let $\eta^\star(\beta)$ be the threshold of Theorem 9. Then $\eta^\star_{\mathrm{eff}}(\beta) = \eta^\star(\beta)/(1 - \beta)$ increases strictly on $\beta \in [0, 1)$, and*

$$\frac{1}{\eta^\star(0)} - \frac{1 - \beta}{\eta^\star(\beta)} \geq \frac{\beta}{1 + \beta}\lambda_{\min}(H).$$

Proof. By Theorem 9(3), $1/\eta^\star(\beta) = \lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2})$ at every β, also when the threshold is η_{det}, so $1/\eta^\star_{\mathrm{eff}}(\beta) = \lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{A}_\beta\mathcal{L}_0^{-1/2})$ with $\mathcal{A}_\beta = (1 - \beta)\mathcal{B}_\beta$. The term $\mathcal{V}$ cancels in $\mathcal{D}_\beta = \mathcal{B}_0 - \mathcal{A}_\beta$, $\mathcal{D}_\beta(X) = \frac{2\beta}{1+\beta}HXH + \frac{\beta}{1-\beta^2}[H, [H, X]]$, which multiplies entry (j, k) in the eigenbasis of H by $m_{jk}(\beta) = \frac{2\beta}{1+\beta}\lambda_j\lambda_k + \frac{\beta}{1-\beta^2}(\lambda_j - \lambda_k)^2$. Both coefficients increase strictly on $[0, 1)$, so $m_{jk}(\beta_2) > m_{jk}(\beta_1)$ for $\beta_1 < \beta_2$ since $\lambda_j\lambda_k > 0$. $\mathcal{L}_0$ multiplies the same entry by $\lambda_j + \lambda_k$, so $\mathcal{L}_0^{-1/2}(\mathcal{A}_{\beta_1} - \mathcal{A}_{\beta_2})\mathcal{L}_0^{-1/2}$ multiplies it by $(m_{jk}(\beta_2) - m_{jk}(\beta_1))/(\lambda_j + \lambda_k) > 0$; it is positive definite on the finite-dimensional space $(\mathbb{S}^P, \langle\cdot,\cdot\rangle_F)$, and Weyl's inequality makes $\lambda_{\max}$ strictly smaller at β_2. For the bound, $m_{jk}(\beta) - \frac{\beta}{1+\beta}(\lambda_j^2 + \lambda_k^2) = \frac{\beta^2(\lambda_j - \lambda_k)^2}{1-\beta^2} \geq 0$ and $\lambda_j^2 + \lambda_k^2 \geq \lambda_{\min}(H)(\lambda_j + \lambda_k)$, so $\mathcal{L}_0^{-1/2}\mathcal{D}_\beta\mathcal{L}_0^{-1/2} \succeq \frac{\beta}{1+\beta}\lambda_{\min}(H)\, I$, and Weyl's inequality gives the bound. □

The small-step analysis of momentum SGD of Yuan et al. [2016] makes stability depend on (η, β) only through $\eta_{\mathrm{eff}} = \eta/(1 - \beta)$, and the leading-order condition of Andreyev et al. [2026, Theorem 4.1], in the noise-dominated regime and for small η, is the SGD condition at η_{eff}. At the threshold this equivalence fails, as it already does without noise, where $\eta^\star_{\mathrm{eff}} = 2(1 + \beta)/((1 - \beta)\lambda_1)$. Written in η_{eff}, (4) reads $\frac{\eta_{\mathrm{eff}}p}{2}\sum_k \lambda_k/\big(1 - \frac{1}{2}\eta_{\mathrm{eff}}\lambda_k(\kappa_\beta^{-1} + p(1 + \kappa_{\mathrm{feat}}))\big) = 1$, where β enters only through $\kappa_\beta^{-1} = \frac{1-\beta}{1+\beta}$, decreasing to 0; at $\kappa_{\mathrm{feat}} = -1$ the limit as $\beta \to 1$ is $\eta^\star_{\mathrm{eff}} = 2/(p\,\mathrm{tr}\,H)$. The secular roots of the Gaussian cells of Phase 1 at $P = 64$ and $B = 64$ give $\eta^\star_{\mathrm{eff}} = 1.927$ at $\beta = 0$ and 7.948 at $\beta = 0.9$, a factor 4.12; `paper/checks/second-moment-renewal.py` computes these roots and, separately, checks the monotonicity and the bound on a non-commuting model with $P = 3$, $N = 5$ and $B = 2$.

For the loss of mini-batch SGD with momentum on least squares, for large n and d under orthogonal invariance of the data, Lee et al. [2022] write a Volterra equation for the loss whose kernel norm $\|\mathcal{K}\| < 1$ they state, citing Asmussen [2003, Proposition 7.4], to be necessary for its solution to converge; their Theorem 1, the concentration of SGD with momentum on that solution over a fixed horizon, assumes $\gamma < (1 + \Delta)/(\zeta\sigma_{\max}^2)$, which is $\eta\lambda_{\max} < 1 + \beta$ in the dictionary below, and in the proof of their Proposition 2 they put the norm in closed form, $\|\mathcal{K}\| = \frac{1}{n}\sum_j \frac{(1-\zeta)\gamma\sigma_j^2(1+\Delta)}{(1-\Delta)(1+\Delta+\Omega_j)}$ with $\Omega_j = 1 - \gamma\zeta\sigma_j^2 + \Delta$. Their update sums the B sample gradients and (2) averages them (in their notation β is the batch size and Δ the momentum); with $\eta = \gamma B$, $\beta = \Delta$, $\lambda_j = \sigma_j^2/n$ and $\zeta = B/n$, this is the left side of (4) with $\alpha_{\mathrm{diag}} = 1$ and a factor $1 - \zeta$; in the model of this note with $B = \zeta n$ the difference $\alpha_{\mathrm{diag}} - 1 = \kappa_\beta(1 + \kappa_{\mathrm{feat}})/B$ is $O(1/n)$; their data are orthogonally

invariant and the note's have independent coordinates, two classes whose common members are Gaussian, and their statement holds with high probability as $n, d \to \infty$ while this one is exact at finite P; what coincides is the two formulas. Their Proposition 2 gives two conditions that together imply $\|\mathcal{K}\| < 1$, $\eta\lambda_{\max} < 1 + \beta$ (half the deterministic threshold) and a trace bound, which their convergence-rate analysis uses; locating the root of $\|\mathcal{K}\| = 1$ is not its aim. In 6 of the 30 cells of Phase 1 ($P = 64$, $\beta = 0$, $B \in \{16, 64\}$, two of them in the pool arm scored by `p4`) that root lies above $(1+\beta)/\lambda_1$, outside the hypothesis of their Theorem 1; there the identity is with their formula, not with their theorem on SGD with momentum. The two objects differ: their kernel norm governs the loss in the high-dimensional limit, (4) the second moment of the parameters at finite P; and the factor $1 - \zeta$ is the pool average of the noise operator of sampling without replacement (paragraph after Theorem 9), so for one realised pool (4) with it is an approximation, which the pool arm of Phase 1 uses as its prediction. The same secular equation is the loss-convergence condition $\widetilde{U}(1) < 1$ of Velikanov et al. [2023, eq. (21) and the paragraph after Proposition 3] for mini-batch SGD with momentum under their spectrally expressible approximation, with their noise strength $\gamma = (N - b)/((N - 1)b)$ of a finite dataset sampled without replacement replaced by $1/B$, their value for an infinite dataset and the factor with replacement here, and parameter $\tau = -(1 + \kappa_{\text{feat}})$. Their Section 5, where Proposition 3 sits, assumes $0 < \tau \leq 1$ and an infinite-dimensional compact H ($\lambda_k \to 0$); τ lies in that range for $\kappa_{\text{feat}} \in [-2, -1)$ ($\tau = 1$ for Rademacher features) and outside it for $\kappa_{\text{feat}} \geq -1$, including the Gaussian case $\tau = -1$ they attribute to Bordelon and Pehlevan, and the second assumption fails at finite P, so at finite P the correspondence is formal; their λ_{crit} bound (Proposition 4) is a one-sided counterpart of the sandwich (5). Theorem 5 shows that the approximation is exact at finite P and B for features with independent coordinates of any kurtosis $\kappa_{\text{feat}} \geq -2$, proves the "if and only if" on the full second-moment operator, and adds the reading of the root as the stochastic ceiling, the two plateaus and the Rademacher separation; Theorem 8, with $\Sigma_z \neq H$, lies outside the spectrally expressible class. At $\beta = 0$ the sandwich gives the batch-size scaling $B^\star \asymp \operatorname{tr} H/\lambda_1$ of Ma et al. [2018]; the component-decoupled data model of Kidambi et al. [2018] and Liu and Belkin [2020] has H_{B_t} commuting with H and no cross-mode term, and Paquette and Paquette [2021], at batch size 1, describe by a Volterra equation (their Theorem 1) the loss of homogenized SGD, a diffusion approximation of a class of stochastic momentum methods that contains heavy-ball with fixed momentum after a rescaling of its parameters with the dimension, in a scaling of small steps and high dimension in which the deterministic ceiling $2(1 + \beta)/\lambda_1$ does not enter; in that limit the homogenized dynamics of heavy-ball with fixed momentum and of SGD have the same loss when the SGD step is their $\gamma^{\text{shb}}/\theta^{\text{shb}}$ (their Theorem 5). Phase 1 of the lab measures Theorem 5 (Section 4.1).

Proposition 11 (A renewal-type equation for the expected loss at finite P)**.** *In the setting of Theorem 5, with $\mathcal{V} = p\mathcal{V}_1$, $\kappa_{\text{feat}} \geq -2$, Σ_ξ diagonal in the eigenbasis of H, $w_0 = w_{-1} = 0$ and $\eta < \eta_{\text{det}}$, let $g_k(0) = 1$, $g_k(1) = 1 + \beta - \eta\lambda_k$ and $g_k(\tau + 1) = (1 + \beta - \eta\lambda_k)g_k(\tau) - \beta g_k(\tau - 1)$ be the impulse response of mode k of deterministic heavy-ball, $\psi_k(\tau) = \eta^2 g_k(\tau)^2$, and h_k the solution of the scalar convolution equation $h_k(\tau) = \psi_k(\tau) + p(1 + \kappa_{\text{feat}})\lambda_k^2 \sum_{u<\tau} \psi_k(\tau - 1 - u)h_k(u)$. Then $\ell_t = \operatorname{tr}(H\mathbb{E}[w_t w_t^\top])$, twice the expected quadratic loss, solves exactly*

$$\ell_t = f_t + \sum_{s=0}^{t-1} K_{t-1-s}\,\ell_s, \qquad K_\tau = p\sum_{k=1}^{P} \lambda_k^2 h_k(\tau), \qquad f_t = \sum_{k=1}^{P} \lambda_k \sigma_{\xi,k}^2 \sum_{\tau<t} h_k(\tau).$$

If $\kappa_{\text{feat}} \geq -1$, then $h_k \geq 0$; if moreover $\eta\lambda_k\alpha_{\text{diag}} < 2(1 + \beta)$ for every k, then $\sum_{\tau\geq 0} K_\tau = \frac{\eta p}{2(1-\beta)} \sum_k \frac{\lambda_k}{1-\eta\lambda_k\alpha_{\text{diag}}/(2(1+\beta))}$, the left side of (4) with $\frac{1}{B}$ replaced by p (as everywhere in this proposition), and ℓ_t is bounded if and only if this sum is below 1. In this model with $B = \zeta n$ and

$n \to \infty$, *the regime of Lee et al. [2022],* $\alpha_{\text{diag}} - 1 = O(1/n)$ *vanishes whatever the kurtosis, and with* $p = (1-\zeta)/B$ *and the dictionary of the previous paragraph their kernel norm is the left side of* (4) *at* $\alpha_{\text{diag}} = 1$*; at finite* P*,* $\alpha_{\text{diag}} = 1$ *holds exactly at* $\kappa_{\text{feat}} = -1$*, where* $h_k = \psi_k$ *and the loss obeys a renewal equation of their form, and for* $\kappa_{\text{feat}} > -1$ *the same with* h_k *in place of* ψ_k*. For* $\kappa_{\text{feat}} \in [-2, -1)$ *the equation is a discrete Volterra equation with a kernel of either sign, not a renewal equation in the probabilistic sense, but* $\lambda_P \mathbb{E}\|w_t\|^2 \le \ell_t \le \lambda_1 \mathbb{E}\|w_t\|^2$*, so Theorem 5 and Corollary 4 give the same conclusion for every* $\kappa_{\text{feat}} \ge -2$*: with every* $\sigma^2_{\xi,k} > 0$ *and* $\eta < 2(1+\beta)/(\lambda_1 \max(\alpha_{\text{diag}}, 1))$*,* ℓ_t *is bounded if and only if the left side of* (4) *is below* 1*; and the Neumann series* $h_k = \sum_{n\ge 0} c_k^n \psi_k^{*(n+1)}$ *converges absolutely, with* $\sum_\tau h_k(\tau) = \Psi_k/(1 - c_k\Psi_k)$*, when* $|c_k|\Psi_k < 1$*, that is* $\eta\lambda_k(2 - \alpha_{\text{diag}}) < 2(1+\beta)$*, in which case* $\sum_\tau K_\tau$ *is again the left side of* (4)*. That bound is sufficient, not necessary, for* h_k *to be summable; where* h_k *is not summable the left side can still be below* 1 *(for* $P = 1$*,* $\beta = 0$*,* $B = 1$*,* $\kappa_{\text{feat}} = -2$*,* $\lambda_1 = 1$ *and* $\eta = 1.5$ *it is* 0.75 *while* h_1 *grows geometrically), and the boundedness of* ℓ_t *comes from Theorem 5, not from a kernel sum.*

Proof. With $s_t = (w_t, w_{t-1})$ and $Q_t = \mathbb{E}[s_t s_t^\top]$, $Q_{t+1} = \mathcal{T}_0(Q_t) + \Delta(Q_t) + G$ (proof of Corollary 4), so from $Q_0 = 0$, $Q_t = \sum_{s<t} \mathcal{T}_0^{t-1-s}(\Delta(Q_s) + G)$. The $(1,1)$ block of $\mathcal{T}_0^\tau(\text{blkdiag}(Y, 0))$ is $(\bar{F}^\tau)_{11} Y (\bar{F}^\tau)_{11}^\top$, with $(\bar{F}^\tau)_{11} = \text{diag}(g_k(\tau))$ in the eigenbasis of H. By Theorem 5, $\mathcal{V}(X)_{kk} = p\lambda_k \operatorname{tr}(HX) + p(1+\kappa_{\text{feat}})\lambda_k^2 X_{kk}$, so $x_t = (\Sigma_{0,t})_{kk}$ solves $x_t = \sum_{s<t} \psi_k(t-1-s)\big(\sigma^2_{\xi,k} + p\lambda_k \ell_s + p(1+\kappa_{\text{feat}})\lambda_k^2 x_s\big)$, whose solution is $x_t = \sum_{s<t} h_k(t-1-s)(\sigma^2_{\xi,k} + p\lambda_k \ell_s)$; multiplying by λ_k and summing over k gives the equation. For $\kappa_{\text{feat}} \ge -1$ the coefficient $c_k = p(1+\kappa_{\text{feat}})\lambda_k^2$ is non-negative, so $h_k \ge 0$, and $\sum_\tau h_k(\tau) = \Psi_k/(1 - c_k\Psi_k)$ with $\Psi_k = \sum_\tau \psi_k(\tau) = \frac{\eta}{2(1-\beta)\lambda_k(1 - \eta\lambda_k/(2(1+\beta)))}$ (Yule–Walker equations of the second-order autoregression), finite when $c_k\Psi_k < 1$, which is $\eta\lambda_k\alpha_{\text{diag}} < 2(1+\beta)$; then $p\lambda_k^2 \sum_\tau h_k(\tau) = \frac{\eta p}{2(1-\beta)} \frac{\lambda_k}{1 - \eta\lambda_k\alpha_{\text{diag}}/(2(1+\beta))}$. A renewal equation with a non-negative kernel and a forcing that increases to a positive limit has a bounded solution if and only if the kernel sums to less than 1 [Asmussen, 2003, Chapter V]. The identity is checked against the exact second-moment recursion for $\kappa_{\text{feat}} \in \{-2, -1, 0, 2\}$ in `paper/checks/second-moment-renewal.py`. □

Remark 12 (When the propagator bound holds up to the threshold). For $\kappa_{\text{feat}} \in [-2, -1)$, $|c_k|\Psi_k < 1$ holds for every k at $\eta = \eta^\star$ if and only if $\mu^\star > (2 - \alpha_{\text{diag}})\lambda_1$. Indeed $|c_k|\Psi_k < 1$ is $\eta\lambda_k(2 - \alpha_{\text{diag}}) < 2(1+\beta)$, so the mode $k = 1$ binds; with $f(\mu) = p\kappa_\beta \sum_k \lambda_k/(\mu - \alpha_{\text{diag}}\lambda_k)$, which equals 1 at $\mu^\star$ and decreases strictly for $\mu > \max(\alpha_{\text{diag}}, 0)\lambda_1$, the condition is $f((2 - \alpha_{\text{diag}})\lambda_1) > 1$, and since $1 - \alpha_{\text{diag}} = p\kappa_\beta|1 + \kappa_{\text{feat}}|$ the term $k = 1$ of f there is $1/(2|1 + \kappa_{\text{feat}}|)$, which gives $\sum_{k\ge 2} \frac{p\kappa_\beta\lambda_k}{(2-\alpha_{\text{diag}})\lambda_1 - \alpha_{\text{diag}}\lambda_k} > 1 - \frac{1}{2|1+\kappa_{\text{feat}}|}$. The lower bound of (5), with $\frac{1}{B}$ replaced by p, gives it whenever $\min(\alpha_{\text{diag}}, 0)\lambda_1 + p\kappa_\beta \operatorname{tr} H > (2 - \alpha_{\text{diag}})\lambda_1$. On the ten Rademacher cells of Phase 1 it holds in 9 and fails at $P = 64$, $\beta = 0$, $B = 64$ (`paper/checks/second-moment-renewal.py`).

Theorem 13 (Modal interpolation identity, fixed-point divergence criteria, and Batch Sharpness at the threshold)**.** *In the setting of Theorem 9, let* $\eta^\star = \eta_{\text{crit}}^{\text{EoSS},P}(\beta, B) = 1/\lambda_{\max}\big(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}\big) \in (0, \eta_{\text{det}}]$ *and let* $\Sigma_0^\star \in \mathbb{S}_+^P \setminus \{0\}$ *(*$\operatorname{tr}\Sigma_0^\star = 1$*) satisfy* $(\mathcal{L}_0 - \eta^\star\mathcal{B}_\beta)(\Sigma_0^\star) = 0$ *(for* $\eta^\star < \eta_{\text{det}}$*, equivalently* $\mathcal{K}_{\eta^\star,\beta}(\Sigma_0^\star) = \Sigma_0^\star$*).*

1. (Modal interpolation law and one-sided Batch Sharpness instability criteria at a fixed checkpoint.) *With the modal curvature and curvature-noise ratios on* $\Sigma_0^\star$*,*

$$a_\star \triangleq \frac{\operatorname{tr}(H^2\Sigma_0^\star)}{\operatorname{tr}(H\Sigma_0^\star)} \in [\lambda_P, \lambda_1], \qquad v_\star \triangleq \frac{\operatorname{tr}(\mathcal{V}(\Sigma_0^\star))}{\operatorname{tr}(H\Sigma_0^\star)} = \frac{\operatorname{tr}\left((\mathbb{E}[H_{B_t}^2] - H^2)\Sigma_0^\star\right)}{\operatorname{tr}(H\Sigma_0^\star)} \ge 0, \tag{6}$$

and the modal second-moment curvature $\mathcal{C}_{\text{mod}}(\Sigma_0^\star) \triangleq \frac{\operatorname{tr}(\mathbb{E}[H_{B_t}^2]\Sigma_0^\star)}{\operatorname{tr}(H\Sigma_0^\star)} = a_\star + v_\star$*, one has (extending*

Proposition 7(2) and the 1D formula of Andreyev et al. [2026, eq. 20] to $\Sigma_0^\star \in \mathbb{S}_+^P$)

$$\frac{1}{\eta^\star} = \frac{a_\star}{2(1+\beta)} + \frac{v_\star}{2(1-\beta)}, \qquad \mathcal{C}_{\mathrm{mod}}(\Sigma_0^\star) = \frac{2(1-\beta)}{\eta^\star} + \frac{2\beta}{1+\beta} a_\star = \frac{2(1+\beta)}{\eta^\star} - \frac{2\beta}{1-\beta} v_\star \in \Big[\frac{2(1-\beta)}{\eta^\star}, \frac{2(1+\beta)}{\eta^\star}\Big]. \tag{7}$$

For every $S \in \mathbb{S}_+^P \setminus \{0\}$ commuting with H ($[H,S]=0$), setting $a(S) \triangleq \frac{\mathrm{tr}(H^2 S)}{\mathrm{tr}(HS)}$ and $v(S) \triangleq \frac{\mathrm{tr}(S^{1/2}\mathcal{V}(S^{1/2}))}{\mathrm{tr}(HS)}$ ($\bar{a} \triangleq a(I_P) = \frac{\mathrm{tr}(H^2)}{\mathrm{tr}\,H}$, $\bar{v} \triangleq v(I_P) = \frac{\mathbb{E}\|\Delta H_i\|_F^2}{B\,\mathrm{tr}\,H}$, $v(u_1 u_1^\top) = \frac{\nu_{11,B}^2}{\lambda_1}$) gives $\frac{1}{\eta^\star} \geq \frac{a(S)}{2(1+\beta)} + \frac{v(S)}{2(1-\beta)}$, so $\frac{a(S)}{2(1+\beta)} + \frac{v(S)}{2(1-\beta)} \geq \frac{1}{\eta}$ is a valid sufficient instability criterion in the sense of Andreyev et al. [2026, Definition A.1] under the hypotheses of Corollary 4 ($\Sigma_\xi \succ 0$ and either ξ_t independent of H_{B_t} or $\mathbb{E}[w_0] = \mathbb{E}[w_{-1}] = 0$). In particular: (a) for any checkpoint θ_0 and any $H_B(\theta_0) \in \mathbb{S}^P$ with $\Pr(g_B(\theta_0) \neq 0) = 1$, setting $R_B(\theta_0) \triangleq \frac{g_B(\theta_0)^\top H_B(\theta_0) g_B(\theta_0)}{\|g_B(\theta_0)\|_2^2}$ and $W_B(\theta_0) \triangleq \frac{\|g_B(\theta_0)\|_2^2}{\mathbb{E}_B\|g_B(\theta_0)\|_2^2}$, the ratio-of-expectations $\overline{\mathrm{BS}}(\theta_0) \triangleq \frac{\mathbb{E}_B[g_B(\theta_0)^\top H_B(\theta_0) g_B(\theta_0)]}{\mathbb{E}_B\|g_B(\theta_0)\|_2^2}$ and the pointwise expectation-of-the-ratio $\mathrm{BS}(\theta_0) \triangleq \mathbb{E}_B[R_B(\theta_0)]$ of Andreyev et al. [2026, Definition 3.1] satisfy the exact covariance identity $\overline{\mathrm{BS}}(\theta_0) - \mathrm{BS}(\theta_0) = \mathrm{Cov}_B(R_B(\theta_0), W_B(\theta_0))$, so $|\overline{\mathrm{BS}}(\theta_0) - \mathrm{BS}(\theta_0)| \leq \sigma_{\mathrm{BS}}(\theta_0)\,\mathrm{CV}_{g^2}(\theta_0)$ with $\sigma_{\mathrm{BS}}(\theta_0) \triangleq (\mathrm{Var}_B(R_B(\theta_0)))^{1/2}$ and $\mathrm{CV}_{g^2}(\theta_0) \triangleq \frac{(\mathrm{Var}_B(\|g_B(\theta_0)\|_2^2))^{1/2}}{\mathbb{E}_B\|g_B(\theta_0)\|_2^2}$; when $\mathbb{E}[g_B(\theta_0) g_B(\theta_0)^\top \mid H_B] = \sigma_g^2 H_B(\theta_0)$ ($\sigma_g > 0$), $\overline{\mathrm{BS}}(\theta_0) = \bar{a} + \bar{v}$, and with noise share $\theta_{\mathrm{noise}} \triangleq \frac{\bar{v}}{\bar{a}+\bar{v}} \in [0,1]$,

$$\overline{\mathrm{BS}}(\theta_0) \geq \mathrm{BS}(\theta_0) - \sigma_{\mathrm{BS}}(\theta_0)\,\mathrm{CV}_{g^2}(\theta_0) \geq \frac{2(1+\beta)}{\eta\left(1 + \frac{2\beta}{1-\beta}\theta_{\mathrm{noise}}\right)} \quad \Big(\text{in particular } \geq \frac{2(1+\beta)}{\eta} \text{ without } \theta_{\mathrm{noise}}, \text{ or } \bar{v} \geq \tfrac{2}{\eta} \tag{8}$$

whose threshold is $\frac{2}{\eta}$ at $\beta = 0$, $\frac{2(1+\beta)}{\eta}$ at $\theta_{\mathrm{noise}} = 0$, and $\frac{2(1-\beta)}{\eta}$ at $\theta_{\mathrm{noise}} = 1$; and (b) in the random-feature model of Theorem 5 ($H_B = \frac{1}{B} Z Z^\top$, $Z = [z_{i_1}, \ldots, z_{i_B}]$), $v_\star = \frac{\mathrm{tr}\,H}{B} + \frac{1+\kappa_{\mathrm{feat}}}{B} a_\star$, $\frac{2(1+\beta)}{\eta^\star} = \alpha_{\mathrm{diag}} a_\star + \kappa_\beta \frac{\mathrm{tr}\,H}{B}$, and for every batch B and every $g_B = \frac{1}{B} Z r \in \mathrm{range}(Z) \setminus \{0\}$ with arbitrary $r \in \mathbb{R}^B$ and $u \triangleq r/\|r\|_2 \in \mathbb{S}^{B-1}$ (requiring no independence between r and Z, and holding in particular for $g_B(\theta) = \frac{1}{B}\sum_{b=1}^B r_{i_b}(\theta) z_{i_b}$ at any fixed θ on the rank-one Gauss–Newton surrogate $H_i = z_i z_i^\top$), the normalised Gram matrix $G_Z \triangleq (\mathrm{tr}\,H)^{-1} Z^\top Z \in \mathbb{S}_+^B$ satisfies the exact pathwise identity $\frac{g_B^\top H_B g_B}{\|g_B\|_2^2} - \frac{\mathrm{tr}\,H}{B} = \frac{\mathrm{tr}\,H}{B}\big(u^\top (G_Z - I_B) u + \frac{\|(I_B - uu^\top)(G_Z - I_B)u\|_2^2}{u^\top G_Z u}\big)$ and $\big|\frac{g_B^\top H_B g_B}{\|g_B\|_2^2} - \frac{\mathrm{tr}\,H}{B}\big| \leq \frac{\mathrm{tr}\,H}{B}\|G_Z - I_B\|_{\mathrm{op}} \leq \frac{\mathrm{tr}\,H}{B}\|G_Z - I_B\|_F$, with $\mathbb{E}\|(I_B - uu^\top)(G_Z - I_B)u\|_2^2 = \big(B - 1 + \kappa_{\mathrm{feat}}(1 - \sum_{b=1}^B u_b^4)\big)\frac{\mathrm{tr}(H^2)}{(\mathrm{tr}\,H)^2}$ for every deterministic $u \in \mathbb{S}^{B-1}$ and $\mathbb{E}\|G_Z - I_B\|_F^2 = B(B+1+\kappa_{\mathrm{feat}})\frac{\mathrm{tr}(H^2)}{(\mathrm{tr}\,H)^2} \triangleq \delta_{\mathrm{Gram}}^2 \leq B(B+1+\kappa_{\mathrm{feat}})\frac{\lambda_1}{\mathrm{tr}\,H}$; hence the pointwise expectation-of-the-ratio Batch Sharpness $\mathrm{BS}(\theta) = \mathbb{E}_B[\frac{g_B(\theta)^\top H_B g_B(\theta)}{\|g_B(\theta)\|_2^2}]$ (with $\Pr(g_B(\theta) \neq 0) = 1$) satisfies $\mathrm{BS}(\theta) = \frac{\mathrm{tr}\,H}{B}\big(1 + \mathbb{E}_B[\frac{\|(I_B - uu^\top)(G_Z - I_B)u\|_2^2}{u^\top G_Z u}]\big) \geq \frac{\mathrm{tr}\,H}{B}$ for every $B \geq 1$ whenever $u(\theta)$ is independent of G_Z (with equality at $B = 1$ for all $\kappa_{\mathrm{feat}} \geq -2$), and for arbitrary $r(\theta)$,

$$\Big|\mathrm{BS}(\theta) - \frac{\mathrm{tr}\,H}{B}\Big| \leq \frac{\mathrm{tr}\,H}{B}\,\delta_{\mathrm{Gram}} \quad \forall\,\theta \qquad (\delta_{\mathrm{Gram}} = 0 \text{ at } B = 1,\ \kappa_{\mathrm{feat}} = -2), \tag{9}$$

so in the small-batch regime $\frac{B^2\lambda_1}{\mathrm{tr}\,H} \to 0$, $\frac{\mathrm{BS}(\theta)}{2(1-\beta)/\eta^\star} = 1 + O(\delta_{\mathrm{Gram}} + \frac{B\lambda_1}{\mathrm{tr}\,H}) \to 1$ uniformly in θ, and for $\kappa_{\mathrm{feat}} \geq -1$, $\mathrm{BS}(\theta) \geq \frac{2(1-\beta)}{\eta}(1 + \delta_{\mathrm{Gram}}) \implies \frac{\mathrm{tr}\,H}{B} \geq \frac{2(1-\beta)}{\eta} \implies \eta \geq \eta^\star \implies \rho(\mathcal{T}) \geq 1$ is a valid sufficient instability criterion in the sense of Andreyev et al. [2026, Definition A.1] stated directly for $\mathrm{BS}(\theta)$.

2. (Global Krein–Rutman pole, stationary Batch Sharpness limit as $\eta \uparrow \eta^\star$, and the two SGDM plateaus.) *Equip $\mathbb{S}^P$ with the η-independent inner product $\langle X, Y\rangle_{\mathcal{L}_0} \triangleq \langle X, \mathcal{L}_0(Y)\rangle_F = 2\operatorname{tr}(XHY)$, in which $\mathcal{L}_0^{-1}\mathcal{B}_\beta$ is self-adjoint with top eigenvalue $1/\eta^\star$. Assume $\Sigma_\xi \succ 0$ and that $1/\eta^\star$ is a simple eigenvalue of $\mathcal{L}_0^{-1}\mathcal{B}_\beta$ with spectral gap $\Delta_{\mathcal{G}} \triangleq 1/\eta^\star - \lambda_2(\mathcal{L}_0^{-1}\mathcal{B}_\beta) > 0$. Then for every $\eta \in (0, \eta^\star)$, the stationary covariance $\Sigma_0(\eta)$ of Theorem 2(2) decomposes orthogonally in $(\mathbb{S}^P, \langle\cdot,\cdot\rangle_{\mathcal{L}_0})$ as*

$$\Sigma_0(\eta) = \frac{\eta\ \operatorname{tr}(\Sigma_0^\star \Sigma_\xi)}{2(1-\beta)\ \operatorname{tr}\big(H(\Sigma_0^\star)^2\big)\ (1-\eta/\eta^\star)}\,\Sigma_0^\star + R_\perp(\eta), \qquad \|R_\perp(\eta)\|_{\mathcal{L}_0} \le \frac{\|\mathcal{L}_0^{-1/2}(\Sigma_\xi)\|_F}{(1-\beta)\,\Delta_{\mathcal{G}}}. \tag{10}$$

When $\eta^\star < \eta_{\mathrm{det}}$, $1 - \rho(\mathcal{K}_{\eta,\beta}) = d_\star(1-\eta/\eta^\star) + O((\eta^\star-\eta)^2)$ with $d_\star \triangleq \eta^\star \partial_\eta \rho(\mathcal{K}_{\eta,\beta})\big|_{\eta=\eta^\star} = \frac{2(1-\beta)\operatorname{tr}(H(\Sigma_0^\star)^2)}{\eta^\star \langle \Sigma_0^\star, \mathcal{V}(\Sigma_0^\star)\rangle_F} > 0$. Consequently $\Sigma_0(\eta)/\operatorname{tr}\Sigma_0(\eta) \to \Sigma_0^\star$ as $\eta \uparrow \eta^\star$ for every $\Sigma_\xi \succ 0$, and on the linearised gradient $g_{B_t}(w_t) = H_{B_t} w_t + \xi_t$ with (H_{B_t}, ξ_t) independent of w_t, $\mathbb{E}\|H_{B_t}\|_{\mathrm{op}}^3 < \infty$ and $\mathbb{E}[\|H_{B_t}\|_{\mathrm{op}}\|\xi_t\|_2^2] < \infty$, the stationary ratio-of-expectations Batch Sharpness satisfies

$$\lim_{\eta\uparrow\eta^\star} \overline{\mathrm{BS}}(\eta) \triangleq \lim_{\eta\uparrow\eta^\star} \frac{\mathbb{E}\big[g_{B_t}(w_t)^\top H_{B_t} g_{B_t}(w_t)\big]}{\mathbb{E}\big[\|g_{B_t}(w_t)\|_2^2\big]} = \mathrm{BS}^\star(\Sigma_0^\star) \triangleq \frac{\operatorname{tr}\big(\mathbb{E}[H_{B_t}^3]\,\Sigma_0^\star\big)}{\operatorname{tr}\big(\mathbb{E}[H_{B_t}^2]\,\Sigma_0^\star\big)}. \tag{11}$$

Whenever $H_{B_t} \succeq 0$ almost surely (in particular when $H_i \succeq 0$ for all i), $\mathrm{BS}^\star(\Sigma_0^\star) \ge \mathcal{C}_{\mathrm{mod}}(\Sigma_0^\star) \ge \frac{2(1-\beta)}{\eta^\star}$ (and pointwise $\frac{w^\top H_{B_t}^3 w}{w^\top H_{B_t}^2 w} \ge \frac{w^\top H_{B_t}^2 w}{w^\top H_{B_t} w}$ on $\{H_{B_t} w \neq 0\}$), and if $H_{B_t} \preceq M_{\max} I_P$ almost surely, $\mathrm{BS}^\star(\Sigma_0^\star) \le M_{\max}$. In the random-feature model of Theorem 5 with $\mu_6 \triangleq \max_k \mathbb{E}[\omega_{i,k}^6] < \infty$, let $q_{\mathrm{feat}} \triangleq 1 + \frac{1+\kappa_{\mathrm{feat}}}{B}$, $r_{\mathrm{feat}} \triangleq 1 + \frac{3(1+\kappa_{\mathrm{feat}})}{B} + \frac{\mathbb{E}[\omega_{i,k}^6] - 11 - 6\kappa_{\mathrm{feat}}}{B^2}$, and $m_{2,\star} \triangleq \frac{\operatorname{tr}(H^3\Sigma_0^\star)}{\operatorname{tr}(H\Sigma_0^\star)}$. Then $\alpha_{\mathrm{diag}}(m_{2,\star} - a_\star^2) = \frac{\kappa_\beta}{B}\big((\operatorname{tr} H)a_\star - \operatorname{tr}(H^2)\big)$ and, for identically distributed $\omega_{i,k}$,

$$\mathrm{BS}^\star(\Sigma_0^\star) = \mathcal{C}_{\mathrm{mod}}(\Sigma_0^\star) + \frac{\frac{q_{\mathrm{feat}}}{B}\operatorname{tr}(H^2) + q_{\mathrm{feat}}^2(m_{2,\star} - a_\star^2) + (r_{\mathrm{feat}} - q_{\mathrm{feat}}^2) m_{2,\star}}{\frac{\operatorname{tr} H}{B} + q_{\mathrm{feat}} a_\star}, \tag{12}$$

so both $\mathcal{C}_{\mathrm{mod}}(\Sigma_0^\star)$ and $\mathrm{BS}^\star(\Sigma_0^\star)$ converge to the two SGDM plateaus of Andreyev et al. [2026]: $\frac{\mathrm{BS}^\star(\Sigma_0^\star)}{2(1-\beta)/\eta^\star} = 1 + O(\frac{B\lambda_1}{\operatorname{tr} H}) \to 1$ as $\frac{B\lambda_1}{\operatorname{tr} H} \to 0$, and $\frac{\mathrm{BS}^\star(\Sigma_0^\star)}{2(1+\beta)/\eta^\star} = 1 + O(\frac{\kappa_\beta \operatorname{tr} H}{B\lambda_1}) \to 1$ as $\frac{\kappa_\beta \operatorname{tr} H}{B\lambda_1} \to 0$.

3. (Suppression of $\mathrm{BS}^\star$ relative to $2(1-\beta)/\eta^\star$ under flat-mode curvature noise when $H_{B_t} \not\succeq 0$.) *In the setting of Theorem 8 ($P \ge 2$, where H_{B_t} can be indefinite) with $\max_m(\lambda_m + s_m) \le M_0 < \infty$ and $\widetilde{\kappa}_6 \triangleq \max_m \frac{|\mathbb{E}[(\omega_{i,m}^2 - 1)^3]|}{2+\kappa_{\mathrm{feat}}} < \infty$ ($0/0 \triangleq 0$: $\widetilde{\kappa}_6 = 0$ for Rademacher, 4 for Gaussian), $|\mathrm{BS}^\star(\Sigma_0^\star)| \le \big(3 + \frac{\max(P+\kappa_{\mathrm{feat}}, 2(P-1)+\widetilde{\kappa}_6)}{B}\big) M_0$ for every spectrum, whereas if a mode k has $\lambda_k \downarrow 0$ at fixed $s_k > 0$ while some $j \neq k$ has $s_j > 0$, then $\eta^\star = O(\sqrt{\lambda_k})$ by Theorem 8(3) and*

$$\frac{|\mathrm{BS}^\star(\Sigma_0^\star)|}{2(1-\beta)/\eta^\star} = O\big(\sqrt{\lambda_k}\big) \longrightarrow 0 \qquad (\lambda_k \downarrow 0). \tag{13}$$

Proof in Appendix A.

Theorem 14 (Saturation in a slow-fast model: equilibrium at $\rho(\mathcal{K}_{\eta,\beta}) = 1$ against a Krylov–Bogoliubov secant equilibrium)**.** *Fix $\eta > 0$, $\beta \in [0, 1)$ and $\Sigma_\xi \succ 0$.*

1. (Fast-epoch, quasi-stationary, and coupled stochastic progressive-sharpening barrier at $\rho(\mathcal{K}_{\eta,\beta}) = 1$.) *Let $y \in [y_{\min}, y_{\max}] \subset \mathbb{R}$ parametrize C^1 families $H(y) \succ 0$ and $\Delta H_i(y) = H_i(y) - H(y)$ in Theorem 9, with threshold $\eta^\star(y) \triangleq \eta_{\mathrm{crit}}^{\mathrm{EoSS},P}(\beta, B; y) = 1/\lambda_{\max}\big(\mathcal{L}_0(y)^{-1/2}\mathcal{B}_\beta(y)\mathcal{L}_0(y)^{-1/2}\big)$ and*

spectral radius $r(y) \triangleq \rho(\mathcal{K}_{\eta,\beta}(y))$. *Suppose* $\eta/\eta^\star(y)$ *is strictly increasing on* $[y_{\min}, y_{\max}]$ *and crosses* 1 *at* $y^\star \in (y_{\min}, y_{\max})$ *with* $\eta < \eta_{\det}(y^\star)$, *simple top eigenvalue* $1/\eta^\star(y^\star) = 1/\eta$, *unit-trace eigenmatrix* $\Sigma_0^\star \in \mathbb{S}_+^P$, *and* $g_0 \triangleq -\partial_y(\eta/\eta^\star(y))\big|_{y=y^\star} < 0$ *(equivalently* $g_\star \triangleq r'(y^\star) > 0$*). For the fast recursion (2) at frozen* y *and a stochastic slow gradient* $g_{y,t}(w_t, y) = -v_{\mathrm{PS}} + \frac{1}{2} w_t^\top \Phi_{B_t}(y) w_t + \xi_{y,t}$ *with drive* $v_{\mathrm{PS}} > 0$, $\mathbb{E}[\Phi_{B_t}(y) \mid \mathcal{F}_{t-1}] = \Phi(y) \in \mathbb{S}^P$ C^1 *in* y *(*$\Phi(y) = \partial_y H(y)$ *for* $\mathcal{L}(w, y) = \mathcal{L}_{\mathrm{slow}}(y) + \frac{1}{2} w^\top H(y) w$*) and* $\mathbb{E}[\xi_{y,t} \mid \mathcal{F}_{t-1}] = 0$, *the* K*-step fast-epoch expected slow gradient* $G_{\mathrm{slow},K}(y) \triangleq \mathbb{E}[\frac{1}{K} \sum_{t=1}^K g_{y,t}(w_t, y)]$ *satisfies: (a) for every* $y \geq y^\star$ *where* $\Phi(y) \succ 0$, $G_{\mathrm{slow},K}(y) \geq -v_{\mathrm{PS}} + \frac{1}{8} \lambda_{\min}(\Phi(y)) \big(c(y)(K+1) - 2\mathbb{E}\|w_0\|^2/K\big) \to +\infty$ *as* $K \to \infty$ *with* $c(y) > 0$ *from Corollary 4; (b) for every* $y \in [y_{\min}, y^\star)$, $\lim_{K\to\infty} G_{\mathrm{slow},K}(y)$ *equals the quasi-stationary slow drift*

$$G_{\mathrm{slow}}(y) \triangleq -v_{\mathrm{PS}} + \frac{1}{2} \operatorname{tr}\big(\Phi(y)\, \Sigma_0(y)\big) = -v_{\mathrm{PS}} + \frac{\bar{c}_\star\, \sigma_\xi^2}{1 - \eta/\eta^\star(y)} + O(1) \qquad (y \uparrow y^\star), \tag{14}$$

where $\Sigma_0(y)$ *is the stationary covariance of Theorem 2(2),* $\sigma_\xi^2 \triangleq \operatorname{tr}(\Sigma_0^\star \Sigma_\xi) > 0$, *and* $\bar{c}_\star \triangleq \frac{\eta \operatorname{tr}(\Phi(y^\star)\Sigma_0^\star)}{4(1-\beta) \operatorname{tr}(H(y^\star)(\Sigma_0^\star)^2)}$. *If* $\operatorname{tr}(\Phi(y^\star)\Sigma_0^\star) > 0$, *then* $\lim_{y\uparrow y^\star} G_{\mathrm{slow}}(y) = +\infty$*: for every* $v_{\mathrm{PS}} > \frac{1}{2} \operatorname{tr}(\Phi(y_{\min})\Sigma_0(y_{\min}))$ *there is a quasi-stationary equilibrium* $y_{\mathrm{eq}} \in (y_{\min}, y^\star)$ *with* $G_{\mathrm{slow}}(y_{\mathrm{eq}}) = 0$, *and as* $\sigma_\xi^2/v_{\mathrm{PS}} \to 0$ *(writing* $\Sigma_\xi = \sigma_\xi^2 \widehat{\Sigma}_\xi$ *with* $\operatorname{tr}(\Sigma_0^\star \widehat{\Sigma}_\xi) = 1$*), the unique equilibrium near* $y^\star$ *satisfies* $G'_{\mathrm{slow}}(y_{\mathrm{eq}}) > 0$ *and, with* $c_\star \triangleq d_\star \bar{c}_\star = \frac{\operatorname{tr}(\Phi(y^\star)\Sigma_0^\star)}{2\langle \Sigma_0^\star, \mathcal{V}_{y^\star}(\Sigma_0^\star)\rangle_F} > 0$,

$$\frac{\eta^\star(y_{\mathrm{eq}}) - \eta}{\eta} = \frac{\bar{c}_\star\, \sigma_\xi^2}{v_{\mathrm{PS}}} + O\left(\frac{\sigma_\xi^4}{v_{\mathrm{PS}}^2}\right) > 0, \qquad 1 - \rho\big(\mathcal{K}_{\eta,\beta}(y_{\mathrm{eq}})\big) = \frac{c_\star\, \sigma_\xi^2}{v_{\mathrm{PS}}} + O\left(\frac{\sigma_\xi^4}{v_{\mathrm{PS}}^2}\right) > 0; \tag{15}$$

and (c) if $\Phi(y) \succ 0$ *on* $[y_{\min}, y_{\max}]$ *and* $v_{\mathrm{PS}} > \frac{1}{2} \operatorname{tr}(\Phi(y_{\min})\Sigma_0(y_{\min}))$, *then for* $G_{\mathrm{slow},K}$ *evaluated from* $w_0 = w_{-1} = 0$ *and every finite* $K > \max_{y\in[y^\star, y_{\max}]} \frac{8 v_{\mathrm{PS}}}{\lambda_{\min}(\Phi(y)) c(y)}$, *every root of* $G_{\mathrm{slow},K}$ *on* $[y_{\min}, y_{\max}]$ *lies strictly in the stable region* $(y_{\min}, y^\star)$*; moreover, on any compact interval* $[y_-, y_+] \subset (y_{\min}, y^\star)$ *containing a root* y_{eq} *of* G_{slow} *on which* $G'_{\mathrm{slow}}(y) > 0$ *(such as the equilibrium near* $y^\star$ *in (b), or any compact subinterval containing the globally unique root* $y_{\mathrm{eq}} \in (y_{\min}, y^\star)$ *when* $y \mapsto \operatorname{tr}(\Phi(y)\Sigma_0(y))$ *has positive derivative on* $[y_{\min}, y^\star)$*), for all sufficiently large* K *there is a unique root* $y_{\mathrm{eq},K} \in [y_-, y_+]$ *of* $G_{\mathrm{slow},K}$ *with* $y_{\mathrm{eq},K} \to y_{\mathrm{eq}}$ *as* $K \to \infty$, *and with* $0 < m_0 \leq \min_{y\in[y_-,y_+]} G'_{\mathrm{slow},K}(y) \leq \max_{y\in[y_-,y_+]} G'_{\mathrm{slow},K}(y) \triangleq L_K < \infty$, $\operatorname{Var}(\frac{1}{K} \sum_{t=1}^K g_{y,t}(w_{k,t}, y_k) \mid \mathcal{F}_{k-1}) \leq s_K^2$, *and any* $\mathcal{F}_{k-1}$*-measurable epoch initial state satisfying* $\mathbb{E}[\|w_{k,0}\|^2 + \|w_{k,-1}\|^2 \mid \mathcal{F}_{k-1}] \leq R_0^2$ *(*$R_0 = 0$ *when each epoch resets* $w_{k,0} = w_{k,-1} = 0$*;* $R_0 < \infty$ *under continuous carry-over* $(w_{k,0}, w_{k,-1}) = (w_{k-1,K}, w_{k-1,K-1})$ *on* $[y_-, y_+]$ *since* $\sup_{y\in[y_-,y_+]} \rho(\mathcal{T}_y) < 1$*), there is a constant* $C_+ < \infty$ *independent of* K *such that the coupled stochastic epoch recursion* $y_{k+1} = \Pi_{[y_-,y_+]}\big(y_k - \eta_y \frac{1}{K} \sum_{t=1}^K g_{y,t}(w_{k,t}, y_k)\big)$ *for every slow step size* $0 < \eta_y \leq 1/L_K$ *satisfies* $\mathbb{E}[(y_k - y_{\mathrm{eq},K})^2] \leq (1 - \eta_y m_0)^k \mathbb{E}[(y_0 - y_{\mathrm{eq},K})^2] + \frac{\eta_y s_K^2}{m_0} + \frac{2C_+^2 R_0^4}{m_0^2 K^2}$ *for all* $k \geq 1$.

2. (Self-consistent Krylov–Bogoliubov secant equilibrium above the tangent threshold.) *Suppose the tangent curvature* $H_{\tan} \succ 0$ *is supercritical (*$\eta_{\tan}^\star < \eta < \eta_{\det}(H_{\tan})$*), and under a Krylov–Bogoliubov secant linearisation [Krylov and Bogoliubov, 1947] parameterized by the excursion variance* $\gamma = \operatorname{tr}\Sigma_0 \in [0, \gamma_{\max}]$, *the* C^1 *secant family* $\big(H_{\mathrm{sec}}(\gamma), \Delta H_i^{\mathrm{sec}}(\gamma)\big)$ *satisfies* $\eta < \eta_{\det}(H_{\mathrm{sec}}(\gamma))$ *and has* $\eta/\eta_{\mathrm{sec}}^\star(\gamma)$ *strictly decreasing from* $\eta/\eta_{\tan}^\star > 1$ *at* $\gamma = 0$ *to* $\eta/\eta_{\mathrm{sec}}^\star(\gamma_{\max}) < 1$, *crossing* 1 *at* $\gamma^\star \in (0, \gamma_{\max})$ *with a simple top eigenvalue, unit-trace eigenmatrix* $\Sigma_0^\star(\gamma^\star)$, *and* $\partial_\gamma \eta_{\mathrm{sec}}^\star(\gamma^\star) > 0$. *Then for every* $\Sigma_\xi \succ 0$ *with* $\operatorname{tr}\Sigma_0^{\mathrm{sec}}(\gamma_{\max}) \leq \gamma_{\max}$ *(in*

particular for all small $\sigma_\xi^2 = \operatorname{tr}(\Sigma_0^\star(\gamma^\star)\Sigma_\xi) > 0$*) there is a self-consistent excursion variance* $\gamma_{\mathrm{eq}} \in (\gamma^\star, \gamma_{\max}]$ *solving* $\operatorname{tr}\Sigma_0^{\mathrm{sec}}(\gamma_{\mathrm{eq}}) = \gamma_{\mathrm{eq}}$*, with* $\eta_{\mathrm{sec}}^\star(\gamma_{\mathrm{eq}}) > \eta > \eta_{\mathrm{tan}}^\star$*; as* $\sigma_\xi^2 \to 0$*,* γ_{eq} *is unique on* $(\gamma^\star, \gamma_{\max}]$ *and satisfies*

$$1 - \frac{\eta}{\eta_{\mathrm{sec}}^\star(\gamma_{\mathrm{eq}})} = \frac{\eta\,\sigma_\xi^2}{2(1-\beta)\,\gamma^\star\,\operatorname{tr}\left(H_{\mathrm{sec}}(\gamma^\star)\Sigma_0^\star(\gamma^\star)^2\right)}\left(1 + O(\sigma_\xi^2)\right) > 0. \tag{16}$$

Proof in Appendix A.

What Theorems 13 and 14 answer and what they do not. (i) The Batch Sharpness of Andreyev et al. [2026, Definition 3.1] is the expectation, over the minibatch at a fixed checkpoint, of the ratio $g_B^\top H_B g_B / \|g_B\|^2$. Theorem 13(2) is about a stationary ratio of expectations, averaged also over the iterate w_t of the linearised recursion, and part (1)(a) about a ratio of expectations at a fixed checkpoint, which differs from the expectation of the ratio by $\operatorname{Cov}_B(R_B, W_B)$, at most $\sigma_{\mathrm{BS}}\mathrm{CV}_{g^2}$, and which equals $\bar{a} + \bar{v}$ under the hypothesis $\mathbb{E}[g_B g_B^\top \mid H_B] = \sigma_g^2 H_B$, which rank-one per-sample curvatures aligned with the per-sample gradients satisfy and the linearised gradient $H_B w$ of the quadratic model in general does not. Part (1)(b) reaches the expectation of the ratio only in the random-feature model with $H_i = z_i z_i^\top$ and $g_i \propto z_i$, and is informative only when $B^2\lambda_1 \ll \operatorname{tr} H$. (ii) In the general setting the theorem is a bridge, an identity at the threshold and the two limits of a stationary Batch Sharpness, together with a sufficient instability criterion through the variational test at matrices S that commute with H; its Batch Sharpness form needs the noise share θ_{noise} besides $\overline{\mathrm{BS}}$, so it is not a threshold on Batch Sharpness alone. (iii) Theorem 14 proves saturation in a quasi-stationary slow-fast model: the curvature is frozen within an epoch, the slow gradient has the form $-v_{\mathrm{PS}} + \frac{1}{2}w^\top \Phi w$, and the fast state restarts at each epoch or is carried over with bounded second moment; it is not a statement about the training dynamics. The $P = 33$ runs of Section 4.4, which train above the $|H|$ reading, are outside the hypotheses of its part (1), since H is indefinite at 90–100 % of their window checkpoints, and behave as its secant part (2) describes; whether they satisfy the hypotheses of part (2) was not tested. The variational criterion, Batch Sharpness limits and saturation law are checked numerically in `paper/checks/bridge-theorems.py`.

4 Measurements

4.1 The stochastic ceiling in $\mathbb{R}^P$: Gaussian against Rademacher features (Phase 1)

Phase 1 measures Theorem 5 on the quadratic model it is about: spectrum $\lambda_k = k^{-1/2}$, $P \in \{64, 256\}$, $\beta \in \{0, 0.9\}$, $B \in \{4, 16, 64\}$ ($B \le 16$ at $P = 256$), isotropic additive noise $\sigma_\xi = 0.01$, and three feature families, Gaussian i.i.d. ($\kappa_{\mathrm{feat}} = 0$), Rademacher i.i.d. ($\kappa_{\mathrm{feat}} = -2$, $\omega^4 \equiv 1$) and a finite pool of n Gaussian features ($n = 512$ at $P = 64$, 1024 at $P = 256$) sampled without replacement, as in Lee et al. [2022], for which the prediction uses the eigenvalues of the pool covariance and the factor $1 - \zeta$, $\zeta = B/n$, a prediction that replaces the noise operator of the realised pool by its average over pool draws (Section 3, after Theorem 9). The empirical ceiling is found by bisection on η (12 iterations, 8 streams of 1000 steps, a learning rate being declared unstable when the mean-square energy over the streams grows by more than 2.5× between the second and the last quarter), 8 seeds per cell; the median over seeds of the bisection half-width is below 0.05 % in every cell. Four predictions were written before the run (`labs/mean-square-boundary/README.md`, Phase 1): `p1`, every i.i.d. cell inside the sandwich within 3 % (the registered text has the sandwich without the terms $\max(\alpha_{\mathrm{diag}}, 0)$ and $\min(\alpha_{\mathrm{diag}}, 0)$ of (5), an interval that is empty when $\alpha_{\mathrm{diag}} < 0$; `p1` is scored

with (5), a deviation written after the registration, and fails under either form); `p2`, the secular root within 5 % median error in every i.i.d. cell; `p3`, in every Rademacher cell with $\kappa_\beta \operatorname{tr} H/(B\lambda_1) \geq 1$ the one-dimensional ceiling exceeds the empirical one by more than 2× while the root stays within 5 %; `p4`, in the pool arm with $\zeta \leq 1/8$ the root of $\|\mathcal{K}\| = 1$ of Lee et al. [2022] within 8 %. Table 1 has the 30 cells.

Table 1: Phase 1 (`raw-phase1.json`, 8 seeds per cell). $\rho_B = \kappa_\beta \operatorname{tr} H/(B\lambda_1)$; 1D is the ceiling of Proposition 1 along u_1 ($\nu_{11}^2 = \lambda_1^2(2 + \kappa_{\text{feat}})/B$); lower, root and upper are the sandwich (5) and the secular root (4) (pool: with the factor $1 - \zeta$ and the pool eigenvalues, seed-averaged; this uses the noise operator averaged over pool draws, and the exact threshold of one realised pool is that of Theorem 9); emp. is the median empirical ceiling; err is the median over seeds of the per-seed relative error of the root (not the error of the median ceiling), Lee the same for the root with $\alpha_{\text{diag}} = 1$; in is the criterion of `p1`, inside the sandwich within 3 % plus the bisection half-width; 1D/emp is the mean over seeds of the per-seed ratio of the one-dimensional ceiling to the empirical one (pool: $\nu_{11}^2 = \lambda_1^2(2 + \kappa_{\text{feat}})(1 - \zeta)/B$ with the pool λ_1, seed-averaged). The entry 2.0399 of the Gaussian $P = 64$, $\beta = 0$, $B = 64$ cell is censored: the bisection reached its cap $1.02 \times 2 = 2.04$ to within its resolution, and the entries marked † are computed against it, not against a measured ceiling.

P	β	B	feat.	ρ_B	1D	lower	root	upper	emp.	err (%)	Lee (%)	1D/emp	in
64	0	4	Gauss.	3.7	1.3333	0.4081	0.4897	0.5479	0.4894	0.34	2.35	2.72	yes
64	0	4	Radem.	3.7	2.0000	0.4545	0.5123	0.5479	0.5135	0.22	2.45	3.89	yes
64	0	4	pool	3.4	1.2645	0.4052	0.4887	0.5533	0.4896	0.33	2.38	2.58	yes
64	0	16	Gauss.	0.9	1.7778	1.0126	1.4468	1.7778	1.4862	2.65	0.47	1.20	yes
64	0	16	Radem.	0.9	2.0000	1.0810	1.5299	2.0000	1.5544	1.57	4.26	1.29	yes
64	0	16	pool	0.9	1.7422	1.0156	1.4341	1.7422	1.4421	1.00	1.97	1.20	yes
64	0	64	Gauss.	0.2	1.9394	1.6080	1.9268	1.9394	2.0399	5.54†	4.11†	0.96†	no
64	0	64	Radem.	0.2	2.0000	1.6494	1.9861	2.0000	2.0004	0.71	2.22	1.00	yes
64	0	64	pool	0.2	1.8870	1.6050	1.8778	1.8870	1.9408	3.82	2.53	0.97	yes
64	0.9	4	Gauss.	69.4	0.3619	0.0506	0.0533	0.0548	0.0532	0.44	2.45	6.80	yes
64	0.9	4	Radem.	69.4	3.8000	0.0548	0.0558	0.0579	0.0559	0.39	2.49	68.04	yes
64	0.9	4	pool	67.4	0.3565	0.0509	0.0536	0.0552	0.0538	0.97	1.54	6.60	yes
64	0.9	16	Gauss.	17.3	1.1259	0.1946	0.2103	0.2191	0.2113	0.59	1.81	5.30	yes
64	0.9	16	Radem.	17.3	3.8000	0.2191	0.2199	0.2215	0.2196	0.78	2.06	17.26	yes
64	0.9	16	pool	16.6	1.1002	0.1989	0.2155	0.2256	0.2170	1.69	1.55	5.07	yes
64	0.9	64	Gauss.	4.3	2.3843	0.6747	0.7948	0.8766	0.7971	0.77	1.99	2.99	yes
64	0.9	64	Radem.	4.3	3.8000	0.7543	0.8315	0.8766	0.8428	1.44	3.54	4.53	yes
64	0.9	64	pool	4.2	2.4203	0.7454	0.8899	1.0012	0.9027	1.44	1.05	2.68	yes
256	0	4	Gauss.	7.6	1.3333	0.2249	0.2531	0.2617	0.2544	0.53	0.15	5.24	yes
256	0	4	Radem.	7.6	2.0000	0.2383	0.2565	0.2617	0.2578	0.49	1.15	7.76	yes
256	0	4	pool	7.3	1.2734	0.2241	0.2528	0.2627	0.2534	0.19	0.68	5.03	yes
256	0	16	Gauss.	1.9	1.7778	0.6727	0.9268	1.0467	0.9269	0.31	0.79	1.92	yes
256	0	16	Radem.	1.9	2.0000	0.7022	0.9416	1.0467	0.9450	0.37	1.15	2.12	yes
256	0	16	pool	1.8	1.7219	0.6708	0.9232	1.0624	0.9220	0.25	0.77	1.86	yes
256	0.9	4	Gauss.	145.2	0.3619	0.0252	0.0260	0.0262	0.0261	0.68	0.26	13.86	yes
256	0.9	4	Radem.	145.2	3.8000	0.0262	0.0263	0.0269	0.0264	0.49	1.13	143.68	yes
256	0.9	4	pool	136.9	0.3422	0.0252	0.0260	0.0262	0.0262	0.82	0.49	13.06	yes
256	0.9	16	Gauss.	36.3	1.1259	0.0987	0.1034	0.1047	0.1039	0.44	0.22	10.85	yes
256	0.9	16	Radem.	36.3	3.8000	0.1047	0.1048	0.1052	0.1051	0.64	0.99	36.15	yes
256	0.9	16	pool	35.1	1.1017	0.0999	0.1047	0.1062	0.1049	0.39	0.43	10.49	yes

Outcomes: `p3` and `p4` hold, `p1` and `p2` fail, both on the same cell, and are recorded as failed. In

29 of the 30 cells the median empirical ceiling lies inside the sandwich within that tolerance, and strictly inside in 27: the Rademacher cell at $P = 64$, $\beta = 0$, $B = 64$ is 0.02 % above its upper bound 2, 1.6 times the median bisection half-width of that cell (0.012 %) and within the tolerance of `p1`, and the pool cell at the same (P, β, B) is 2.85 % above its seed-averaged upper bound 1.8870, the effect of the failed cell at a smaller size. The secular root predicts the median with median error between 0.19 % and 3.82 %, the upper end from that pool cell (0.19–2.65 % in the other 28 cells) (the Lee root, without the kurtosis term, between 0.15 % and 4.26 %); the bisection resolution is 0.05 %, so these errors are the finite-horizon bias of the divergence criterion and the seed-to-seed spread, not resolution. The Rademacher separation is as predicted: at $\beta = 0.9$ the one-dimensional ceiling along u_1, which is $2(1 + \beta)/\lambda_1 = 3.8$ for every B because $\nu_{11}^2 = 0$, exceeds the measured ceiling by 68× ($P = 64$, $B = 4$), 17× ($B = 16$), 4.5× ($B = 64$), 144× ($P = 256$, $B = 4$) and 36× ($B = 16$), while the secular root is within 1.5 % in all five cells; the cross-mode couplings $\lambda_1\lambda_k/B$ set the ceiling, and a probe restricted to the ray along u_1 cannot see them. The two plateaus are visible along B at $\beta = 0$: at $B = 4$ the ceiling is $\operatorname{tr} H/B$-dominated ($\rho_B = 3.7$, 7.6) and at $B = 64$ it is within 4 % of the deterministic $2/\lambda_1$.

The failed cell is $P = 64$, $\beta = 0$, $B = 64$ with Gaussian features: the measured ceiling, 2.0399, is above the deterministic ceiling $2/\lambda_1 = 2$ (the bisection cap is 1.02× that value), which no mean-square ceiling can be. The diagnosis is the gap between mean-square and almost-sure stability of a random-coefficient recursion: along u_1 the multiplier $1 - \eta(\lambda_1 + \zeta_t)$ has, at $\eta = 2.00$–2.04 and $\operatorname{Var}(\zeta_t) = 2\lambda_1^2/B$, a negative Lyapunov exponent $\mathbb{E}\log|1-\eta(\lambda_1+\zeta_t)|$ while its second moment grows, so the typical trajectory decays (over the χ^2_{64} law of ζ_t the exponent is -0.072, -0.051, -0.030 at $\eta =$ 2.00, 2.02, 2.04 and $+0.031$ at 2.10, `paper/checks/evidence-ranges.mjs`) and the mean square diverges through rare paths that 8 streams do not sample. For the homogeneous recursion along u_1 at $\beta = 0$ the multipliers are i.i.d., so $\mathbb{E}[x_t^4]/\mathbb{E}[x_t^2]^2 = (m_4/m_2^2)^t$ with $m_q = \mathbb{E}|1-\eta(\lambda_1+\zeta_t)|^q$; over the χ^2_{64} law $m_4/m_2^2 = 1.498$ at $\eta = 1.94$, so the ratio is e^{404} at $t = 1000$, and the relative standard error of a mean of x_t^2 over 8 paths is of order e^{202} (`paper/checks/second-moment-renewal.py`). Resampling targets the second moment directly. Carry S unit vectors $u_{t,s} \in \mathbb{R}^{2P}$, apply an independent step $y_{t,s} = F_{t,s}u_{t,s}$, record $\hat\mu_t = \frac{1}{S}\sum_s \|y_{t,s}\|^2$, and draw $u_{t+1,1}, \dots, u_{t+1,S}$ from $\{y_{t,s}/\|y_{t,s}\|\}$ with probabilities $\|y_{t,s}\|^2/(S\hat\mu_t)$; with $\widehat{M}_t = \prod_{\tau<t}\hat\mu_\tau$ and $\widehat{Q}_t = \widehat{M}_t \frac{1}{S}\sum_s u_{t,s}u_{t,s}^\top$, $\mathbb{E}[\widehat{Q}_{t+1} \mid \mathcal{F}_t] = \mathcal{T}(\widehat{Q}_t)$, so $\mathbb{E}\widehat{Q}_t = \mathcal{T}^t(\widehat{Q}_0)$ for every S and t (checked by exact enumeration in the same script). Along u_1 at $\beta = 0$ the $\hat\mu_\tau$ are i.i.d. with mean m_2 and variance $(m_4 - m_2^2)/S$: the relative variance of $\widehat{M}_t$ is $(1 + (m_4/m_2^2 - 1)/S)^t - 1$, still e^{60} at $S = 8$ and $t = 1000$, but $\frac{1}{T}\sum_{\tau<T}\hat\mu_\tau$ estimates m_2 with variance $(m_4 - m_2^2)/(ST)$, and its crossing of 1 along u_1 is the one-dimensional mean-square ceiling 1.9394 of this cell. On the full operator of this cell, with $S = 8$ and $K = 64$ independent blocks of $2m = 16$ steps (the budget of one Phase 1 cell), the ratio $\widehat{R} = \overline{\widehat{M}_{2m}}/\overline{\widehat{M}_m}$ of block means tends as $K \to \infty$ to $\operatorname{tr}\mathcal{T}^{2m}(Q_0)/\operatorname{tr}\mathcal{T}^m(Q_0)$, which by the exact recursion crosses 1 at 1.926417 for $m = 8$ (`paper/checks/second-moment-renewal.py`) and at the secular root as m grows. A 12-step bisection of $\widehat{R} = 1$ on 8 seeds gives a median of 1.9330 (range 1.9200–1.9400), 0.32 % above the secular root 1.9268, where the sample paths of Phase 1 gave 2.003 with the cap raised; the three predictions written before the run (median within 1 % of the root, every seed within 3 %, median below 2.003) hold (`paper/checks/phase1-resampling-estimator.py`, `phase1-resampling-estimator.json`). The Rademacher cell at the same (P, β, B) has $\nu_{11}^2 = 0$, no such gap, and matches the root within 0.71 %; the Gaussian cell at $B = 16$ shows the same effect at a smaller size (2.65 %, the empirical value above the root). Proposition 7(1) makes this quantitative: along u_1 the mean-square and almost-sure thresholds sit at $\frac{2}{\lambda_1}(1 \mp \nu_{11,B}^2/\lambda_1^2)$ to first order (1.9375 and 2.0625 here; exactly 1.9394 and, by quadrature over the χ^2_{64} density, 2.0688, `paper/checks/evidence-ranges.mjs`), and the measured 2.0399 is the bisection cap $1.02 \times 2 = 2.04$ to within the bisection resolution, a censored

observation above the mean-square ceiling. A post-hoc rerun of this cell with the cap raised to 2.3, not a pre-registered prediction (`evidence/mean-square-boundary/phase1-diag-as-vs-ms.json`, written by `run-phase1-cap-diagnostic.ts`), gives a sample-path threshold of median 2.003 (range 1.963–2.067) for Gaussian and 2.000 (1.976–2.013) for Rademacher features. For Gaussian features the median lies between the mean-square ceiling and the one-dimensional almost-sure root, whose order Proposition 7(1) gives, near the middle of that interval; a sample-path harness of this length does not estimate the almost-sure root; for Rademacher features both thresholds are 2. Theorem 5 is a statement about the second moment; where the multiplicative noise along u_1 dominates and B is large, a sample-path harness measures a threshold above the mean-square one (here between the mean-square ceiling and the one-dimensional almost-sure root), and the resampling estimator above, which targets $\mathbb{E}\|w_t\|^2$ itself, measures the theorem there.

The secular root against the exact second-moment operator (Phase 3). Phase 1 compares Theorem 5 with sample paths, whose threshold can lie above the mean-square one where the multiplicative noise along u_1 is large. Phase 3 compares it with the mean-square threshold itself: the spectral radius of the second-moment operator $\mathcal{T}$, computed by the matrix-free power iteration from the identity of Lemma 3, with one application of $\mathcal{T}$ in $O(P^2)$ operations in the eigenbasis of H for random-feature minibatches (checked against the exact expectation by enumeration on a non-symmetric input; the operator uses the closed form of $\mathcal{V}$ of Theorem 5, so this phase tests the diagonal closure, the secular reduction and the code, and the enumeration tests that closed form). On the spectrum $\lambda_k = k^{-1/2}$ of Phase 1, with $P \in \{64, 256, 1024\}$, $\beta \in \{0, 0.9\}$, $B \in \{4, 16, 64\}$ and Gaussian and Rademacher features (36 cells), $\rho(\mathcal{T})$ is below 1 at $(1 - 10^{-3})\eta^\star$ and above 1 at $(1 + 10^{-3})\eta^\star$ in every cell, $\eta^\star$ the secular root; the crossing located from $\rho(\mathcal{T}(\eta^\star)) - 1$ and the local slope is within 6.11×10^{-10} (relative) of $\eta^\star$ in every cell (six cells at $P = 64$, $\beta = 0.9$, $B \in \{4, 16, 64\}$ for both feature families have `allConverged` false, the conjunction of four convergence flags; a rerun, `check-phase3-convergence.ts`, shows that the flag that fails is the agreement test of the Gelfand estimate on the antisymmetric matrices, while the three symmetric iterations, which locate the crossing, converge in 248–275 steps from a cold start (the run, warm-started from the iterate at $(1 - 10^{-3})\eta^\star$, recorded 175–208 at the root); these cells have crossing relative errors between 3.2×10^{-14} and 5.1×10^{-13} and $|\rho(\mathcal{T}(\eta^\star)) - 1| \le 3.8 \times 10^{-14}$), and the radius on the antisymmetric matrices is 0.65–0.99 at $\eta^\star$, below the symmetric radius 1, as Lemma 3 requires. Three predictions were written before the full run (`raw-phase3.json`, field `predictionTexts`): `q1`, in every cell the radius is below 1 at $(1-10^{-3})\eta^\star$ and above 1 at $(1+10^{-3})\eta^\star$; `q2`, the crossing within 10^{-8} (relative) of $\eta^\star$; `q3`, the radius on the antisymmetric matrices at $\eta^\star$ strictly below 1. All three hold. They were written after a four-cell prototype at $P \le 256$, which the raw file records; after them a $P = 16$ debugging run, in which the ratio of norms did not converge on the antisymmetric part at $\beta = 0.9$, led to replacing the antisymmetric estimator by the Gelfand estimate over a fixed horizon before the full run, so `q3` was seen at $P = 16$ first; the $P = 1024$ cells, beyond the reach of the Phase 1 simulation, were not run before them (`raw-phase3.json`, field `provenanceNote`).

4.2 Trace and the ceiling in $\mathbb{R}^P$ on two networks (Phase 2)

Phase 2 asks whether the random-feature model of Theorem 5 describes real per-sample Hessians. The checkpoints are two networks of the companion manuscript: a 64–48–10 network ($P = 3\,610$) and a 256–48–10 network ($P = 12\,826$), both with tanh hidden units, squared loss and $N = 64$ samples of a synthetic teacher–student regression with ten outputs, with $\lambda_1 = 98.677$ and 148.353 at Θ_0 from analytical Hessian-vector products [Pearlmutter, 1994]; $\eta_{\text{crit}}^{\text{quad}} = 2(1 + \beta)/\lambda_1$ is the deterministic threshold. Predictions were written before the run (`raw-phase2.json`, field

predictionTexts); of the three reported here q1 and q2 hold and q3 fails. The others concern the estimator of the companion manuscript.

Trace and noise-rank ratio (q1). The trace is exact, $\sum_i e_i^\top H e_i$ from P analytical Hessian-vector products: $\operatorname{tr} H = 601.74$ at $P = 3\,610$ and 901.72 at $P = 12\,826$, so $\operatorname{tr} H/\lambda_1 = 6.10$ and 6.08 and $\rho_B = 7.24$ and 7.22 at $\beta = 0.9$, $B = 16$. One hundred shifted power iterations give Rayleigh quotients 0.012 and -0.072, which are upper bounds on $\lambda_{\min}$: at $P = 12\,826$ the Hessian is indefinite ($\lambda_{\min} \le -0.072 = -4.9 \times 10^{-4}\lambda_1$); at $P = 3\,610$ the value certifies nothing about the sign, and the dense spectrum of Section 4.3 has $\lambda_{\min} = -0.0833$ with 1 528 negative eigenvalues. Both Hessians are indefinite, so the lower ends of the trace brackets below are approximations (Remark 16; the guaranteed lower end at $P = 3\,610$ is 0.00411). The effective feature kurtosis read from ν_{11} ($\nu_{11}^2 = \lambda_1^2(2 + \kappa_{\text{feat}})/B$) is -1.16 and -0.32, $\alpha_{\text{diag}} = 0.81$ and 1.80. With λ_1, λ_2 and $\operatorname{tr} H$ alone the secular root of Theorem 5 is bracketed (Appendix A, trace bracket): $[0.00492, 0.00521]$ and $[0.00308, 0.00338]$, which is 3.7× and 2.5× below the untethered ($\mu = 0$) one-dimensional ceiling $\eta_{\text{crit}}^{\text{EoSS}}(0.9, 16) = 2(1+\beta)\lambda_1/(\lambda_1^2 + \kappa_\beta \nu_B^2) = 0.01928$ and 0.008564.

Heavy-ball SGD with real minibatches in $\mathbb{R}^P$ (q2, q3). The iteration is $w_{t+1} = w_t + \beta(w_t - w_{t-1}) - \eta\,(g_{B_t}(w_t) - g_{B_t}(\Theta_0))$ with minibatches of 16 of the 64 samples drawn with replacement: the curvature noise is the one of the network and the mean gradient is removed, so Θ_0 is a fixed point and the question is its linear stability under that noise. Every stream starts at distance 10^{-6} in the range of H (a Gaussian start would put part of its energy in the null space of H, where heavy-ball keeps it constant; at $P = 12\,826$, where the input dimension 256 exceeds the 64 samples, first-layer weights orthogonal to every input span a null space of every H_i of dimension at least $48 \cdot 192 = 9\,216$), and the statistic is the one of Phase 1: the log ratio of the energy in the fourth and second quarters of a run, summed over 8 streams of 1 000 steps, bisected over η in 14 steps from 1.1× the quadratic threshold, 4 seeds. The median values are 0.002707 at $P = 3\,610$ and 4.90×10^{-5} at $P = 12\,826$ (bisection half-widths 0.05 % and 1.8 %): 7.1× and 175× below the one-dimensional ceiling and 14× and 523× below the quadratic threshold. q2 (a factor of at least 1.5 below the one-dimensional ceiling) holds; q3 (inside the trace bracket of the random-feature model within 10 %) fails, by 1.8× at $P = 3\,610$ and 63× at $P = 12\,826$.

Growth grid (post hoc, no prediction registered). On a fixed grid of learning rates, 4 streams, 1 000 and 4 000 steps, the log growth at $P = 3\,610$ is negative at both lengths for 0.002–$0.01 \times \eta_{\text{crit}}^{\text{quad}}$ (-0.99 to -0.22), is -0.88 then $+0.69$ at 0.02×, -0.42 then $+5.8$ at 0.05×, and $+2.7$ at 0.1× with the excursion guard tripped at 4 000 steps; at $P = 12\,826$ it is $+1.0$ then $+10.2$ at 0.002× and $+6.0$ at 0.005× with the guard tripped at 4 000 steps. A diffusive process gives the same window ratio at any length; the growth here increases with the length, so it is exponential, and the bisection value is the learning rate at which it reaches the ratio 2.5 within 1 000 steps, not a boundary. This finite-horizon statistic changes sign between 0.01× and 0.02× the quadratic threshold at $P = 3\,610$ (3.9×10^{-4} to 7.7×10^{-4}, 25–50× below the one-dimensional ceiling) and below 0.002× at $P = 12\,826$ (under 5.2×10^{-5}, more than 167× below). It is not a mean-square stability boundary: the mean Hessian has negative eigenvalues at both checkpoints (1 528 at $P = 3\,610$, Section 4.3; $\lambda_{\min} \le -0.072$ at $P = 12\,826$), so the mean of the untethered iteration has a root above 1 at every $\eta > 0$ and no learning rate is mean-square stable; the grid measures how fast energy escapes along the negative directions within 1 000–4 000 steps.

What this means for the ceiling. The one-dimensional ceiling of Proposition 1 and the secular root of Theorem 5 are statements about models in which the curvature noise entering a mode is proportional to that mode's curvature ($z = H^{1/2}\omega$ puts noise $\lambda_k\omega_k^2$ on mode k and none on a flat mode). On these networks the per-sample Hessians H_i inject curvature noise into directions where H itself is nearly flat; Theorem 8 makes this quantitative: with a curvature-noise covariance $\Sigma_z \neq H$ the ceiling is at most $2(1-\beta)B/\operatorname{tr}(H^{-1}\Sigma_z^2)$ when every $d_k \geq 0$, and for every $\kappa_{\text{feat}} \geq -2$ and $P \geq 2$ a mode with $\lambda_k \to 0$ at fixed noise s_k drives it to zero at least as fast as $\sqrt{\lambda_k}$, since $\eta^\star \leq 2(1-\beta)B\sqrt{\lambda_j\lambda_k}/(s_j s_k)$ for any other mode j; such a direction has little restoring force and receives energy at every learning rate. At these two checkpoints, however, the mean Hessian also has negative directions, and Phase 4 (Section 4.3) shows at $P = 3\,610$ that once a tether lifts them the exact threshold is 1.23× below the one-dimensional ceiling, with no flat-mode collapse down to a smallest tethered eigenvalue of $\frac{1}{128}$: the growth measured here at 25–50× and more than 167× below that ceiling comes from the negative directions, not from curvature noise on positive flat modes. The excursions of actual training at the learning rates practitioners use come from the nonlinearity (the loss is bounded below and the mean gradient is not zero), not from a linear ceiling. In $\mathbb{R}^P$ the linear cap at these checkpoints is one to two orders of magnitude below the one-dimensional ceiling. Theorem 5 is unaffected: Phase 1 measures it on the model it is about, and Phase 2 measures where the model stops applying.

Table 2: Phase 2 (`raw-phase2.json`; $\beta = 0.9$, $B = 16$, 4 seeds, 8 streams, 1 000 steps). 1D is the ceiling of Proposition 1 along u_1; the bracket is the trace bracket of the secular root (Appendix A) with κ_{feat} read from ν_{11}; emp. is the median learning rate at which the energy of the linearised minibatch iteration grows by the Phase 1 ratio within the run, which is not a stability boundary (see the text).

P	$\operatorname{tr} H$	$\operatorname{tr} H/\lambda_1$	ρ_B	$\lambda_{\min}$ (upper bd.)	$\eta_{\text{crit}}^{\text{quad}}$	1D	bracket	emp.	1D/emp
3 610	601.74	6.10	7.24	0.012	0.038509	0.019278	[0.00492, 0.00521]	0.002707	7.1
12 826	901.72	6.08	7.22	−0.072	0.025615	0.008564	[0.00308, 0.00338]	0.0000490	175

4.3 The exact threshold of Theorem 9 on tethered network Hessians (Phase 4)

Phase 2 measured an iteration for which the theory has no finite target: the full eigendecomposition of the mean Hessian of the $P = 3\,610$ network gives $\lambda_{\min} = -0.0833$ with 1 528 of its 3 610 eigenvalues negative (prediction `t2` below), and the $P = 33$ network of the companion manuscript (6 inputs, 4 tanh hidden units, one output, squared loss, $N = 64$ samples of a synthetic teacher–student regression) has $\lambda_{\min} = -0.0080$ with two negative eigenvalues. A tether $\frac{\mu}{2}\|w - \Theta_0\|^2$ with $\mu = g + \max(0, -\lambda_{\min}(H))$ makes $H + \mu I \succ 0$ with smallest eigenvalue g and leaves ΔH_i, hence $\mathcal{V}$, unchanged, and Theorem 9(3) then gives one finite threshold $\eta^\star(\mu) = 1/\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2})$. Phase 4 computes it by Lanczos from the real per-sample Hessians: at $P = 33$ dense matrices from central differences of per-sample gradients ($\epsilon = 10^{-5}$, symmetrised), $\beta \in \{0, 0.9\}$; at $P = 3\,610$ exact per-sample Hessian-vector products [Pearlmutter, 1994], $\beta = 0.9$; $B = 16$ with replacement, $g \in \{128, 32, 8, 2, \frac{1}{2}, \frac{1}{8}, \frac{1}{32}, \frac{1}{128}\}$. Six predictions were written before the run (`raw-phase4.json`); `t1`–`t4` and `t6` hold and `t5` fails.

The formula against the exact operator (`t1`). In all 16 cells at $P = 33$ the spectral radius of the second-moment operator $\mathcal{T}$ of the tethered iteration is below 1 at $(1 - 10^{-3})\eta^\star$ (largest 0.99988515) and above 1 at $(1 + 10^{-3})\eta^\star$ (smallest 1.00011489), and a floating-point Collatz–

Wielandt bracket of the last power iterate decides each side. A post-hoc run with the certified bracket of Section 4.5 (`run-phase4-certified-bracket-p33.ts`, output `phase4-certified-bracket-p33.json`; run on 1 November 2026 after the expectation, both values inside the bracket and $\eta_{\mathrm{low}}/\eta_{\mathrm{up}} \geq 0.98$, was written in the header of the runner and committed in the lab) gives in all 16 cells an interval $[\eta_{\mathrm{low}}, \eta_{\mathrm{up}}]$ that contains the stored $\eta^\star$ and a dense reference, with $\eta_{\mathrm{low}}/\eta_{\mathrm{up}} \geq 0.99931$; both offsets $(1 \mp 10^{-3})\eta^\star$ lie outside it, so through Theorem 9 each side of `t1` is decided with every rounding error bounded, for the computed per-sample Hessians (central differences) and not for the exact network Hessians. At $P = 3\,610$, $g = 2$, the power-iteration radii are 0.99982427 and 1.00017600 (439 and 237 steps, converged to 10^{-12}), but the last iterate is not positive definite in floating point, at $\pm 10^{-3}$ and at the fallback offset $\pm 10^{-2}$, so the bracket does not apply (the implementation checks `c2` and `c3` fail): at that size the crossing rests on the power iteration alone.

Sample paths (`t3`, `t4`). With the statistic of Phases 1 and 2, the median sample-path ceiling of the linearised tethered iteration lies at 1.0008–1.190 $\eta^\star$ at $P = 33$ and at 1.081–1.257 $\eta^\star$ at $P = 3\,610$, never below the mean-square threshold, as Jensen's inequality requires of the almost-sure one (Proposition 7). The network iteration started at distance 10^{-6} from the checkpoint gives the same median ceiling as the linearised one to the printed digits in 23 of 24 cells and 1.0053× it in the remaining one ($P = 3\,610$, $g = 32$).

The tether sweep (`t5`, `t6`). The fraction $f^\star = \eta^\star(\lambda_1 + \mu)/(2(1 + \beta))$ of the deterministic threshold rises with μ on both networks: from 0.891 to 0.9998 ($P = 33$, $\beta = 0$), from 0.171 to 0.9965 ($P = 33$, $\beta = 0.9$) and from 0.4065 to 0.835 ($P = 3\,610$). $\eta^\star$ itself has an interior maximum, at $g = 8$ ($P = 33$, $\beta = 0.9$, $\eta^\star = 0.1941$) and at $g = 32$ ($P = 3\,610$, $\eta^\star = 0.01788$). The flat-mode collapse of Theorem 8 was predicted at $P = 3\,610$ between $g = \frac{1}{32}$ and $\frac{1}{128}$ as a log–log slope of $\eta^\star$ against g in $[0.4, 1.1]$; $\eta^\star$ goes from 0.0156894 to 0.0156408, a slope of 0.0022, and `t5` fails. On this grid the critical eigenmatrix lies mostly on the 175 rigid modes (eigenvalues above $2|\lambda_{\min}|$) and on the positive near-flat block (64 % and 27 % of its weight at $g = \frac{1}{32}$, 6 % on the negative directions), and a diagonal upper bound computed after the run places the collapse below $g \approx 6 \times 10^{-4}$, outside the grid. At the smallest tether the threshold is 1.23× below the one-dimensional ceiling (0.019278) at $P = 3\,610$ and 2.11× below it at $P = 33$, $\beta = 0.9$. In Phase 2 the untethered sample-path ceiling at $P = 3\,610$ was 0.002707, 7.1× below that ceiling; with the 1 528 negative directions lifted by $\mu \geq 0.0833$ it is 0.0156. On this checkpoint, then, most of the gap of Phase 2 comes from the negative directions of H, which a mean-square analysis of a fixed point does not cover, and not from curvature noise on positive flat modes.

4.4 Does training saturate the threshold? (Phases 5 and 5b)

Theorem 9 is a statement about a fixed point. Phases 5 and 5b evaluate it at the iterates of a training run, the first of the questions left open in Section 3. The $P = 33$ network is trained from a fresh initialisation ($W_1 \sim N(0, 1/d_{\mathrm{in}})$, $W_2 \sim N(0, 1/d_{\mathrm{hidden}})$, zero biases) by heavy-ball SGD $\theta_{t+1} = \theta_t + \beta(\theta_t - \theta_{t-1}) - \eta(g_{B_t}(\theta_t) + \mu_{\mathrm{train}}\theta_t)$ with $\beta = 0.9$, $B = 16$ drawn with replacement, 20 000 steps, $\eta \in \{0.01, 0.03, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5\}$ and weight decay $\mu_{\mathrm{train}} \in \{0, 10^{-3}\}$, three seeds per cell, 48 runs. A run is divergent if its full loss becomes non-finite or exceeds 10^3; none did. Every 200 steps the Hessian H of the training objective and the 64 per-sample Hessians are computed at θ_t, and Theorem 9 is evaluated with mean curvature $|H|$ (the eigenvalues of H replaced by their absolute values) and noise $\Delta H_i = H_i - H$; we write $\eta^\star_{|H|}(\theta_t)$. H is indefinite at most checkpoints of the runs with $\eta \geq 0.15$, so $|H|$ is a choice of the diagnostic, not the mean curvature of the iteration. The diagnostic is quasi-static: given θ_t the minibatch Hessian has the law of Theorem 9,

Table 3: Phase 4 (`raw-phase4.json`; $B = 16$). g is the smallest eigenvalue of $H + \mu I$; 1D is the one-dimensional ceiling of Proposition 1 of the tethered iteration; $\eta^\star$ is Theorem 9(3) by Lanczos; $f^\star = \eta^\star/\eta^{(\mu)}_{\mathrm{crit}}(\beta)$; lin. and net. are the median sample-path ceilings of the linearised and of the network iteration over $\eta^\star$.

P	β	g	$\eta^{(\mu)}_{\mathrm{crit}}(\beta)$	1D	$\eta^\star$	$f^\star$	lin./$\eta^\star$	net./$\eta^\star$
33	0	128	0.0149509	0.0149484	0.014948331	0.9998	1.001	1.001
33	0	2	0.257374	0.244788	0.24286101	0.9436	1.107	1.107
33	0	$\frac{1}{128}$	0.346103	0.316663	0.30841041	0.8911	1.124	1.124
33	0.9	128	0.0284068	0.0283135	0.028307600	0.9965	1.003	1.003
33	0.9	8	0.275946	0.210477	0.19406782	0.7033	1.104	1.104
33	0.9	2	0.489010	0.247371	0.18081066	0.3697	1.190	1.190
33	0.9	$\frac{1}{128}$	0.657597	0.237703	0.11242063	0.1710	1.183	1.183
3 610	0.9	128	0.0167578	0.0140951	0.013993390	0.8350	1.081	1.081
3 610	0.9	32	0.0290608	0.0185325	0.017880637	0.6153	1.196	1.202
3 610	0.9	2	0.0377132	0.0192735	0.016781922	0.4450	1.257	1.257
3 610	0.9	$\frac{1}{128}$	0.0384739	0.0192781	0.015640812	0.4065	1.204	1.204

but the drift of θ_t is outside the theorem. Per run the statistic is the median $m_\star$ of $\eta^\star_{|H|}(\theta_t)$ over the 51 checkpoints of steps 10 000–20 000, and $r = m_\star/\eta$; $r < 1$ means that the run trains above the mean-square threshold of its own iterates.

Phase 5: registered, then read as exploratory. Phase 5 (seeds 1–3) registered five hypotheses and six predictions before any threshold was computed: H1, that $\eta^\star_{|H|}(\theta_t)$ descends to η and stays there ($r \in [0.8, 1.25]$ in every run whose λ_1 has fallen below 0.8 times that of the control), against H2, that $r < 0.5$ throughout, and H3, that $\eta^\star = 0$ because $H + \mu_{\mathrm{train}} I$ is indefinite, with H4 and H5 on the mean-square and almost-sure thresholds at the final point. The registration was committed before any threshold was computed; a pre-commitment file with the hashes of the registration, the evaluator, the runner, the code package and the job list was written at 23:25:29 UTC on 28 October 2026, after the 48 runs had been launched (23:15:35) and when 22 of them had finished, and the author declares that no run output had been read at that time. As computed, neither H1 nor H2 holds at either μ_{train}; `p4` holds and `p1`, `p2`, `p3`, `p5`, `p6` fail. An implementation check failed as well: at $(1 \mp 10^{-3})\eta^\star_{|H|}(\theta_T)$ the Collatz–Wielandt bracket of the second-moment operator leaves at least one side undecided in 12 of 48 runs (21 of 96 sides). The diagnosis, written after the runs, is a resolution limit: on the 9 undecided lower sides at $\mu_{\mathrm{train}} = 0$, near-flat modes put the Perron eigenvalue of the operator 1.46×10^{-9}–9.38×10^{-6} below 1 (by inverse iteration), and its eigenmatrix is too ill-conditioned for a double-precision test to separate the radius from 1; on no side does a certificate contradict the side. The registration stops the reading of the predictions at a failed check, and the author decided not to read the Phase 5 predictions as outcomes. Phase 5 is reported as run and is used as the exploratory run of Phase 5b.

Phase 5b: a within-dataset replication. Phase 5b repeats Phase 5 without any change to the network, the data, the training, the grid or the criteria, on three new seeds (1401–1403), and tests six predictions `q1`–`q6` written after reading Phase 5 and restricted to the cells where its three

seeds agreed. The initial point and the minibatch stream depend on the seed only, so Phase 5b tests whether the structure of Phase 5 reproduces for new $(\theta_0, \{B_t\})$ streams on the same data and teacher; it is not a generalisation across datasets. A threshold read by a prediction with Lanczos relative residual above 10^{-10}, a failed check of the almost-sure threshold or of the batch-sharpness identity, or a contradicting certificate suspends the predictions that read it, and Phase 5b counts as a replication only if all six hold and none is suspended. The registration, the evaluator and the hashes of the code package and job list were committed at 02:22 UTC on 29 October, before the first run started (02:33). All 48 runs finished, no check suspends a prediction, and the six hold (Table 4). The certificate check at the final point (`c5b`, the reading of Phase 5 that accepts either certificate) fails on one run ($\mu_{\text{train}} = 0$, $\eta = 0.4$, seed 1402: radii $1 - 6.3 \times 10^{-7}$ and $1 - 5.5 \times 10^{-7}$ at the two sides after 200 000 power steps), the resolution limit of Phase 5; the other 94 sides are decided, and no prediction reads that check. By the power-iteration check `c5` of Phase 5, 12 runs (21 sides) are undecided.

Table 4: Phase 5b (`raw-phase5b.json`, field `phase5b`; seeds 1401–1403, $\beta = 0.9$, $B = 16$). Each prediction was written before the run and holds or fails as stated, with no tolerance; the last column gives the observed value nearest to each bound. $m_\star$ and r as in the text; n_{sat} counts hidden units with $s_h(\theta_t) < 10^{-4}$; e_{GN} and g_{GN} are the window medians of $|\eta^\star_{\text{GN}}/\eta^\star_{|H|} - 1|$ and of $\eta^\star_{\text{GN}}/\eta^\star_{|H|}$, with the Gauss–Newton per-sample Hessians in place of the full ones; b is the largest window median of batch sharpness (expectation of the ratio and ratio of expectations) and of λ_1, each times $\eta/(2(1+\beta))$; f_- is the fraction of the window with $\lambda_{\min}(H) < 0$.

	prediction (domain)	outcome	observed
`q1`	$\mu_{\text{train}} = 10^{-3}$, every η: $m_\star \in [0.07, 0.18]$, $\max_\eta m_\star / \min_\eta m_\star \le 2.5$ per seed	holds	$m_\star$ 0.088–0.154; ratio ≤ 1.69
`q2`	$\mu_{\text{train}} = 10^{-3}$: $r(0.1) > 1.15$, $r(0.15) \in [0.80, 1.15]$, $r(0.3) < 0.45$, r decreasing over 0.15–0.4	holds	1.285; 0.840–0.963; ≤ 0.378
`q3`	$\mu_{\text{train}} = 0$, $\eta = 0.5$: median $n_{\text{sat}} \ge 1$, $r \in [1.2, 1.5]$; $n_{\text{sat}} = 0$ at $\eta \le 0.2$ and median 0 at 10^{-3}	holds	$n_{\text{sat}} = 4$, r 1.356–1.360
`q4`	$e_{\text{GN}} < 0.03$ at $\eta \le 0.03$, < 0.12 at $\eta = 0.1$; $g_{\text{GN}} \ge 1.5$ at 10^{-3}, $\eta \ge 0.3$	holds	0.0235; 0.0918; ≥ 1.670
`q5`	$\eta \ge 0.15$: $b < 0.38$; $\eta \le 0.3$: $b < 0.25$	holds	≤ 0.353; ≤ 0.193
`q6`	$\eta \ge 0.15$: $f_- \ge 0.85$	holds	$\ge 46/51 = 0.902$

What the replication measures. With weight decay 10^{-3}, the threshold at the iterates does not follow the learning rate: $m_\star$ stays between 0.088 and 0.154 over a fifty-fold range of η, while the window median of λ_1 falls from 5.82–5.84 at $\eta = 0.01$ to 2.36–2.51 at $\eta = 0.5$. The runs cross it between $\eta = 0.1$ ($r \approx 1.29$) and $\eta = 0.15$ ($r = 0.84$–0.96) and above it train at 2.6–2.9× ($\eta = 0.3$), 4.0–4.5× ($\eta = 0.4$) and 4.1–5.1× ($\eta = 0.5$) the $|H|$ reading of their own iterates without diverging, so H1 does not describe these runs: the linear mean-square criterion does not act as a cap that training saturates, and what bounds the iterates beyond it is outside the linearisation. Not diverging is not training well: at $\mu_{\text{train}} = 10^{-3}$ the window median of the data loss is 0.0025–0.0028 at $\eta \le 0.03$, 0.05 at $\eta = 0.2$, 0.23–0.37 at $\eta = 0.3$ and 1.55–1.71 at $\eta = 0.5$. Without weight decay, at $\eta = 0.5$ four of the hidden units saturate and the threshold returns above the learning rate ($r = 1.36$). The Gauss–Newton form of the criterion agrees with the full one to 2.4 % at small learning rates and overestimates it by at least 1.67× in the regime above the threshold, which is where a Gauss–Newton estimate of a safe step would be used. The one-dimensional quantities stay off their deterministic edge ($b \le 0.353$), so in these runs neither batch sharpness nor λ_1 locates the

boundary that the operator of Theorem 9 gives, and the Hessian has a negative eigenvalue at 90–100 % of the window checkpoints once $\eta \geq 0.15$. Three bounds held by small margins: $r(0.15)$ by 0.040, e_{GN} by 0.0065 and b by 0.027. These are measurements on one network with 33 parameters, one dataset of 64 samples and three new seeds; whether the same structure holds on other data and at larger P is not tested here.

4.5 A certified bracket of the Gauss–Newton threshold (Phase 6)

The object is the heavy-ball iteration whose per-sample curvature is the Gauss–Newton matrix $J_i^\top J_i$ plus the tether $\mu_{\mathrm{train}} I$, at a frozen iterate θ_t, restricted to the row space of the stacked Jacobian ($R = N d_{\mathrm{out}} = 640$ dimensions at $P = 3\,610$ and $12\,826$). Each $J_i^\top J_i$ leaves that row space invariant and vanishes on its complement, where at $\mu_{\mathrm{train}} = 10^{-3}$ the iteration is a noiseless scalar heavy ball with threshold $2(1+\beta)/\mu_{\mathrm{train}} = 3\,800$; the row-space threshold is therefore the threshold $\eta^\star_{\mathrm{GN}}$ of the whole Gauss–Newton-plus-tether iteration. It is a surrogate: at the same checkpoints the Hessian of the loss is indefinite and the second moment of the exact linearisation grows (measured in the pilot runs below, not reported here), and nothing below changes that.

The bracket. Both sides come from Theorem 9, $\eta^\star = 1/\lambda_{\max}(G)$ with $G = L_0^{-1/2} B_\beta L_0^{-1/2}$, and are computed in double precision with every rounding error bounded [Higham, 2002, Chapter 3]. *Upper side:* any unit vector y gives $\langle y, Gy\rangle \leq \lambda_{\max}(G)$, so the Rayleigh quotient q of the top Ritz vector of a short Lanczos run on G (with full reorthogonalisation; 30 steps, then 60 and 120 when the Ritz residual or the bracket is wide), minus a forward error bound δ_q, gives $\eta_{\mathrm{up}} = 1/(q - \delta_q) \geq \eta^\star$; no convergence is needed. *Lower side:* the operator $K(\eta)$ of Theorem 9 preserves the cone $\mathbb{S}^R_+$, so a witness $X \succ 0$ with $X - K(\eta)(X) \succ 0$ gives $\rho(K(\eta)) < 1$ [Vandergraft, 1968, Theorem 4.2] and hence $\eta < \eta^\star$. The witnesses are the positive part of $L_0^{-1/2}(y)$ plus εI for $\varepsilon = 10^{-1}, \ldots, 10^{-9}$; positive definiteness of $X - K(\eta)(X)$ in exact arithmetic is proved by a floating-point Cholesky factorisation of the computed matrix shifted by its forming error and by Rump's bound [Rump, 2006], and η_{low} is the largest rate that passes in a 50-step bisection below η_{up}. A rate at which every witness fails gives no lower bound, never a number. Checkpoints are regenerated by rerunning training, and the stored loss trace is compared bit for bit before any bracket is computed.

Stage 1: $P = 33$, against the dense exact value. On every window checkpoint of the 48 Phase 5b runs (Gauss–Newton at $\mu_{\mathrm{train}} = 10^{-3}$, at $\mu_{\mathrm{train}} = 0$ the same object after removing the modes with $|\lambda + \mu_{\mathrm{train}}| < 10^{-8}\lambda_{\mathrm{top}}$, a numerical kernel, and $|H|$; 9 600 brackets over all 4 800 checkpoints, 2 448 of them in the window), the regenerated runs match the stored loss traces bit for bit, no bracket excludes the dense exact threshold, and every bracket has $\eta_{\mathrm{low}}/\eta_{\mathrm{up}} \geq 0.98$ (smallest 0.9934 for Gauss–Newton, 0.980 for $|H|$); at $\mu_{\mathrm{train}} = 0$, 94 of the stored Lanczos values of Phase 5b lie outside the bracket, all where that Lanczos had not converged. The window medians of $\eta^\star_{\mathrm{GN}}/\eta$ at $\mu_{\mathrm{train}} = 10^{-3}$ fall through 1 as η grows, per seed 11.9–12.0 at $\eta = 0.01$, 1.29–1.30 at 0.1, 0.99–1.44 at 0.15, 0.85–0.92 at 0.2, 0.68–0.86 at 0.3 and 0.48–0.76 at 0.4–0.5: on this network the Gauss–Newton threshold does not cap training at the largest rates.

Stage 2: $P = 3\,610$ and $12\,826$. The checkpoints come from exploratory pilot runs of the companion lab: the two networks of Section 4.2 trained as in Section 4.4 (heavy-ball, $\beta = 0.9$, $B = 16$ with replacement, 20 000 steps) on datasets drawn with pilot seeds, with tanh, softsign $z/(1+|z|)$ and leaky ReLU of slope 0.1 hidden units, which recorded at each window checkpoint the threshold $\eta^\star_{\mathrm{GN},64}$ of the Gauss–Newton iteration restricted to its top 64 modes, an upper bound

of $\eta^\star_{\mathrm{GN}}$ because a restriction to a subspace can only raise the threshold. Seed 0 of every pilot cell with $\mu_{\mathrm{train}} = 10^{-3}$ and a finite run (30 runs; η from 0.001 to 0.1 at $P = 3\,610$ and to 0.05 at $P = 12\,826$, leaky ReLU only up to 0.01), at the 11 window checkpoints $10\,000, 11\,000, \ldots, 20\,000$. The checkpoint list was committed before the stage 2 batch was run; the reference is the same certified bracket with Lanczos run to a target Ritz residual of 10^{-10} (reached at 232 of the 330 checkpoints within 200 steps; largest residual 1.6×10^{-6}). All 30 regenerations matched. At all 330 checkpoints the bracket and the reference overlap with no tolerance, the two upper bounds agree to 7×10^{-6} relative, and $\eta_{\mathrm{low}}/\eta_{\mathrm{up}} \geq 0.982$ (median 0.9994); the row space has full rank 640 at every checkpoint. The top-64 compression of the pilot overstated the threshold by 1.7–8.9 % ($\eta_{\mathrm{up}}/\eta^\star_{\mathrm{GN},64} = 0.919$–$0.983$).

Reading each checkpoint against the band $[0.8\,\eta, 1.25\,\eta]$ of the Phase 5 registration (*inside* when $0.8\,\eta \leq \eta_{\mathrm{low}}$ and $\eta_{\mathrm{up}} \leq 1.25\,\eta$, *below* when $\eta_{\mathrm{up}} < 0.8\,\eta$, *above* when $\eta_{\mathrm{low}} > 1.25\,\eta$): the 12 runs at $\eta \leq 0.003$ read above at every checkpoint (η_{up}/η between 1.6 and 27), as expected near a minimum. Of the 18 runs at $\eta \geq 0.01$, 16 read inside at most of their checkpoints, with window medians of $\eta^\star_{\mathrm{GN}}/\eta$ between 0.81 and 1.10; the other two, tanh and softsign at $P = 3\,610$, $\eta = 0.01$, read above (1.46 and 1.81). Over these 198 checkpoints the readings are 161 inside, 23 above and 14 below, and no run reads below at more than 4 of its 11 checkpoints; 13 of the 30 runs read inside at 80 % or more of their checkpoints. Unlike $P = 33$, where the ratio keeps falling to 0.48–0.76, on these two networks the window median of $\eta^\star_{\mathrm{GN}}/\eta$ stays within the band (0.81–1.10) at every finite rate from $\eta = 0.02$ upwards, for both bounded activations; single checkpoints go down to 0.46.

What Phase 6 answers. Three answers that the earlier phases did not give, and that we have not found in the cited literature. (i) The exact mean-square threshold of heavy-ball SGD with per-sample curvatures that do not commute can be bracketed from both sides, with every rounding error bounded for the computed Gauss–Newton factors, at training checkpoints of networks with thousands of parameters and without forming a Hessian: at $R = 640$ the operator G acts on a space of dimension 205 120, and all 330 brackets have relative width below 1.8 % (median 0.06 %). (ii) For the Gauss–Newton surrogate, the saturation half of the open question of Andreyev et al. [2026] has a measured, exploratory answer that depends on the network (Theorem 14 gives a model in which both outcomes occur). At $P = 33$ training goes past the threshold at the largest rates (window medians 0.48–0.76); at $P = 3\,610$ and 12 826, for both bounded activations and every finite rate from $\eta = 0.02$, the window median of $\eta^\star_{\mathrm{GN}}/\eta$ lies in $[0.81, 1.10]$, so training runs at the certified threshold of the surrogate, a quantity the pilot bounded only from above, by $\eta^\star_{\mathrm{GN},64}$. At the smallest rates ($\eta \leq 0.003$) the threshold sits 1.6–27 times above η: saturation is reached only once η is large enough. (iii) At training checkpoints the top-64 compression overstates the threshold by 1.7–8.9 %, a gap that had not been measured. None of the three is a statement about the loss linearisation, whose second moment grows at the same checkpoints.

Status. The bracket is a measurement of the surrogate at a checkpoint, certified for the stored factors and $\mathrm{fl}(\lambda_k + \mu_{\mathrm{train}})$; the distance from those factors to the Gauss–Newton operator is reported beside it (relative residuals below 9×10^{-10} and 1.4×10^{-14}). Stages 1 and 2 are exploratory and read only pilot checkpoints and the Phase 5b runs, one seed per cell at the larger sizes; the expectations written before stage 2 are not registered predictions, and they are reported as they came out: the top-64 compression within 5 % ($\eta_{\mathrm{up}}/\eta^\star_{\mathrm{GN},64} \geq 0.95$) at 90 % of the checkpoints holds narrowly (90.6 %); every run with $\eta \geq 0.01$ inside at most of its checkpoints fails, on the two $P = 3\,610$ runs at $\eta = 0.01$; no run below at most of its checkpoints holds. Stage 2 passes the validity and adoption rules of the plan (every bracket overlaps its reference with no tolerance, $\eta_{\mathrm{low}}/\eta_{\mathrm{up}} \geq 0.98$ at every

checkpoint), so the larger network ($P = 34\,186$) may be bracketed next. The band question is left to a registration on unseen seeds and datasets. A larger network ($P = 34\,186$, $R = 2\,560$) has not been bracketed.

5 Scope and limitations

All statements are about the linearised dynamics around a fixed point of a quadratic model: i.i.d. minibatch Hessians with mean H, additive noise independent of the past, and stationarity. The criterion is a mean-square one; between the mean-square and the almost-sure thresholds of Proposition 7 the second moment diverges while typical trajectories decay. Not answered here: the dynamics beyond the quadratic model, which is what bounds the training runs of Section 4.4 above the criterion, and a threshold on the batch sharpness of Definition 3.1 of Andreyev et al. [2026] alone: Theorem 13 gives an identity at the threshold and a sufficient criterion that also uses the noise share θ_{noise}, and Theorem 14 proves saturation in a slow-fast model, not for the training dynamics. On the two networks of Section 4.2 the random-feature model of Theorem 5 does not hold (`q3` fails by 1.8× and 63×), and the untethered network iteration has no mean-square stable learning rate, because the mean Hessian is indefinite at both checkpoints; its growth within 1 000 steps changes sign between 0.01× and 0.02× the deterministic threshold at $P = 3\,610$ and below 0.002× at $P = 12\,826$. Theorem 9 was evaluated on network Hessians only with a tether that lifts the negative eigenvalues of the mean Hessian (Section 4.3); at $P = 33$ the threshold is enclosed in all 16 cells by the certified bracket of Section 4.5 (a post-hoc run), for the computed per-sample Hessians; at $P = 3\,610$ the crossing rests on power iteration alone. Section 4.4 evaluates it at training iterates with $|H|$ in place of an indefinite H, as a quasi-static diagnostic of a moving point, on one network with 33 parameters and one dataset of 64 samples; it measures a replication on new seeds, not a generalisation across data or size. Section 4.5 brackets the threshold of the Gauss–Newton surrogate, not of the loss linearisation, whose second moment grows at the same checkpoints; it is exploratory, with one seed per cell at the larger sizes.

6 Related work

Cohen et al. [2021] described the edge of stability for full-batch training. Andreyev and Beneventano [2024] described its stochastic counterpart, in which the batch sharpness rather than the full-batch sharpness stabilizes, and Andreyev et al. [2026] extended it to momentum; their one-dimensional mean-square threshold is Proposition 1. The linear-stability account of Wu et al. [2018] contains its $\beta = 0$ case; see also the break-even analysis of Jastrzebski et al. [2020]. The exact second-moment equation of Section 3 sits in the line of stability analyses of SGD with momentum on quadratics: Kidambi et al. [2018] and Liu and Belkin [2020] with component-decoupled data, Ma et al. [2018] for the batch-size scaling of SGD without momentum, Mulayoff and Michaeli [2024] for the exact mean-square stability threshold of SGD without momentum on quadratic losses, Paquette and Paquette [2021] (batch size 1, through homogenized SGD) and Lee et al. [2022] (mini-batches) in high-dimensional limits through Volterra equations, with the exact high-dimensional dynamics of SGD on least squares by homogenization [Paquette et al., 2022], and the power-law random features model analysed for one-pass SGD [Paquette et al., 2024] and for stochastic momentum at small batch [Ferbach et al., 2025], where SGD with fixed momentum has the scaling-law exponents of SGD and dimension-adapted Nesterov acceleration improves them, and Gitman et al. [2019] for the stationary covariance with additive noise, and Yuan et al. [2016] for the equivalence of momentum SGD with SGD at step $\eta/(1-\beta)$ for small constant steps. At $\beta = 0$, $B = 1$ and Gaussian features, Theorem 5 is the mean-square condition of the LMS filter with independent Gaussian inputs [Horowitz and Senne, 1981, Feuer and Weinstein, 1985], whose second-moment recursion has the same diagonal closure and rank-one secular structure; the Kronecker form of that recursion is standard in adaptive filtering [Sayed, 2008]. For momentum LMS with independent Gaussian

inputs, which is Theorem 5 at $B = 1$ and $\kappa_{\text{feat}} = 0$ with $\eta = 2\mu$ and $\beta = \alpha$ in their notation, Roy and Shynk [1990] close the diagonal second-moment recursion (their (A6a)) and obtain from one Schur–Cohn test, $D(1) > 0$, a necessary condition for mean-square convergence, $Q(1) > 0$ (their (31)–(33)); evaluated from their (A10), $Q(1) > 0$ reads $\sum_k \mu\lambda_k(1+\alpha)/(1-\alpha^2-2\mu\lambda_k) < 1$, which is (4) in that case (their printed stability condition 3 omits the factor $1+\alpha$, which their misadjustment (36) carries). Theorem 5 shows that for $\alpha \in [0,1)$ and $2\mu\lambda_k < 1-\alpha^2$ this condition is also sufficient, and extends it to minibatches and to non-Gaussian independent features. Tugay and Tanik [1989] also analysed momentum LMS, and Sharma et al. [1998] give small-step asymptotics (an ODE limit and the asymptotic distribution) for every $\alpha \in (-1,1)$. Jin et al. [2026] prove L_p exponential stability and tracking and regret bounds for a normalized momentum LMS with dependent, nonstationary regressors, under a sufficient bound on the step size and a momentum $\beta \le C_\beta\mu$ that vanishes with the step; they do not give a threshold. Theorem 5 has the form of the convergence condition of Velikanov et al. [2023], made exact for independent-coordinate features (with their parameter τ outside the range $0 < \tau \le 1$ of their convergence analysis for $\kappa_{\text{feat}} \ge -1$, Gaussian features included, and their infinite-dimensional H replaced by a finite P), and is a finite-dimensional counterpart, with a kurtosis term, of the kernel-norm condition that Lee et al. [2022] state in a high-dimensional limit. Theorem 2 extends the exact mean-square analysis of SGD by Mulayoff and Michaeli [2024], whose form it takes at $\beta = 0$ with sampling with replacement, to heavy-ball momentum with non-commuting curvature noise; the threshold of Theorem 9 is also Theorem F.1 of Cai et al. [2026], obtained concurrently and independently; Mulayoff and Stich [2026] show that beyond linearization a single unstable batch can make SGD diverge in expectation. Theorem 2 gives the stability boundary of the Schur-complement expansion of Andreyev et al. [2026, Appendix D] exactly, and Corollary 4 gives, for $\rho(\mathcal{T}) \ge 1$ and $\Sigma_\xi \succ 0$, linear growth of the second moment; its probabilistic conclusion has the form of their Definition A.1 but is also met below $\rho(\mathcal{T}) = 1$, so it does not settle the valid instability criterion they list as open for SGD with momentum. The secular structure of Theorem 5 (cone-preserving operator, rank-one coupling, Krein–Rutman) is also that of Song et al. [2026] for full-batch zeroth-order methods, whose randomness is the search direction rather than the minibatch; Sharma et al. [2026] give the exact mean-square criterion of outer Nesterov momentum on stochastic pseudo-gradients, which at one local step covers SGD with Nesterov momentum.

7 Conclusion

For heavy-ball SGD on a quadratic with i.i.d. minibatch Hessians, the mean-square stability boundary is decided at every $\eta > 0$ and $\beta \in [0,1)$ by the stationary equation of Theorem 2, without a small-η, noise-dominated or commutation assumption, and beyond it the second moment diverges (Corollary 4). For random-feature minibatches the boundary is the root of the secular equation of Theorem 5, whose two limits take the values of the two batch-sharpness plateaus, through λ_1 and $\operatorname{tr} H/B$ (Corollary 6); for arbitrary per-sample Hessians it is the reciprocal of the top eigenvalue of the operator of Theorem 9; and when the curvature noise is not proportional to the curvature, a nearly flat mode with curvature noise lowers it without bound (Theorem 8). The secular root matches the spectral radius of the exact second-moment operator within 6.11×10^{-10} relative in 36 cells and the sample-path ceiling in 29 of 30 cells; on the two network checkpoints the random-feature model does not hold, and the untethered linearised iteration, with an indefinite mean Hessian, has no mean-square stable learning rate. With a tether that lifts the negative eigenvalues of the mean Hessian, the threshold of Theorem 9 computed from the real per-sample Hessians is where the spectral radius of the exact operator crosses 1, within a relative $\pm 10^{-3}$ in η, and it lies

1.23× below the one-dimensional ceiling at $P = 3\,610$. At the iterates of heavy-ball training of a 33-parameter network with weight decay the threshold does not follow the learning rate; the runs cross its $|H|$ reading between $\eta = 0.1$ and 0.15 and train up to 5.1× above it without diverging, so on that network the linear mean-square criterion is not a cap that training saturates. At the threshold an exact identity ties the criterion to the two batch-sharpness plateaus, and in a quasi-stationary slow-fast model saturation occurs at a distance $\Theta(\sigma_\xi^2/v_{\mathrm{PS}})$ below the threshold, while a secant equilibrium lies above the tangent one (Theorems 13 and 14). The threshold of Theorem 9 can be bracketed from both sides with every rounding error bounded for the computed factors and without forming a Hessian: on the Gauss–Newton surrogate of networks with 3 610 and 12 826 parameters (205 120-dimensional operator) all 330 brackets have relative width below 1.8 %, and there, unlike at $P = 33$, the window medians of the threshold stay at 0.81–1.10 times the learning rate from $\eta = 0.02$ upwards.

Open questions: (i) whether the structure of Section 4.4 holds on other data and at larger P, and what bounds the iterates above the criterion; (ii) the dynamics beyond the quadratic model, where the loss is bounded below and the mean gradient is not zero, which is where the excursions of training at practical learning rates come from (Section 4.2); (iii) a threshold on the batch sharpness of Andreyev et al. [2026, Definition 3.1] alone, and saturation for the training dynamics rather than for a slow-fast model; (iv) the almost-sure threshold for $\beta \in (0, 1)$ with unbounded curvature noise, where $\det M_t = \beta$ removes the singular multiplier of $\beta = 0$ and the proof of Proposition 7 lacks only the return of its projective coordinate after large excursions (heuristically, the steps back after a near-singular multiplier v accumulate $\log v$, as the singular integral does at $\beta = 0$); (v) a measurement of the mean-square boundary on networks with an estimator of $\mathbb{E}\|w_t\|^2$, such as the resampling estimator of Section 4.1, instead of sample paths, whose threshold can lie above the mean-square one where the multiplicative noise along u_1 dominates (Section 4.1). Adaptive (state-dependent) preconditioners and orthogonalised updates are not covered; a fixed preconditioner reduces to the theorems (Section 3, after Theorem 9).

Reproducibility

Code, raw files and tests are in the repository that accompanies this paper, https://github.com/apis-os/hb-sgd-mean-square, lab `labs/mean-square-boundary`, whose `README.md` maps the theorem letters used in the code to the numbering of this paper. Phase 1: `solve-secular-eoss-threshold.ts`, `simulate-heavy-ball-feature-noise-energy.ts`, `measure-empirical-eoss-ceiling.ts`, `compute-symmetric-eigenvalues-jacobi.ts`, `evaluate-phase1-rademacher-cells.ts`, `run-phase1.ts` (8 seeds per cell, Node 26.5, 43 min on one thread), and `run-phase1-cap-diagnostic.ts` for the post-hoc diagnostic of its failed cell. Phase 2: `compute-exact-hessian-trace.ts`, `compute-top-eigenpair-power-iteration.ts`, `compute-secular-ceiling-bracket-from-trace.ts`, `simulate-network-minibatch-heavy-ball-energy.ts`, `measure-network-minibatch-heavy-ball-ceiling.ts`, `evaluate-phase2-trace-ceiling-cells.ts`, `run-phase2.ts`. Phase 3: `run-phase3.ts`. Phase 4: `run-phase4.ts`, `evaluate-phase4-tethered-mean-square-cells.ts`, `create-phase4-sample-hessian-provider.ts`, and `run-phase4-certified-bracket-p33.ts` with its output `phase4-certified-bracket-p33.json` for the certified brackets at $P = 33$. Phase 5: `run-phase5.ts` (one run per process, 48 runs, one CPU each), `run-phase5-aggregate.ts`, `run-phase5-c5-perron.ts` for the certificate diagnosis; Phase 5b: `evaluate-phase5b-predictions.ts`, run records in `evidence/mean-square-boundary/phase5b-runs/` and the job list `phase5b-jobs.json`; the registration texts, the pre-commitments and package manifests of Phases 5 and 5b and their

commits are in the companion lab, not in this repository. Phase 6: plan `phase6-plan-certified-mean-square-bracket.md`, `compute-certified-rayleigh-upper-threshold.ts`, `certify-cone-collatz-wielandt-at-rate.ts`, `verify-positive-definite-shifted-cholesky.ts`, `compute-adaptive-certified-threshold-bracket.ts`, `run-phase6-stage1.ts`, `run-phase6-stage2.ts`, `summarize-phase6-stage1.ts`, `summarize-phase6-stage2.ts`, the checkpoint list `phase6-checkpoints.json` (committed before the runs), raw records in `phase6-stage1/` and `phase6-stage2/`, the pilot records `phase6-pilot/` that stage 2 regenerates, and the summaries `phase6-stage1-summary.json` and `phase6-stage2-summary.json`. The Phase 4 row workers use a WebAssembly `f64x2` product kernel shipped as `dense-f64-matmul.wasm` and checked bit for bit against the sequential product; the increment-stream block of the Phase 5 checkpoint records belongs to the companion manuscript and is not computed in the repository. Tests `tests/phase1-secular-ceiling.test.ts`, `tests/phase2-trace-ceiling.test.ts`, `tests/phase3-operator-radius.test.ts`, `tests/phase4-tethered-thm-g6.test.ts`, `tests/phase5b-evaluator.test.ts`, `tests/phase6-certified-bracket.test.ts`, `tests/phase6-stage2-summary.test.ts` and `tests/phase4-certified-bracket-p33.test.ts` re-execute reduced versions of Phases 1–3 against `raw-phase1.json`, `raw-phase2.json` and `raw-phase3.json`, including the failed predictions; check the Phase 4 kernels and a resumed row, the Phase 5b evaluator on the Phase 5b run records, the regeneration of Phase 6 checkpoints from those runs and from the pilot records, the Phase 6 stage 2 summary, and the certified brackets at $P = 33$ against `raw-phase4.json`; `tests/phase4-5-raw-files.test.ts` recomputes the predictions and checks stored in `raw-phase4.json` from its cell records and the decisions stored in `raw-phase5.json` and `raw-phase5b.json` from the 96 run records. The identities and limits of Theorem 13 and the saturation law of Theorem 14 are checked in `paper/checks/bridge-theorems.py`. The numerical checks cited in the text are in `paper/checks/`, `paper/check-schur-factor.py` and the lab's `check-phase3-convergence.ts`.

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A Proofs of the results of Section 3

Proof of Theorem 2. Let $A = (1+\beta)I - \eta H$ and $\Sigma_1 = \mathbb{E}[w_{t+1}w_t^\top]$. (1) Since $(H_{B_t} - H, \xi_t)$ is independent of $\mathcal{F}_{t-1}$ with zero mean, $\mathbb{E}[w_{t+1} \mid \mathcal{F}_{t-1}] = Aw_t - \beta w_{t-1}$; multiplying on the right by $w_t^\top$, $\Sigma_1 = A\Sigma_0 - \beta\Sigma_1^\top$, and transposing ($A$ symmetric), $\Sigma_1^\top = \Sigma_0 A - \beta\Sigma_1$. Hence $\Sigma_1 = \frac{A\Sigma_0 - \beta\Sigma_0 A}{1-\beta^2}$, $\Sigma_1 + \Sigma_1^\top = \frac{A\Sigma_0 + \Sigma_0 A}{1+\beta} = 2\Sigma_0 - \frac{\eta}{1+\beta}(H\Sigma_0 + \Sigma_0 H)$, and $D_\Theta = 2\Sigma_0 - \Sigma_1 - \Sigma_1^\top$. (2) Expand $\Sigma_0 = \mathbb{E}[w_{t+1}w_{t+1}^\top]$ with $w_{t+1} = Aw_t - \beta w_{t-1} - \eta(H_{B_t} - H)w_t - \eta\xi_t$; the cross terms vanish because $(H_{B_t} - H, \xi_t)$ is independent of (w_t, w_{t-1}) and $\mathbb{E}[w_t] = 0$ (this covers H_{B_t} and ξ_t from the same minibatch), leaving $\Sigma_0 = A\Sigma_0 A + \beta^2\Sigma_0 - \beta(A\Sigma_1 + \Sigma_1^\top A) + \eta^2\mathcal{V}(\Sigma_0) + \eta^2\Sigma_\xi$. With Σ_1 from (1) and $A^2\Sigma_0 + \Sigma_0 A^2 - 2\beta A\Sigma_0 A = [A,[A,\Sigma_0]] + 2(1-\beta)A\Sigma_0 A$, $-\beta(A\Sigma_1 + \Sigma_1^\top A) = -\frac{\beta}{1-\beta^2}[A,[A,\Sigma_0]] - \frac{2\beta}{1+\beta}A\Sigma_0 A$, and $[A,[A,\Sigma_0]] = \eta^2[H,[H,\Sigma_0]]$. Collecting, $(1-\beta^2)\Sigma_0 - \frac{1-\beta}{1+\beta}A\Sigma_0 A = -\frac{\beta\eta^2}{1-\beta^2}[H,[H,\Sigma_0]] + \eta^2\mathcal{V}(\Sigma_0) + \eta^2\Sigma_\xi$; expanding $A\Sigma_0 A = (1+\beta)^2\Sigma_0 - \eta(1+\beta)(H\Sigma_0 + \Sigma_0 H) + \eta^2 H\Sigma_0 H$ the left side is $\eta(1-\beta)(H\Sigma_0 + \Sigma_0 H) - \eta^2\frac{1-\beta}{1+\beta}H\Sigma_0 H$, and dividing by $\eta(1-\beta)$ gives (2). (3) Sandwich (2) between $u_j^\top$ and u_k and substitute $(\Sigma_0)_{jk} = \frac{\eta(1+\beta)}{\lambda_j + \lambda_k}C_{jk}$ from (1); at $j = k$ the bracket is $1 - \frac{\eta\lambda_k}{2(1+\beta)}$. For the comparison with the Schur complement of Andreyev et al. [2026]: in the eigenbasis of H, the blocks M_{12}, M_{22} and the mean part of M_{21} act on each entry (j,k) separately through $A_j \otimes A_k$, $A_j = \begin{pmatrix} 1-\eta\lambda_j & -\eta\beta \\ \lambda_j & \beta \end{pmatrix}$, and the curvature noise enters only M_{11}, as $\eta^2\mathcal{V}$, and M_{21}, as $(-\eta\mathcal{V}, -\eta\mathcal{V}, \mathcal{V})$ in the xv, vx, vv rows. Rescaling the vv row and column and adding the third row to the first two gives $\det(I - M_{22})_{jk} = (1-\beta)\big((1-\beta)^2(1+\beta) + \beta\eta(\lambda_j + \lambda_k)\big) > 0$. With $\mathcal{S}_{jk} = \frac{(1-\beta)^2(1+\beta)}{(1-\beta)^2(1+\beta) + \beta\eta(\lambda_j+\lambda_k)}$ and L_{jk} the (j,k) coefficient of $\mathcal{L}$ at $\mathcal{V} = 0$, the identities $\det(I - A_j \otimes A_k) = (1-\beta)^3(1+\beta)L_{jk}$ and $\eta^2 + M_{12}(I - M_{22})^{-1}(-\eta, -\eta, 1)^\top = \mathcal{S}_{jk}\eta^2/(1-\beta)^2$ are rational identities in $(\eta, \beta, \lambda_j, \lambda_k)$; each

entry of $A_j \otimes A_k$ has degree at most $(2,2,1,1)$ in $(\eta, \beta, \lambda_j, \lambda_k)$, so $\det(I - M_{22})_{jk}$ has degree at most $(6,6,3,3)$ and $\operatorname{adj}(I - M_{22})_{jk}$ at most $(4,4,2,2)$; with $s_{jk} = (1-\beta)^2(1+\beta) + \beta\eta(\lambda_j + \lambda_k)$, the script checks $\det(I - M_{22})_{jk} = (1-\beta)s_{jk}$, the first identity in its Schur form $\det(I - M_{22})_{jk}(1 - M_{11}) - M_{12} \operatorname{adj}(I - M_{22}) M_{21} = (1-\beta)^2(1+\beta) \det(I - M_{22})_{jk} L_{jk}/s_{jk}$, and the second identity; multiplied by $s_{jk}(1-\beta)^2(1-\beta^2)$ and by $s_{jk} \det(I - M_{22})_{jk}$ they become polynomial identities of degree at most $(9,15,5,5)$ and $(9,9,4,4)$. They hold in exact rational arithmetic on a $13 \times 22 \times 13 \times 13$ product grid (`paper/check-schur-factor.py`), which has more points than the degree in each variable, so they hold identically. With the Schur determinant formula $\det(I - A_j \otimes A_k) = \det(I - M_{22})_{jk}(1 - T_{\text{slow}})_{jk}$ they give $I - T_{\text{slow}} = \mathcal{S} \circ \mathcal{L}$ entry by entry, the noise term included; the same script checks the operator identity with six non-commuting per-sample Hessians at $P = 2$ (residual 8×10^{-16}) and the leading term $R = \frac{2\beta\eta^2}{(1-\beta)^3} H\Sigma H$ of the remainder on pairs with $\lambda_k/\lambda_j \in \{1/3, 10^{-2}, 10^{-3}, 0\}$, where it separates from the double-commutator reading (ratio 0.999995 at $\eta = 10^{-7}$, $\beta = 0.9$). □

Lemma 15 (The remainder of the Schur-complement expansion)**.** *In the setting of Theorem 2 with $H \succeq 0$, let $T_{\text{slow}}(\eta, \beta)$ be the Schur complement of Andreyev et al. [2026, Appendix D], $\eta_{\text{eff}} = \eta/(1-\beta)$, $K_H(X) = HX + XH$, $G(X) = \mathbb{E}[H_{B_t} X H_{B_t}] = HXH + \mathcal{V}(X)$, and define their remainder by $T_{\text{slow}} = I - \eta_{\text{eff}} K_H + \eta_{\text{eff}}^2 G + R$. Write $\mathcal{L} = \eta_{\text{eff}} K_H + \mathcal{L}_2$ for $\eta/(1-\beta)$ times the left side of Theorem 2(2), so that $\mathcal{L}_2(X) = -\frac{\eta^2}{1-\beta^2} HXH + \frac{\eta^2\beta}{(1-\beta)^2(1-\beta^2)}[H,[H,X]] - \eta_{\text{eff}}^2 \mathcal{V}(X)$, and $c = \frac{\beta\eta}{(1-\beta)^2(1+\beta)}$. Then, exactly,*

$$R(\Sigma) = \frac{2\beta\eta^2}{(1-\beta)^3} H\Sigma H + R_3(\Sigma), \qquad R_3 = c\, K_H \circ \mathcal{L}_2 - c^2 K_H^2 \circ \mathcal{S} \circ \mathcal{L},$$

and, in the Frobenius and in the spectral norm, $\|R_3(\Sigma)\| \le 2c\|H\|\,\|\mathcal{L}_2(\Sigma)\| + 4c^2\|H\|^2\|\mathcal{L}(\Sigma)\|$. The order is not uniform in β: $R_3 = O\big(\beta\eta^3(\|H\|^3 + \|H\|\|\mathcal{V}\|)/(1-\beta)^4 + \beta^2\eta^3\|H\|^3/(1-\beta)^5\big)$, so R_3 is small against the leading term when $\eta\|H\| \ll 1-\beta$, $\eta\|\mathcal{V}\| \ll (1-\beta)\|H\|$ and $\beta\eta\|H\| \ll (1-\beta)^2$, and it is not small when $\beta\eta\|H\| \gtrsim (1-\beta)^2$. If $\lambda_k = 0$ and Σ is symmetric, the (j,k) entry is $R(\Sigma)_{jk} = \eta_{\text{eff}}^2(\mathcal{S}_{jk} - 1)\mathcal{V}(\Sigma)_{jk}$ for every η; it vanishes when $\mathcal{V}(\Sigma)_{jk} = 0$ (no curvature noise, or $\Sigma_z = H$ in Theorem 8, where $\mathcal{V}(\Sigma)_{jk} = \lambda_j\lambda_k\Sigma_{jk}/B$), but not in the flat-mode setting of Theorem 8 with $s_k > 0$, where $\mathcal{V}(\Sigma)_{jk} = s_j s_k \Sigma_{jk}/B$.

Proof. By the identity $I - T_{\text{slow}} = \mathcal{S} \circ \mathcal{L}$ with $\mathcal{S}^{-1} = I + cK_H$, $\mathcal{S} = I - cK_H \circ \mathcal{S}$, so $\mathcal{S} \circ \mathcal{L} = \mathcal{L} - cK_H\mathcal{L} + c^2K_H^2\mathcal{S}\mathcal{L} = \eta_{\text{eff}}K_H + \mathcal{L}_2 - c\eta_{\text{eff}}K_H^2 - cK_H\mathcal{L}_2 + c^2K_H^2\mathcal{S}\mathcal{L}$. Hence $R = T_{\text{slow}} - I + \eta_{\text{eff}}K_H - \eta_{\text{eff}}^2 G = -\mathcal{L}_2 - \eta_{\text{eff}}^2(H\Sigma H + \mathcal{V}) + c\eta_{\text{eff}}K_H^2 + R_3$. The $\mathcal{V}$ terms cancel. With $a = c\,\eta_{\text{eff}} = \frac{\eta^2\beta}{(1-\beta)^2(1-\beta^2)}$, the coefficient of the double commutator in $\mathcal{L}_2$, and $K_H^2(\Sigma) - [H,[H,\Sigma]] = 4H\Sigma H$, the remaining terms are $\big(\frac{\eta^2}{1-\beta^2} - \eta_{\text{eff}}^2 + 4a\big)H\Sigma H = \frac{2\beta\eta^2}{(1-\beta)^3} H\Sigma H$. For the bound, $\|K_H\| \le 2\|H\|$ in both norms. In the eigenbasis $\mathcal{S}$ multiplies the (j,k) entry by $\mathcal{S}_{jk} = 1/(1 + c\lambda_j + c\lambda_k) = 1/(a_j + a_k)$ with $a_j = \frac{1}{2} + c\lambda_j > 0$ for $H \succeq 0$: these multipliers lie in $(0,1]$, so $\|\mathcal{S}\| \le 1$ in the Frobenius norm, and they form a positive semidefinite Cauchy matrix with diagonal at most 1, so by Schur's theorem on Hadamard multipliers $\mathcal{S}$ is also a contraction in the spectral norm. The orders follow from $c = O(\beta\eta/(1-\beta)^2)$, $\|\mathcal{L}\| = O(\eta\|H\|/(1-\beta))$ and $\|\mathcal{L}_2\| = O\big(\eta^2(\|H\|^2 + \|\mathcal{V}\|)/(1-\beta)^2 + \beta\eta^2\|H\|^2/(1-\beta)^3\big)$ (the term $-\frac{\eta^2}{1-\beta^2} HXH$ and the curvature noise carry no factor β, the double commutator carries the largest power of $(1-\beta)^{-1}$), which give $c\|H\|\|\mathcal{L}_2\| = O\big(\beta\eta^3\|H\|(\|H\|^2 + \|\mathcal{V}\|)/(1-\beta)^4 + \beta^2\eta^3\|H\|^3/(1-\beta)^5\big)$ against the leading $2\beta\eta^2\|H\|^2/(1-\beta)^3$, a ratio of order $\eta\|H\|/(1-\beta) + \eta\|\mathcal{V}\|/((1-\beta)\|H\|) + \beta\eta\|H\|/(1-\beta)^2$. For the last claim, at $\lambda_k = 0$ the leading term $\lambda_j\lambda_k\Sigma_{jk}$ is zero, $(K_H X)_{jk} = \lambda_j X_{jk}$, $\mathcal{S}_{jk} = 1/(1 + c\lambda_j)$, $\mathcal{L}_2(\Sigma)_{jk} = a\lambda_j^2\Sigma_{jk} - \eta_{\text{eff}}^2\mathcal{V}(\Sigma)_{jk}$ and $\mathcal{L}(\Sigma)_{jk} = \eta_{\text{eff}}\lambda_j(1 + c\lambda_j)\Sigma_{jk} - \eta_{\text{eff}}^2\mathcal{V}(\Sigma)_{jk}$ (using

$a = c\eta_{\text{eff}}$), so $(R_3)_{jk} = c\lambda_j\big(a\lambda_j^2\Sigma_{jk} - \eta_{\text{eff}}^2\mathcal{V}_{jk}\big) - \frac{c^2\lambda_j^2}{1+c\lambda_j}\big(\eta_{\text{eff}}\lambda_j(1+c\lambda_j)\Sigma_{jk} - \eta_{\text{eff}}^2\mathcal{V}_{jk}\big) = -\eta_{\text{eff}}^2\frac{c\lambda_j}{1+c\lambda_j}\mathcal{V}_{jk} = \eta_{\text{eff}}^2(\mathcal{S}_{jk} - 1)\mathcal{V}(\Sigma)_{jk}$; the Σ_{jk} terms cancel because $a = c\eta_{\text{eff}}$. □

Proof of Theorem 5. Entries of $\mathcal{V}$. In the eigenbasis $(H_{B_t} - H)_{lm} = \frac{\sqrt{\lambda_l\lambda_m}}{B}\sum_b(\omega_{b,l}\omega_{b,m} - \delta_{lm})$ and $(\mathcal{V}(X))_{jk} = \sum_{l,m}\mathbb{E}[(H_{B_t} - H)_{jl}(H_{B_t} - H)_{mk}]X_{lm}$. With independent zero-mean unit-variance coordinates, $\mathrm{Cov}(\omega_j\omega_l, \omega_m\omega_k) \neq 0$ only when $\{j,l\} = \{m,k\}$ as multisets: for $j = k$ it is 1 when $l = m \neq k$ and $2 + \kappa_{\text{feat}}$ when $l = m = k$, so $(\mathcal{V}(X))_{kk} = \frac{\lambda_k}{B}\operatorname{tr}(HX) + \frac{1+\kappa_{\text{feat}}}{B}\lambda_k^2X_{kk}$; for $j \neq k$ it is 1 only at $(l,m) = (k,j)$ (the case $l = m = j$ gives $\mathbb{E}[(\omega_j^2 - 1)\omega_j]\mathbb{E}[\omega_k] = 0$), so $(\mathcal{V}(X))_{jk} = \frac{\lambda_j\lambda_k}{B}X_{kj}$ for any X, which is $\frac{\lambda_j\lambda_k}{B}X_{jk}$ on the symmetric matrices where Σ_0 and $\mathcal{T}$ live (on full matrices, e.g. in a basis of single entries, the transpose matters); the B i.i.d. draws divide by B. *Diagonal system.* The (k,k) entry of Theorem 2(2), where the double commutator vanishes, reads $2\lambda_k\gamma_{0,k} - \frac{\eta}{1+\beta}\lambda_k^2\gamma_{0,k} - \frac{\eta}{1-\beta}\big(\frac{\lambda_k}{B}\sum_j\lambda_j\gamma_{0,j} + \frac{1+\kappa_{\text{feat}}}{B}\lambda_k^2\gamma_{0,k}\big) = \frac{\eta}{1-\beta}\sigma_{\xi,k}^2$; substituting $\gamma_{0,j} = \frac{\eta(1+\beta)}{2\lambda_j}c_j$ and dividing by $\eta(1+\beta)$ gives (3). The matrix $\mathcal{M} = \alpha_{\text{diag}}\Lambda + \frac{\kappa_\beta}{B}\lambda\mathbf{1}^\top$ is similar to the symmetric $\widetilde{\mathcal{M}} = \Lambda^{-1/2}\mathcal{M}\Lambda^{1/2} = \alpha_{\text{diag}}\Lambda + \frac{\kappa_\beta}{B}\sqrt{\lambda}\sqrt{\lambda}^\top$, whose diagonal entries $\lambda_k(1 + \kappa_\beta(2 + \kappa_{\text{feat}})/B) \geq \lambda_k$ and off-diagonal entries $\kappa_\beta\sqrt{\lambda_j\lambda_k}/B$ are non-negative for every $\kappa_{\text{feat}} \geq -2$, so $I - \frac{\eta}{2(1+\beta)}\mathcal{M}$ is a nonsingular M-matrix, with a positive solution for every positive right-hand side, if and only if $\frac{\eta}{2(1+\beta)}\lambda_{\max}(\widetilde{\mathcal{M}}) < 1$ [Berman and Plemmons, 1994, Ch. 6]; the rank-one determinant lemma $\det(\mu I - \alpha_{\text{diag}}\Lambda - \frac{\kappa_\beta}{B}\sqrt{\lambda}\sqrt{\lambda}^\top) = \det(\mu I - \alpha_{\text{diag}}\Lambda)\big(1 - \frac{\kappa_\beta}{B}\sum_k\frac{\lambda_k}{\mu - \alpha_{\text{diag}}\lambda_k}\big)$ identifies $\lambda_{\max}(\widetilde{\mathcal{M}})$ as the unique root above $\max(\alpha_{\text{diag}}\lambda_1, 0)$ (the sum decreases to 0 there; the top eigenvalue lies in that interval by interlacing), and $\mu = 2(1+\beta)/\eta$ gives (4). *The full operator.* Let $\mathcal{T}(Q) = \mathbb{E}[F_tQF_t^\top]$ on $\mathbb{S}^{2P}$ with $F_t = \begin{pmatrix}(1+\beta)I - \eta H_{B_t} & -\beta I\\ I & 0\end{pmatrix}$ and $G = \operatorname{blkdiag}(\eta^2\Sigma_\xi, 0)$. By Lemma 3 it suffices to work on $\mathbb{S}^{2P}$. A numerical check of $\rho(\mathcal{T})$ is safest on the $n(n+1)/2$-dimensional symmetric basis, with $\mathcal{T}$ in Kraus form, or by the orbit of the identity: a formula for $\mathcal{V}$ that is wrong only on the antisymmetric part leaves the theorem intact but can move the radius computed on the full basis (on $\mathbb{S}^{2P}$ the antisymmetric part is excluded, and by Lemma 3 its radius is at most the symmetric one). *Only if:* let Q be any stationary second moment of (w_t, w_{t-1}). Its mean m is zero automatically: taking expectations in (2) gives $m = (1+\beta)m - \eta Hm - \beta m$, so $\eta Hm = 0$ and $m = 0$ since $H \succ 0$; Theorem 2, which assumes $\mathbb{E}[w_t] = 0$, therefore applies. The blocks of Q satisfy Theorem 2 and its diagonal solves (3), $c = Nc + b$ with $N = \frac{\eta}{2(1+\beta)}\mathcal{M} \geq 0$ entrywise and $b = \sigma_\xi^2/(1-\beta^2) > 0$, so $c \geq b > 0$ and $Nc = c - b < c$ componentwise; by the Collatz–Wielandt formula $\rho(N) \leq \max_k(Nc)_k/c_k < 1$, i.e. $\frac{\eta}{2(1+\beta)}\lambda_{\max}(\widetilde{\mathcal{M}}) < 1$. *If:* for $\eta < \eta_{\text{crit}}^{\text{EoSS},P}$ construct Q from the stationary equations with Σ_ξ diagonal in the eigenbasis ($\rho(\mathcal{T})$ does not depend on Σ_ξ). The diagonal of Σ_0 is the positive solution of (3); each off-diagonal entry satisfies $L_{jk}(\eta)(\Sigma_0)_{jk} = 0$ with $L_{jk}(\eta) = (\lambda_j + \lambda_k) - \frac{\eta\lambda_j\lambda_k}{1+\beta}(1 + \frac{\kappa_\beta}{B}) + \frac{\eta\beta(\lambda_j - \lambda_k)^2}{(1-\beta)(1-\beta^2)}$, and $(\Sigma_0)_{jk} = 0$ solves it whatever the sign of L_{jk} (its positivity below the ceiling follows from $\rho(\mathcal{T}) < 1$, each off-diagonal pair spanning an invariant subspace, and is not used). So $\Sigma_0 = \operatorname{diag}(\gamma_{0,k})$, $\Sigma_1 = \frac{A}{1+\beta}\Sigma_0 = \operatorname{diag}(\rho_{1,k}\gamma_{0,k})$ with $\rho_{1,k} = 1 - \frac{\eta\lambda_k}{1+\beta} \in (-1,1)$, and $Q = \begin{pmatrix}\Sigma_0 & \Sigma_1\\ \Sigma_1^\top & \Sigma_0\end{pmatrix} \succ 0$ (its Schur complement is $\operatorname{diag}(\gamma_{0,k}(1-\rho_{1,k}^2))$); the block equations of Theorem 2 are the blocks of $Q = \mathcal{T}(Q) + G$, so $(I - \mathcal{T})(Q) = G \succeq 0$. Since G is only semidefinite, perturb: $(I-\mathcal{T})(Q - \varepsilon G) = (1-\varepsilon)G + \varepsilon\mathcal{T}(G)$ with $\mathcal{T}(G) = \eta^2\mathbb{E}\big[\binom{A_t}{I}\Sigma_\xi(A_t \;\; I)\big]$, $A_t = (1+\beta)I - \eta H_{B_t}$, whose quadratic form at $(x,y) \neq 0$ is at least $\eta^2[(1-\varepsilon)x^\top\Sigma_\xi x + \varepsilon(Ax+y)^\top\Sigma_\xi(Ax+y)] > 0$ by Jensen's inequality in A_t (if $x = 0$ then $y \neq 0$); for $0 < \varepsilon < \min(1, \lambda_{\min}(Q)/(\eta^2\|\Sigma_\xi\|_{\text{op}}))$ also $Q_\varepsilon = Q - \varepsilon G \succ 0$. A positive operator on $\mathbb{S}_+^{2P}$ with $Q_\varepsilon \succ 0$ and $Q_\varepsilon - \mathcal{T}(Q_\varepsilon) \succ 0$ has $\rho(\mathcal{T}) < 1$: some $\theta < 1$ gives $\mathcal{T}(Q_\varepsilon) \preceq \theta Q_\varepsilon$, hence $\mathcal{T}^n(Q_\varepsilon) \preceq \theta^n Q_\varepsilon$, and every positive semidefinite X is

dominated by a multiple of Q_ε. *Sandwich.* In the secular equation every denominator $\mu^\star - \alpha_{\mathrm{diag}}\lambda_k$ is positive and lies between $\mu^\star - \max(\alpha_{\mathrm{diag}},0)\lambda_1$ and $\mu^\star - \min(\alpha_{\mathrm{diag}},0)\lambda_1$; replacing all of them by the smaller value gives $1 \le \kappa_\beta \operatorname{tr} H/(B(\mu^\star - \max(\alpha_{\mathrm{diag}},0)\lambda_1))$, the upper bound, replacing them by the larger value gives the second lower bound, and keeping $k=1$ only gives $\mu^\star \ge \alpha_{\mathrm{diag}}\lambda_1 + \kappa_\beta\lambda_1/B$. For the large-batch statement of the corollary, write $\mu^\star = \alpha_{\mathrm{diag}}\lambda_1 + \delta$, $\delta > 0$ when $\alpha_{\mathrm{diag}} > 0$; separating $k=1$ in (4), $\varepsilon_B\lambda_1/\delta = 1 - \varepsilon_B T(\mu^\star)$, which is the exact identity. Each denominator of $T(\mu^\star)$ is $\alpha_{\mathrm{diag}}(\lambda_1-\lambda_k)+\delta \ge \alpha_{\mathrm{diag}}(\lambda_1-\lambda_k)$, so $0 \le \varepsilon_B T(\mu^\star) \le \varepsilon_B S_\perp/\alpha_{\mathrm{diag}} \le \frac12$ and $\delta = \varepsilon_B\lambda_1(1+r)$ with $0 \le r = \frac{\varepsilon_B T}{1-\varepsilon_B T} \le 2\varepsilon_B S_\perp/\alpha_{\mathrm{diag}}$. Then $\mu^\star/\lambda_1 = 1+x$ with $x = \varepsilon_B(2+\kappa_{\mathrm{feat}}+r) \ge 0$, and $\frac{\eta\lambda_1}{2(1+\beta)} = \frac{1}{1+x} = 1 - \varepsilon_B(2+\kappa_{\mathrm{feat}}) - \varepsilon_B r + \frac{x^2}{1+x}$, so $\varrho = \frac{x^2}{1+x} - \varepsilon_B r \in [-2\varepsilon_B^2 S_\perp/\alpha_{\mathrm{diag}},\, x^2]$. With $\lambda_1 = \lambda_2$ the $k=2$ term of (4) is λ_1/δ and doubles the first-order shift. The corollary inserts the bounds (at $\kappa_{\mathrm{feat}} = -2$, $\alpha_{\mathrm{diag}} = 1 - \kappa_\beta/B$ and the second lower bound is $\lambda_1 + \kappa_\beta \operatorname{tr} H_\perp/B$ for $B \le \kappa_\beta$) and $\operatorname{Var}(u_1^\top H_{B_t} u_1) = \lambda_1^2(2+\kappa_{\mathrm{feat}})/B$, which vanishes for Rademacher features. For $P=1$, (4) reads $\eta\lambda_1\alpha_{\mathrm{diag}}/(2(1+\beta)) + \eta\lambda_1\kappa_\beta/(2(1+\beta)B) = 1$, i.e. $\eta = 2(1+\beta)\lambda_1/(\lambda_1^2 + \kappa_\beta\nu_B^2)$ with $\nu_B^2 = \lambda_1^2(2+\kappa_{\mathrm{feat}})/B$, which is Proposition 1. □

Proof of Proposition 7. (1) Write $f = \eta\lambda_1/(2(1+\beta))$, $\varphi_1 = 1+\beta-\eta\lambda_1$, $\varphi_2 = -\beta$ and $\rho_1 = \varphi_1/(1-\varphi_2) = 1-2f$. An additive excitation of variance $\sigma^2 > 0$ independent of the past does not change the second-moment operator; with it, squaring the recursion and taking stationary expectations gives $\gamma_0 = \big(\varphi_1^2 + \varphi_2^2 + 2\varphi_1\varphi_2\rho_1 + \eta^2\nu_{11,B}^2\big)\gamma_0 + \eta^2\sigma^2$, with $\rho_1 = \varphi_1/(1-\varphi_2)$ from the lag-one equation. The coefficient of the noise-free part equals $1 - 4(1-\beta^2)f(1-f)$, and $\eta^2\nu_{11,B}^2 = 4(1+\beta)^2 f^2 \nu_{11,B}^2/\lambda_1^2$, so a finite positive γ_0 exists if and only if $f\big(1+\kappa_\beta\nu_{11,B}^2/\lambda_1^2\big) < 1$, which by Corollary 4 is the mean-square threshold of the homogeneous recursion as well. The identity $(1+x)^{-1} = 1 - x + x^2/(1+x)$ with $0 \le x^2/(1+x) \le x^2$ for $x \ge 0$ is the formula for ϱ.

Let $\varepsilon = \nu_{11,B}^2/\lambda_1^2$, $\mu_3 = m_{3,B}/\lambda_1^3$ and $\mu_u = \mathbb{E}u_t = 2\kappa_\beta\delta$. On $|u| \le 1/2$ the third derivative of $\log(1+\xi)$ is $2/(1+\xi)^3$, hence at most 16, and Taylor's theorem with the Lagrange remainder gives $\log|1+u| = u - \frac12 u^2 + R(u)$ with $|R(u)| \le \frac83|u|^3$. At $\beta = 0$ the companion factor is the scalar $-(1+u_t)$, so $\gamma_{\mathrm{Lyap}} = \mathbb{E}\log|1+u_t|$ as soon as $|u_t| \le 1/2$ almost surely. Then $\gamma_{\mathrm{Lyap}} = \mathbb{E}u_t - \frac12\mathbb{E}u_t^2 + \theta$ with $|\theta| \le \frac83\mathbb{E}|u_t|^3$, $\mathbb{E}u_t = 2\delta$ and $\mathbb{E}u_t^2 = 4\delta^2 + 4(1+\delta)^2\varepsilon$, hence $\gamma_{\mathrm{Lyap}} = 2\delta - 2\varepsilon - 4\delta\varepsilon - 2\delta^2\varepsilon - 2\delta^2 + \theta$. For $|\delta| \le 1/2$, $|4\delta\varepsilon| \le 2\delta^2 + 2\varepsilon^2$ and $2\delta^2\varepsilon \le \varepsilon^2$. Also $|u_t| \le 2|\delta| + 3|\zeta_t|/\lambda_1$ and $(|a|+|b|)^3 \le 8(|a|^3+|b|^3)$, so $\mathbb{E}|u_t|^3 \le 8(8|\delta|^3 + 27\mu_3)$ and $|\delta|^3 \le \frac12\delta^2$. This is the expansion at $\beta = 0$. *Unbounded ζ_t at $\beta = 0$.* Write $R(u) = \log|1+u| - (u - \frac12u^2)$ and split into $E_1 = \{|u| \le \frac12\}$, $E_2 = \{|u| > \frac12, |1+u| > \frac12\}$ and $E_3 = \{|1+u| \le \frac12\}$. On E_1, $|R| \le \frac83|u|^3$. On E_2, $|\log|1+u|| \le \max(\log 2, |u|) \le 8\log 2\,|u|^3$ and $|u - \frac12u^2| \le |u| + \frac12u^2 \le 5|u|^3$ (both because $|u| > \frac12$), so $|R| \le 11|u|^3$. On E_3, $|u| \ge \frac12$ again gives $|u - \frac12u^2| \le 5|u|^3$, and $\mathbb{E}\big[|\log|1+u_t|| \mathbf{1}_{E_3}\big] = \int_{-1/2}^{1/2}(-\log|v|) f_{1+u}(v)\,dv \le (1+\log 2)M_{\mathrm{sing}}$, since $\int_{-1/2}^{1/2} -\log|v|\,dv = 1 + \log 2$. Adding the three regions gives the stated inequality, and the root follows as before with the extra M_{sing}. For Gaussian ζ_t, $1+u_t$ is normal with mean $1+2\delta \in [\frac34, \frac54]$ and standard deviation $2(1+\delta)\sqrt{\varepsilon} \in [\frac74\sqrt{\varepsilon}, \frac94\sqrt{\varepsilon}]$, and every $|v| \le \frac12$ is at distance at least $\frac14$ from the mean. For $\zeta_t = \lambda_1(\chi_B^2/B - 1)$, $1+u_t = -1 + 2(1+\delta)w$ with $w = \chi_B^2/B$, so $|1+u_t| \le \frac12$ forces $w \in [\frac29, \frac67]$; there Stirling's bound $\Gamma(k) \ge \sqrt{2\pi/k}(k/e)^k$ gives a density of w at most $\sqrt{B/(2\pi w^2)}\exp(-\frac{B}{2}(w - 1 - \log w))$ with $w - 1 - \log w \ge \log\frac76 - \frac17 > 0.0112$, and the change of variable divides by $2(1+\delta) \ge \frac74$.

For $\beta \in (0,1)$ assume $|u_t| \le \varepsilon \le \varepsilon_0(\beta)$ almost surely, and set $K = 2\beta/(1-\beta)$. The matrix $P = \begin{pmatrix} 1 & 1 \\ -1 & -\beta^{-1} \end{pmatrix}$ diagonalizes $M_\star = \begin{pmatrix} -(1+\beta) & -\beta \\ 1 & 0 \end{pmatrix}$ to $\operatorname{diag}(-1, -\beta)$, and $P^{-1}M_tP = \begin{pmatrix} -1-u_t & -u_t \\ \beta u_t & -\beta + \beta u_t \end{pmatrix}$. In the coordinate $r_t = y_{2,t}/y_{1,t}$ one has $y_{1,t+1}/y_{1,t} = -(1+Z_t)$ and

$r_{t+1} = \beta(r_t - u_t(1+r_t))/(1+Z_t)$ with $Z_t = u_t(1+r_t)$. If $|r| \le K\varepsilon$, then $|Z| \le \varepsilon(1+K\varepsilon) \le 1/5$ and

$$|r'| \le \frac{\beta\varepsilon(K+1+K\varepsilon)}{1-\varepsilon(1+K\varepsilon)}.$$

The bound $\varepsilon \le (1-\beta)/(8(1+\beta))$ gives $K\varepsilon = \beta/(4(1+\beta)) \le 1/4$ and $\varepsilon(1+\beta)+K\varepsilon^2 \le 5(1-\beta)/32 < (1-\beta)/2$. Since $\beta(K+1)/K = (1+\beta)/2$, this is $\beta(K+1+K\varepsilon) \le K\big(1-\varepsilon(1+K\varepsilon)\big)$. The ball is therefore forward-invariant. Starting at $r_0 = 0$ yields $|r_t| \le K\varepsilon \le 1/4$ and $|1+Z_t| \ge 4/5$. The identity $r_{t+1} - \beta r_t = -\beta Z_t(1+r_t)/(1+Z_t)$ then gives $|r_{t+1} - \beta r_t| \le 2\beta|u_t|$, because $(1+|r|)^2/(4/5) \le 2$. The L^2 triangle inequality produces $(\mathbb{E}r_t^2)^{1/2} \le \beta(\mathbb{E}r_{t-1}^2)^{1/2} + 2\beta(\mathbb{E}u_t^2)^{1/2}$, so $\mathbb{E}r_t^2 \le \big(2\beta/(1-\beta)\big)^2 \mathbb{E}u_t^2$.

On $|Z| \le 1/2$, $1/(1+Z) = 1 - Z + Z^2/(1+Z)$ with $|Z^2/(1+Z)| \le 2Z^2$, and $Z = u(1+r)$, so $u(1+r)^2/(1+Z) = u(1+r)^2 - u^2(1+r)^3 + \psi$ with $|\psi| \le 2|u|^3(1+|r|)^4$. Independence of u_t and r_t gives $\mathbb{E}r_{t+1} = \beta\mathbb{E}r_t - \beta\mu_u + \xi_t$ with $|\xi_t| \le C_\beta(\mathbb{E}u_t^2 + \mathbb{E}|u_t|^3)$: the terms that contain a factor r_t are estimated by $|\mathbb{E}r_t| \le C_\beta(\mathbb{E}u_t^2)^{1/2}$ and by Lyapunov's comparison $(\mathbb{E}u_t^2)^{3/2} \le \mathbb{E}|u_t|^3$. From $r_0 = 0$, $\mathbb{E}r_t = -\frac{\beta}{1-\beta}\mu_u + O(\beta^t|\mu_u|) + \rho_t$ with $|\rho_t| \le C_\beta(\mathbb{E}u_t^2 + \mathbb{E}|u_t|^3)$.

Since $|Z_t| \le 1/2$, $\mathbb{E}\log|1+Z_t| = \mathbb{E}Z_t - \frac{1}{2}\mathbb{E}Z_t^2 + O(\mathbb{E}|Z_t|^3)$. Independence gives $\mathbb{E}Z_t = \mu_u(1+\mathbb{E}r_t)$ and $\mathbb{E}Z_t^2 = \mathbb{E}u_t^2\,\mathbb{E}(1+r_t)^2$. Substituting $\mathbb{E}u_t^2 = 4\kappa_\beta^2\delta^2 + 4\kappa_\beta^2(1+\delta)^2\varepsilon$ produces the main terms $2\kappa_\beta\delta - 2\kappa_\beta^2\varepsilon$. The hypothesis $|u_t| \le \varepsilon_0(\beta)$ bounds ε by a constant depending only on β, and $|\delta|\varepsilon \le \frac{1}{2}(\delta^2+\varepsilon^2)$, so every cross term is $O(\delta^2 + \mu_3 + \varepsilon^2)$ with a constant depending only on β. The Cesàro mean of the transient $O(\beta^t|\delta|)$ vanishes. In the coordinates of P every M_t maps the closed cone $C_+ = \{y : |y_2| \le K\varepsilon\, y_1\}$ into $-C_+$ (the ball $|r| \le K\varepsilon$ is forward-invariant for every realisation and $1+Z_t \ge 4/5$), and e_1 is interior to C_+. For any product A of such matrices and any x, with $s = |x_1| + |x_2|/(K\varepsilon)$ both $x + se_1$ and se_1 lie in C_+, and $c \in C_+$ satisfies $\alpha e_1 - c \in C_+$ for $\alpha = c_1 + |c_2|/(K\varepsilon)$, so $|(Ac)_1| \le \alpha|(Ae_1)_1|$ and $\|Ax\| \le c_{K\varepsilon}\|x\|\,|(Ae_1)_1|$ with $c_{K\varepsilon}$ depending only on $K\varepsilon$; with $1 \le \|y_t\|/|y_{1,t}| \le 1 + |r_t|$ this gives $\frac{1}{t}\log\|M_t\cdots M_1\| = \frac{1}{t}\sum_{s<t}\log|1+Z_s| + O(1/t)$. The summands are uniformly bounded, so the expectations converge to γ_{Lyap} [Furstenberg and Kesten, 1960]. This is the expansion for $\beta \in (0,1)$.

Write $\gamma_{\mathrm{Lyap}} = 2\kappa_\beta\delta - 2\kappa_\beta^2\varepsilon + R$ with $|R| \le C_\beta(\delta^2 + \mu_3 + \varepsilon^2)$ on the δ-interval where $|u_t| \le \varepsilon_0(\beta)$. If $|\zeta_t| \le c\lambda_1$ almost surely, then $|u_t| \le 2\kappa_\beta|\delta| + 2\kappa_\beta(1+|\delta|)c$, while $\varepsilon \le c^2$ and $\mu_3 \le c^3$. Fix $A = 2C_\beta(2\kappa_\beta^2+1)/\kappa_\beta$ and $\tau = A(\varepsilon^2+\mu_3)$, and take c small enough that $|\kappa_\beta\varepsilon| + \tau$ lies in the interval and $\tau \le 3\kappa_\beta/(4C_\beta)$. Then $2\kappa_\beta\tau \ge C_\beta(2\kappa_\beta^2\varepsilon^2 + 2\tau^2 + \mu_3 + \varepsilon^2)$, so γ_{Lyap} is nonnegative at $\delta = \kappa_\beta\varepsilon + \tau$ and nonpositive at $\delta = \kappa_\beta\varepsilon - \tau$. For $d = \delta - \kappa_\beta\varepsilon$ with $\tau \le |d| \le \kappa_\beta/(2C_\beta)$, $\delta^2 \le 2d^2 + 2\kappa_\beta^2\varepsilon^2$ gives $|R| \le C_\beta(2d^2 + (2\kappa_\beta^2+1)\varepsilon^2 + \mu_3) < 2\kappa_\beta|d|$, so γ_{Lyap} has the sign of d on that whole range and the almost-sure threshold in it satisfies $|\delta^\star - \kappa_\beta\varepsilon| \le \tau$; no continuity of the limit exponent in δ is used. At the mean-square threshold $\delta = -\kappa_\beta\varepsilon/(1+\kappa_\beta\varepsilon)$ the same expansion has main term $-2\kappa_\beta^2\varepsilon(2+\kappa_\beta\varepsilon)/(1+\kappa_\beta\varepsilon)$, which dominates R for small ε and gives $\gamma_{\mathrm{Lyap}} < 0$. In every case Jensen's inequality yields $\gamma_{\mathrm{Lyap}} \le \frac{1}{2}\log\rho(\mathbb{E}[M_t^{\otimes 2}])$, so $\gamma_{\mathrm{Lyap}} \le 0$ at the mean-square threshold.

At $\beta = 0$ and $|u| \le 1/2$, $f(u) = (1+u)^p$ has third derivative $p(p-1)(p-2)(1+u)^{p-3}$, bounded on $[-1/2, 1/2]$ by a constant C_p. Thus $\mathbb{E}|1+u_t|^p = 1 + p\mathbb{E}u_t + \frac{p(p-1)}{2}\mathbb{E}u_t^2 + \rho$ with $|\rho| \le C_p\mathbb{E}|u_t|^3$. For c_p small enough that $|u_t| \le 1/2$ throughout a neighbourhood of $\delta = -(p-1)\varepsilon$, $\log\mathbb{E}|1+u_t|^p = 2p\delta + 2p(p-1)\varepsilon + O_p(\delta^2 + \mu_3 + \varepsilon^2)$, whose root is $\delta = -(p-1)\varepsilon + O_p(\mu_3 + \varepsilon^2)$.

The third-moment bound does not close the estimate. At $\beta = 0$, $\delta = 0$, $\lambda_1 = 1$ and $\zeta = \pm\frac{1}{2}(1-\varepsilon)$ with equal probability, $\mathbb{E}\log|1+2\zeta| = \frac{1}{2}\log\big(\varepsilon(2-\varepsilon)\big)$ and $-2\mathbb{E}\zeta^2 = -\frac{1}{2}(1-\varepsilon)^2$, so the ratio of the difference to $\mathbb{E}|\zeta|^3$ tends to $-\infty$ as $\varepsilon \downarrow 0$. Rosenthal's inequality supplies $\nu_{11,B}^2 = O(1/B)$ and $m_{3,B} = O(B^{-3/2})$ for an average of i.i.d. third-moment summands, and these rates may be substituted above only under $|u_t| \le \varepsilon_0(\beta)$. For $\lambda_1 = 1$, $B = 64$ and $\beta = 0$, $2/(1+2/64) =$

64/33 rounds to 1.9394, the first-order values $2(1 \mp 2/64)$ are 1.9375 and 2.0625, and the two quadrature roots are the output of `paper/checks/g3-lyapunov-remainder.mjs`. If $\zeta_t = 0$ then $u_t = 2\kappa_\beta\delta$ and both thresholds equal $2(1+\beta)/\lambda_1$. (2) Take the trace of Theorem 2(2) at a learning rate with a stationary second moment: $\operatorname{tr}[H,[H,\Sigma_0]] = 0$ by cyclicity and $\operatorname{tr}\mathcal{V}(\Sigma_0) = \frac{1}{B}\mathbb{E}\operatorname{tr}(H_i\Sigma_0 H_i - H\Sigma_0 H)$ because the B draws are i.i.d.; under the lower bound on $\mathbb{E}[H_i X H_i]$ the left side is at most $\operatorname{tr}(H\Sigma_0)\big(2 - \frac{\eta c_0 \operatorname{tr} H}{(1-\beta)B}\big) - \eta\operatorname{tr}(H^2\Sigma_0)\big(\frac{1}{1+\beta} - \frac{1}{(1-\beta)B}\big)$, and for $B \geq \kappa_\beta$ the last bracket is non-negative, so a positive right side forces $2 > \eta c_0 \operatorname{tr} H/((1-\beta)B)$ for every such η, and the threshold, their supremum, satisfies the non-strict inequality. Multiplying Theorem 2(2) by H^{-1} first, $\operatorname{tr}(H^{-1}[H,[H,\Sigma_0]]) = \operatorname{tr}(H\Sigma_0) - 2\operatorname{tr}(H\Sigma_0) + \operatorname{tr}(H\Sigma_0) = 0$ and $\operatorname{tr}(H^{-1}\mathcal{V}(\Sigma_0)) = \frac{1}{B}\operatorname{tr}(\mathbb{E}[(H_i - H)H^{-1}(H_i - H)]\Sigma_0)$ by cyclicity, which gives the H^{-1} identity; as $\Sigma_0 \succ 0$ and the right side is positive, the bracket cannot be negative semidefinite, which is the $\lambda_{\min}$ condition. □

Remark 16 (Trace bracket of the secular root). With $c = \eta\alpha_{\text{diag}}/(2(1+\beta)) > 0$ and $f(x) = x/(1-cx)$, $f(x)/x$ is increasing on $[0, 1/c)$, so for $0 \leq \lambda_k \leq \lambda_2$ ($k \geq 2$), $\operatorname{tr} H - \lambda_1 \leq \sum_{k\geq 2} f(\lambda_k) \leq (\operatorname{tr} H - \lambda_1) f(\lambda_2)/\lambda_2$. The lower inequality also holds for negative λ_k ($f(x) \geq x$ for $x < 1/c$); the upper one then needs $\operatorname{tr} H$ replaced by $\operatorname{tr} H + \sum_{\lambda_k<0} |\lambda_k|$, since only the non-negative eigenvalues contribute positive terms. Both networks of Section 4.2 have negative eigenvalues and the bracket there uses $\operatorname{tr} H$, so the upper sum, and with it the lower end of the bracket on $\eta^\star$, is formally an approximation at both sizes. At $P = 3\,610$ the dense spectrum of Section 4.3 has 1 528 negative eigenvalues, all $\geq \lambda_{\min} = -0.083330$, so the negative mass is at most 127.33, and $\operatorname{tr} H + 127.33$ in the upper sum gives the guaranteed lower end 0.00411 in place of 0.00492; at $P = 12\,826$, where $\lambda_{\min} \leq -0.072 = -4.9 \times 10^{-4}\lambda_1$, the negative mass was not measured. The left side of (4) is increasing in η, so the root computed with the upper sum is a lower bound on $\eta^\star$ and the root with the lower sum an upper bound: a two-sided bracket from λ_1, λ_2 and $\operatorname{tr} H$ alone, inside the sandwich (5), and tight when the bulk is concentrated near λ_2 or near 0. Phase 2 uses it on the networks, where only λ_1, λ_2 and $\operatorname{tr} H$ are available (`compute-secular-ceiling-bracket-from-trace.ts`).

Proof of Theorem 8. (1) In the eigenbasis $(z_i z_i^\top)_{lm} = \sqrt{s_l s_m}\,\omega_{i,l}\omega_{i,m}$ and $\mathcal{V}(X) = \frac{1}{B}\mathbb{E}[(zz^\top - \Sigma_z)X(zz^\top - \Sigma_z)]$ by independence of the B draws, so $(\mathcal{V}(X))_{jk} = \frac{1}{B}\sum_{l,m}\sqrt{s_j s_l s_m s_k}\operatorname{Cov}(\omega_j\omega_l, \omega_m\omega_k)X_{lm}$. By independence and standardisation the covariance is non-zero only when $\{j,l\} = \{m,k\}$ as multisets: for $j = k$ the pairs $l = m \neq k$ contribute $\frac{s_k s_l}{B}X_{ll}$ and $l = m = k$ contributes $\frac{2+\kappa_{\text{feat}}}{B}s_k^2 X_{kk}$; for $j \neq k$ only $(l,m) = (k,j)$ contributes $\frac{s_j s_k}{B}X_{kj}$, equal to $\frac{s_j s_k}{B}X_{jk}$ for symmetric X. The diagonal of $[H,[H,\Sigma_0]]$ vanishes (Theorem 2), which gives the equation. (2) Dividing row k by $2\lambda_k$ and using $\frac{\eta}{2(1-\beta)} = \frac{\eta\kappa_\beta}{2(1+\beta)}$ gives $M(\eta)\gamma_0 = \frac{\eta}{2(1-\beta)}\Lambda^{-1}\sigma_\xi^2$ with $M(\eta) = I - \frac{\eta}{2(1+\beta)}(\operatorname{diag}(d_k) + \frac{\kappa_\beta}{B}\Lambda^{-1}ss^\top)$, similar through $\Lambda^{1/2}$ to $I - \frac{\eta}{2(1+\beta)}\widetilde{\mathcal{M}}$, whose off-diagonal entries $\frac{\kappa_\beta}{B}\sqrt{\rho_j\rho_k}$ are non-negative and whose diagonal entries $\lambda_k + \kappa_\beta\frac{2+\kappa_{\text{feat}}}{B}\rho_k$ are positive for every $\kappa_{\text{feat}} \geq -2$; $M(\eta)$ is a non-singular M-matrix with non-negative inverse if and only if $\frac{\eta}{2(1+\beta)}\lambda_{\max}(\widetilde{\mathcal{M}}) < 1$, and the rank-one determinant lemma $\det(\mu I - \operatorname{diag}(d_k) - \frac{\kappa_\beta}{B}vv^\top) = \prod_k(\mu - d_k)\big(1 - \frac{\kappa_\beta}{B}\sum_k \frac{\rho_k}{\mu - d_k}\big)$ at $\mu = 2(1+\beta)/\eta$ gives the secular equation. For $k \notin \mathcal{A}$, $v_k = 0$ and $d_k = \lambda_k$, so $\widetilde{\mathcal{M}} = \operatorname{diag}(\lambda_k)_{k\notin\mathcal{A}} \oplus \widetilde{\mathcal{M}}_{\mathcal{A}}$ and $\lambda_{\max}(\widetilde{\mathcal{M}}) = \max\big(\max_{k\notin\mathcal{A}}\lambda_k, \lambda_{\max}(\widetilde{\mathcal{M}}_{\mathcal{A}})\big)$. When $\mathcal{A} \neq \emptyset$, every v_k with $k \in \mathcal{A}$ is positive, so $g(\mu) = \frac{\kappa_\beta}{B}\sum_{k\in\mathcal{A}}\frac{\rho_k}{\mu - d_k}$ decreases strictly from $+\infty$ to 0 on $(\max_{k\in\mathcal{A}} d_k, \infty)$, where the eigenvalues of $\widetilde{\mathcal{M}}_{\mathcal{A}}$ are the solutions of $g(\mu) = 1$; there is exactly one, $\lambda_{\max}(\widetilde{\mathcal{M}}_{\mathcal{A}})$. Its diagonal entries $\lambda_k + \kappa_\beta\frac{2+\kappa_{\text{feat}}}{B}\rho_k = d_k + \frac{\kappa_\beta}{B}\rho_k$ are at least λ_k and at least d_k, so $\lambda_{\max}(\widetilde{\mathcal{M}}_{\mathcal{A}}) \geq \max_{k\in\mathcal{A}}\max(d_k, \lambda_k)$: in $\eta = 2(1+\beta)/\mu$ the root lies in $(0, \min_{k\in\mathcal{A}} 2(1+\beta)/\max(d_k,\lambda_k)]$, is the only root there, and $\eta^\star$ is the smaller of it and $\min_{k\notin\mathcal{A}} 2(1+\beta)/\lambda_k$. The equivalence with $\rho(\mathcal{T}) < 1$ on $\mathbb{S}^{2P}$ is the argument in the proof of Theorem 5 (under "Only if" and "If") with the same Q_ε, since the off-

diagonal equations are decoupled scalar equations solved by $(\Sigma_0)_{jk} = 0$ for co-diagonal Σ_ξ and $\mu^\star \geq \widetilde{\mathcal{M}}_{kk} \geq \lambda_k$ gives the deterministic condition $\eta < 2(1+\beta)/\lambda_k$ for every mode. (3) Weyl's inequality with $\min_k d_k\, I \preceq \mathrm{diag}(d_k) \preceq \max_k d_k\, I$ and $\|v\|^2 = \mathcal{T}_{\mathrm{flat}}$, and the Rayleigh quotients at e_k, give the sandwich; dropping $\min_k d_k$ from the lower bound, which gives $\eta^\star \leq 2(1-\beta)B/\mathcal{T}_{\mathrm{flat}}$, needs $\min_k d_k \geq 0$; for the flat-mode rate, the Rayleigh quotient of $\widetilde{\mathcal{M}}$ at $(e_j+e_k)/\sqrt{2}$, $j \neq k$, is $\frac{1}{2}(\lambda_j+\lambda_k) + \frac{\kappa_\beta}{2B}\big((2+\kappa_{\mathrm{feat}})(\rho_j+\rho_k) + 2\sqrt{\rho_j\rho_k}\big) \geq \frac{\kappa_\beta}{B}\sqrt{\rho_j\rho_k} = \frac{\kappa_\beta s_j s_k}{B\sqrt{\lambda_j\lambda_k}}$ for every $\kappa_{\mathrm{feat}} \geq -2$, so $\mu^\star \geq \frac{\kappa_\beta s_j s_k}{B\sqrt{\lambda_j\lambda_k}}$ and $\eta^\star = 2(1+\beta)/\mu^\star \leq \frac{2(1-\beta)B\sqrt{\lambda_j\lambda_k}}{s_j s_k}$; for $P=1$ with Rademacher features $\widetilde{\mathcal{M}} = \lambda_1$ and $\eta^\star = 2(1+\beta)/\lambda_1$; under the tether $H_i \mapsto H_i + \mu I$ and $H \mapsto H + \mu I$ leave $H_i - H$, hence s_k, unchanged. □

Proof of Theorem 9. (1) At $\mathcal{V} = 0$, $\rho(A_0) < 1$ on $(0, \eta_{\mathrm{det}})$ and the $(1,1)$ block of $\sum_{t\geq 0} A_0^t\, \mathrm{blkdiag}(\eta^2\Sigma_\xi, 0)(A_0^t)^\top \succeq 0$ equals $\frac{1}{1-\beta}\mathcal{N}_{\eta,\beta}(\Sigma_\xi)$, so $\mathcal{L}_{\eta,\beta}^{-1}$ preserves $\mathbb{S}_+^P$ and $[1/\ell_{jk}(\eta,\beta)] = \frac{1}{\eta}\mathcal{N}_{\eta,\beta}(\mathbf{1}\mathbf{1}^\top) \succeq 0$ (only semidefinite: for $H = \lambda I$ it is a multiple of $\mathbf{1}\mathbf{1}^\top$). In the orthonormal basis $\{E_{jj}\}_j \cup \{(E_{jk}+E_{kj})/\sqrt{2}\}_{j<k}$ of $\mathcal{H} = (\mathbb{S}^P, \langle\cdot,\cdot\rangle_F)$, in the eigenbasis of H, $\mathcal{N}_{\eta,\beta}$ is diagonal with entries $\eta/\ell_{jk}(\eta,\beta) > 0$ (for $\eta < \eta_{\mathrm{det}}$, $\ell_{jk} \geq \lambda_j + \lambda_k - \eta\lambda_j\lambda_k/(1+\beta) > \lambda_j(1-\lambda_k/\lambda_1) + \lambda_k(1-\lambda_j/\lambda_1) \geq 0$), so $\mathcal{N}_{\eta,\beta} \succ_{\mathcal{H}} 0$, and $\mathcal{K}_{\eta,\beta} = \frac{1}{1-\beta}\mathcal{N}_{\eta,\beta}\mathcal{V}$ is similar via $\mathcal{N}_{\eta,\beta}^{-1/2}$ to the symmetric operator $\widetilde{\mathcal{K}}_{\eta,\beta} = \frac{1}{1-\beta}\mathcal{N}_{\eta,\beta}^{1/2}\mathcal{V}\mathcal{N}_{\eta,\beta}^{1/2}$ on $\mathcal{H}$, which has a real spectrum and the inertia of $\mathcal{V}$ (for $\Delta H = \mathrm{diag}(1,-1)$, $\mathcal{V}(E_{12}+E_{21}) = -(E_{12}+E_{21})$); since $\mathcal{K}_{\eta,\beta}$ preserves $\mathbb{S}_+^P$, the Krein–Rutman theorem [Krein and Rutman, 1948] gives $\rho(\mathcal{K}_{\eta,\beta}) = \lambda_{\max}(\mathcal{K}_{\eta,\beta})$ with an eigenmatrix $X_\eta \in \mathbb{S}_+^P \setminus \{0\}$. For $\mathcal{T} = \mathcal{T}_0 + \Delta$ on $\mathbb{S}_+^{2P}$, $(I-\mathcal{T}_0)^{-1} = \sum_{t\geq 0}\mathcal{T}_0^t \geq 0$ gives $\rho(\mathcal{T}) < 1 \iff \rho((I-\mathcal{T}_0)^{-1}\Delta) < 1$ and $\rho(\mathcal{T}) = 1 \iff \rho((I-\mathcal{T}_0)^{-1}\Delta) = 1$ [Berman and Plemmons, 1994], and the compression of $(I-\mathcal{T}_0)^{-1}\Delta$ to the $(1,1)$ block is $\mathcal{K}_{\eta,\beta}$; a compression alone does not fix the spectral radius, and the equality $\rho((I-\mathcal{T}_0)^{-1}\Delta) = \rho(\mathcal{K}_{\eta,\beta})$ is the factorization $\mathcal{R} = \mathcal{J}\circ\Pi$, $\Pi\circ\mathcal{J} = \mathcal{K}_{\eta,\beta}$ of Theorem 24(1), whose proof uses the invertibility of $\mathcal{L}_{\eta,\beta}$ established here and Theorem 2(2), and not $\beta > 0$. (2) Substituting $Y = \mathcal{N}_{\eta,\beta}^{-1/2}X$ in the Rayleigh quotient of $\widetilde{\mathcal{K}}_{\eta,\beta}$ gives the displayed formula for $\rho(\mathcal{K}_{\eta,\beta}) = \lambda_{\max}(\widetilde{\mathcal{K}}_{\eta,\beta})$; testing $X = I_P$ gives $\langle I_P, \mathcal{V}(I_P)\rangle_F = \frac{1}{B}\mathbb{E}\|\Delta H_i\|_F^2 > 0$, and since each weight $\ell_{jk}(\eta,\beta)/\eta = (\lambda_j+\lambda_k)/\eta + b_{jk}(\beta)$ is strictly decreasing in η and tends to $+\infty$ as $\eta \downarrow 0$, evaluating at an optimizer X_{η_1} for $0 < \eta_1 < \eta_2 < \eta_{\mathrm{det}}$ gives $\rho(\mathcal{K}_{\eta_2,\beta}) > \rho(\mathcal{K}_{\eta_1,\beta})$ and $\lim_{\eta\downarrow 0}\rho(\mathcal{K}_{\eta,\beta}) = 0$. (3) Let $w_1(\eta) \triangleq \ell_{11}(\eta,\beta)/\eta = \frac{\lambda_1^2}{(1+\beta)\eta}(\eta_{\mathrm{det}} - \eta) \downarrow 0$ while $\ell_{jk}(\eta,\beta)/\eta \geq c_0 > 0$ outside $\mathcal{I}_1 \times \mathcal{I}_1$ ($\mathcal{I}_1 = \{j : \lambda_j = \lambda_1\}$). If $P_1\Delta H_i P_1 \neq 0$ for some i, $X = P_1$ gives $\langle P_1, \mathcal{V}(P_1)\rangle_F = \frac{1}{B}\mathbb{E}\|P_1\Delta H_i P_1\|_F^2 > 0$ and $\rho(\mathcal{K}_{\eta,\beta}) = \Theta(w_1(\eta)^{-1})$. If $P_1\Delta H_i P_1 = 0$ for all i and $\Delta H_i P_1 \neq 0$ for some i, decomposing $X = X_{11} + X_\perp$ gives $P_1\mathcal{V}(X_{11})P_1 = 0$, so $\langle X, \mathcal{V}(X)\rangle_F = 2\langle X_{11}, \mathcal{V}(X_\perp)\rangle_F + \langle X_\perp, \mathcal{V}(X_\perp)\rangle_F$ and $\rho(\mathcal{K}_{\eta,\beta}) = O(w_1(\eta)^{-1/2})$, while testing $X = sP_1 + \mathcal{V}(P_1)$ ($\mathcal{V}(P_1) \neq 0$, $\langle P_1, \mathcal{V}(P_1)\rangle_F = 0$) at $s = w_1(\eta)^{-1/2}$ gives numerator $2s\|\mathcal{V}(P_1)\|_F^2 + O(1)$ and denominator $s^2\|P_1\|_F^2 w_1(\eta) + \Theta(1)$ (the top block of $\mathcal{V}(P_1)$ is $P_1\mathcal{V}(P_1)P_1 = 0$), hence $\rho(\mathcal{K}_{\eta,\beta}) = \Theta(w_1(\eta)^{-1/2})$. If $\Delta H_i P_1 = 0$ for all i, $\Delta H_i X \Delta H_i = \Delta H_i (I-P_1)X(I-P_1)\Delta H_i$ vanishes on $\mathcal{I}_1 \times \mathcal{I}_1$, so the quotient restricts to $\widetilde{X}_{jk} = 0$ on $\mathcal{I}_1 \times \mathcal{I}_1$, where all weights are at least $c_0 > 0$, and $\rho(\mathcal{K}_{\eta,\beta}) = O(1)$. Finally, $\rho(\mathcal{K}_{\eta,\beta}) < 1 \iff \lambda_{\max}(\widetilde{\mathcal{K}}_{\eta,\beta}) < 1 \iff \mathcal{L}_0 - \eta\mathcal{B}_\beta \succ_{\mathcal{H}} 0 \iff \eta\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}) < 1$. For $\eta \geq \eta_{\mathrm{det}}$ the iteration is not mean-square stable: $\mathcal{T} = \mathcal{T}_0 + \Delta$ with Δ preserving the cone $\mathbb{S}_+^{2P}$, so $\rho(\mathcal{T}) \geq \rho(\mathcal{T}_0) = \rho(A_0)^2 \geq 1$ [Berman and Plemmons, 1994, Ch. 1]. And $1/\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}) \leq \eta_{\mathrm{det}}$: for $X = u_1u_1^\top$ the commutator term vanishes and $\langle X, \mathcal{B}_\beta X\rangle_F / \langle X, \mathcal{L}_0 X\rangle_F \geq \frac{\lambda_1^2/(1+\beta)}{2\lambda_1} = 1/\eta_{\mathrm{det}}$. Hence the ceiling is $1/\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2})$ in every case. □

Proof of Theorem 13. (1) From $(\mathcal{L}_0 - \eta^\star\mathcal{B}_\beta)(\Sigma_0^\star) = 0$, taking the trace and using $\mathrm{tr}(H\Sigma_0^\star) > 0$ ($H \succ 0$, $\Sigma_0^\star \in \mathbb{S}_+^P \setminus \{0\}$), $\mathrm{tr}[H,[H,\Sigma_0^\star]] = 0$, $\mathrm{tr}(H\Sigma_0^\star H) = \mathrm{tr}(H^2\Sigma_0^\star)$, and $\mathrm{tr}(\mathcal{V}(\Sigma_0^\star)) = \mathrm{tr}((\mathbb{E}[H_{B_t}^2] - H^2)\Sigma_0^\star) \geq 0$, then dividing by $2\eta^\star\,\mathrm{tr}(H\Sigma_0^\star)$, gives the first identity in (7); $\lambda_P H \preceq H^2 \preceq \lambda_1 H$

gives $a_\star \in [\lambda_P, \lambda_1]$, and multiplying by $2(1 \mp \beta)$ gives the sandwich for $a_\star + v_\star$. For any $S \in \mathbb{S}_+^P \setminus \{0\}$ with $[H, S] = 0$, testing $\frac{1}{\eta^\star} = \max_{X \neq 0} \frac{\langle X, \mathcal{B}_\beta(X)\rangle_F}{\langle X, \mathcal{L}_0(X)\rangle_F}$ at $X = S^{1/2}$ ($[H, X] = 0$) gives $\frac{1}{\eta^\star} \geq \frac{a(S)}{2(1+\beta)} + \frac{v(S)}{2(1-\beta)}$; at $S = I_P$, $\frac{\bar{a}}{2(1+\beta)} + \frac{\bar{v}}{2(1-\beta)} = \frac{1+\frac{2\beta}{1-\beta}\theta_{\text{noise}}}{2(1+\beta)}(\bar{a} + \bar{v})$, and under $\mathbb{E}[g_B(\theta_0) g_B(\theta_0)^\top \mid H_B] = \sigma_g^2 H_B(\theta_0)$ one has $\mathbb{E}[g_B^\top H_B g_B] = \sigma_g^2 \operatorname{tr}(\mathbb{E}[H_B^2])$ and $\mathbb{E}\|g_B\|_2^2 = \sigma_g^2 \operatorname{tr} H$, so $\overline{\mathrm{BS}}(\theta_0) = \bar{a} + \bar{v}$; without that hypothesis, $R_B W_B = g_B^\top H_B g_B / \mathbb{E}_B\|g_B\|_2^2$ and $\mathbb{E}_B[W_B] = 1$ give $\overline{\mathrm{BS}}(\theta_0) - \mathrm{BS}(\theta_0) = \mathbb{E}_B[R_B W_B] - \mathbb{E}_B[R_B]\mathbb{E}_B[W_B] = \mathrm{Cov}_B(R_B, W_B)$, and Cauchy–Schwarz gives $|\overline{\mathrm{BS}}(\theta_0) - \mathrm{BS}(\theta_0)| \leq \sigma_{\mathrm{BS}}(\theta_0)\mathrm{CV}_{g^2}(\theta_0)$. If $\eta^\star \leq \eta < \eta_{\text{det}}$, Theorem 9(2) gives $\rho(\mathcal{K}_{\eta,\beta}) \geq 1$ and Theorem 9(1) $\rho(\mathcal{T}) \geq 1$; if $\eta \geq \eta_{\text{det}}$, $\mathcal{T}(Q) \succeq \bar{F} Q \bar{F}^\top$ with $\bar{F} = \mathbb{E}F_t$ gives $\rho(\mathcal{T}) \geq \rho(\bar{F})^2 \geq 1$. Corollary 4 then yields (8). In Theorem 5, $\Sigma_0^\star = \mathrm{diag}(\gamma_{0,k}^\star)$ is diagonal in the eigenbasis of H and $\operatorname{tr}(\mathcal{V}(\Sigma_0^\star)) = \frac{\operatorname{tr} H}{B}\operatorname{tr}(H\Sigma_0^\star) + \frac{1+\kappa_{\text{feat}}}{B}\operatorname{tr}(H^2\Sigma_0^\star)$. For (b), any $g_B = \frac{1}{B} Z r \neq 0$ ($u = r/\|r\|_2$, $D \triangleq G_Z - I_B$) has $v \triangleq (Z^\top Z)^{1/2} r \neq 0$, $\|g_B\|_2^2 = \frac{1}{B^2}\|v\|_2^2 = \frac{\operatorname{tr} H}{B^2}(1 + u^\top D u)$, and $g_B^\top H_B g_B = \frac{\operatorname{tr} H}{B^3} v^\top G_Z v = \frac{(\operatorname{tr} H)^2}{B^3}(1 + 2u^\top D u + \|Du\|_2^2)$, so $\frac{u^\top G_Z^2 u}{u^\top G_Z u} - 1 = u^\top D u + \frac{\|Du\|_2^2 - (u^\top D u)^2}{u^\top G_Z u} = u^\top D u + \frac{\|(I_B - u u^\top) D u\|_2^2}{u^\top G_Z u}$ (giving $\mathrm{BS}(\theta) \geq \frac{\operatorname{tr} H}{B}$ when $u(\theta)$ is independent of G_Z, since $\mathbb{E}_B[D] = 0$) and $\left|\frac{g_B^\top H_B g_B}{\|g_B\|_2^2} - \frac{\operatorname{tr} H}{B}\right| \leq \frac{\operatorname{tr} H}{B}\|D\|_{\text{op}} \leq \frac{\operatorname{tr} H}{B}\|D\|_F$; since $\mathbb{E}[D_{bb}^2] = \frac{(2+\kappa_{\text{feat}})\operatorname{tr}(H^2)}{(\operatorname{tr} H)^2}$ and $\mathbb{E}[D_{bb'}^2] = \frac{\operatorname{tr}(H^2)}{(\operatorname{tr} H)^2}$ for $b \neq b'$ are mutually orthogonal in L^2, $\mathbb{E}\|Du\|_2^2 = (B + 1 + \kappa_{\text{feat}})\frac{\operatorname{tr}(H^2)}{(\operatorname{tr} H)^2}$, $\mathbb{E}[(u^\top D u)^2] = (2 + \kappa_{\text{feat}}\sum_{b=1}^B u_b^4)\frac{\operatorname{tr}(H^2)}{(\operatorname{tr} H)^2}$, and $\mathbb{E}\|D\|_F \leq (\mathbb{E}\|D\|_F^2)^{1/2} = \delta_{\text{Gram}}$, giving (9) and, via $\frac{2(1+\beta)}{\eta^\star} \geq \kappa_\beta \frac{\operatorname{tr} H}{B}$ when $\kappa_{\text{feat}} \geq -1$ (Theorem 5), the divergence criterion.

(2) Theorem 2(2) reads $(\mathcal{L}_0 - \eta\mathcal{B}_\beta)(\Sigma_0(\eta)) = \frac{\eta}{1-\beta}\Sigma_\xi$, or $(I - \eta\mathcal{L}_0^{-1}\mathcal{B}_\beta)(\Sigma_0(\eta)) = \frac{\eta}{1-\beta}\mathcal{L}_0^{-1}(\Sigma_\xi)$. Since $\mathcal{L}_0^{-1}\mathcal{B}_\beta$ is self-adjoint in $\langle\cdot,\cdot\rangle_{\mathcal{L}_0}$ with simple top eigenpair $(1/\eta^\star, \Sigma_0^\star)$, $\langle \Sigma_0^\star, \mathcal{L}_0^{-1}(\Sigma_\xi)\rangle_{\mathcal{L}_0} = \operatorname{tr}(\Sigma_0^\star \Sigma_\xi)$, and $\|\Sigma_0^\star\|_{\mathcal{L}_0}^2 = 2\operatorname{tr}(H(\Sigma_0^\star)^2) > 0$, orthogonal projection onto $\mathrm{span}(\Sigma_0^\star)$ and $(\Sigma_0^\star)^{\perp_{\mathcal{L}_0}}$ gives (10), where for every eigenvalue $\mu_m \leq 1/\eta^\star - \Delta_{\mathcal{G}}$ ($m \geq 2$), $1 - \eta\mu_m = (1 - \eta/\eta^\star) + \eta\Delta_{\mathcal{G}} \geq \eta\Delta_{\mathcal{G}}$. For $\eta^\star < \eta_{\text{det}}$, differentiating $\mathcal{L}_{\eta,\beta}(\Sigma_0^\star(\eta)) = \frac{\eta}{(1-\beta)r(\eta)}\mathcal{V}(\Sigma_0^\star(\eta))$ at $\eta = \eta^\star$ ($r(\eta^\star) = 1$) against $\Sigma_0^\star$ and using $\eta^\star \partial_\eta \mathcal{L}_{\eta,\beta}\big|_{\eta^\star}(\Sigma_0^\star) = \mathcal{L}_{\eta^\star,\beta}(\Sigma_0^\star) - \mathcal{L}_0(\Sigma_0^\star)$ gives $d_\star$. Since $\mathbb{E}[w_t] = 0$ and (H_{B_t}, ξ_t) is independent of w_t, $\mathbb{E}[w_t^\top H_{B_t}^2 \xi_t] = 0$ and $\mathbb{E}[w_t^\top H_{B_t} \xi_t] = 0$, so $\mathbb{E}[g_{B_t}^\top H_{B_t} g_{B_t}] = \operatorname{tr}(\mathbb{E}[H_{B_t}^3]\Sigma_0(\eta)) + \mathbb{E}[\xi_t^\top H_{B_t} \xi_t]$ and $\mathbb{E}\|g_{B_t}\|_2^2 = \operatorname{tr}(\mathbb{E}[H_{B_t}^2]\Sigma_0(\eta)) + \operatorname{tr}\Sigma_\xi$; dividing by $\operatorname{tr}\Sigma_0(\eta) \to \infty$ gives (11). When $H_{B_t} \succeq 0$ a.s., Cauchy–Schwarz on $(\Sigma_0^\star)^{1/2} H_{B_t}^{1/2}$ and $(\Sigma_0^\star)^{1/2} H_{B_t}^{3/2}$ gives, for each realisation, $\operatorname{tr}(H_{B_t}^2 \Sigma_0^\star) \leq \operatorname{tr}(H_{B_t}\Sigma_0^\star)^{1/2}\operatorname{tr}(H_{B_t}^3\Sigma_0^\star)^{1/2}$, and Cauchy–Schwarz in the probability space then gives $(\operatorname{tr}(\mathbb{E}[H_{B_t}^2]\Sigma_0^\star))^2 \leq \operatorname{tr}(H\Sigma_0^\star)\operatorname{tr}(\mathbb{E}[H_{B_t}^3]\Sigma_0^\star)$. In Theorem 5, expanding $H_{B_t}^3$ over the $B(B-1)(B-2)$ distinct triples, $3B(B-1)$ pairs and B diagonal triples of $H_i = z_i z_i^\top$ gives $\mathrm{diag}(\mathbb{E}[H_{B_t}^3]) = \left[(\frac{\operatorname{tr} H}{B})^2 + \frac{q_{\text{feat}}}{B}\operatorname{tr}(H^2)\right]\Lambda + 2\frac{\operatorname{tr} H}{B} q_{\text{feat}}\Lambda^2 + r_{\text{feat}}\Lambda^3$, while $(\mu^\star - \alpha_{\text{diag}}\lambda_k)\gamma_{0,k}^\star = \frac{\kappa_\beta}{B}\operatorname{tr}(H\Sigma_0^\star)$ ($\mu^\star = 2(1+\beta)/\eta^\star$) multiplied by λ_k and λ_k^2 and summed over k gives $\mu^\star - \alpha_{\text{diag}} a_\star = \frac{\kappa_\beta}{B}\operatorname{tr} H$, $\mu^\star a_\star - \alpha_{\text{diag}} m_{2,\star} = \frac{\kappa_\beta}{B}\operatorname{tr}(H^2)$, $\alpha_{\text{diag}}(m_{2,\star} - a_\star^2) = \frac{\kappa_\beta}{B}((\operatorname{tr} H) a_\star - \operatorname{tr}(H^2))$, and (12). Write N and $D = \frac{\operatorname{tr} H}{B} + q_{\text{feat}} a_\star$ for the numerator and denominator of the fraction in (12). Small batch: $\frac{2(1-\beta)}{\eta^\star} = \frac{\operatorname{tr} H}{B} + \frac{1-\beta}{1+\beta}\alpha_{\text{diag}} a_\star$ with $a_\star \leq \lambda_1$ and $|\frac{1-\beta}{1+\beta}\alpha_{\text{diag}}| \leq 1 + \frac{|1+\kappa_{\text{feat}}|}{B}$, so $\frac{2(1-\beta)}{\eta^\star} = \frac{\operatorname{tr} H}{B}(1 + O(\frac{B\lambda_1}{\operatorname{tr} H}))$, likewise $\mathcal{C}_{\text{mod}} = \frac{2(1-\beta)}{\eta^\star} + O(\lambda_1)$, and $D \geq \frac{\operatorname{tr} H}{B}$ with $N \leq q_{\text{feat}}\lambda_1\frac{\operatorname{tr} H}{B} + O(\lambda_1^2)$ gives $N/D = \frac{\operatorname{tr} H}{B} O(\frac{B\lambda_1}{\operatorname{tr} H})$. Large batch, $\varepsilon \triangleq \frac{\kappa_\beta \operatorname{tr} H}{B\lambda_1} \to 0$: then $B \to \infty$, $\alpha_{\text{diag}}, q_{\text{feat}} \to 1$ and $r_{\text{feat}} - q_{\text{feat}}^2 = O(1/B)$; $\mu^\star > \alpha_{\text{diag}}\lambda_1$ and $\mu^\star - \alpha_{\text{diag}} a_\star = \frac{\kappa_\beta}{B}\operatorname{tr} H$ give $a_\star \geq \lambda_1(1 - \varepsilon/\alpha_{\text{diag}})$ and $\mu^\star = \lambda_1(1 + O(\varepsilon))$; the moment identity gives $m_{2,\star} - a_\star^2 \leq \frac{\kappa_\beta \operatorname{tr} H \lambda_1}{B\alpha_{\text{diag}}} = \lambda_1^2 O(\varepsilon)$; hence $N/D = \lambda_1 O(\varepsilon + \frac{1}{B})$ and $\mathcal{C}_{\text{mod}} = \mu^\star - \frac{2\beta}{1-\beta} v_\star = \mu^\star(1 + O(\varepsilon))$, since $v_\star = \frac{\operatorname{tr} H}{B} + \frac{1+\kappa_{\text{feat}}}{B} a_\star$.

(3) In the eigenbasis of Theorem 8, $\Sigma_0^\star = \mathrm{diag}(c_1^\star, \ldots, c_P^\star)$ is diagonal and $(\mathbb{E}[H_{B_t}^2])_{mm} = \lambda_m^2 + A_m + C_m$ with $A_m \triangleq \frac{s_m}{B}\sum_{q \neq m} s_q \geq 0$ and $C_m \triangleq \frac{2+\kappa_{\text{feat}}}{B} s_m^2 \geq 0$. With $\Delta_{B_t} = H_{B_t} - H$, expanding $\mathbb{E}[H_{B_t}^3] = H^3 + 2H\mathcal{V}(I_P) + \mathcal{V}(H) + \mathbb{E}[\Delta_{B_t}^3]$ gives $0 \leq (H^3 + 2H\mathcal{V}(I_P) + \mathcal{V}(H))_{mm} \leq 3M_0(\mathbb{E}[H_{B_t}^2])_{mm}$

and $|(\mathbb{E}[\Delta_{B_t}^3])_{mm}| = \big|\frac{s_m}{B^2}\sum_{p\neq q,\,p,q\neq m} s_p s_q + \frac{2+\kappa_{\mathrm{feat}}}{B^2} s_m \sum_{q\neq m} s_q^2 + \frac{2(2+\kappa_{\mathrm{feat}})}{B^2} s_m^2 \sum_{q\neq m} s_q + \frac{\mathbb{E}[(\omega_{i,m}^2-1)^3]}{B^2} s_m^3\big| \le \frac{P+\kappa_{\mathrm{feat}}}{B} M_0 A_m + \frac{2(P-1)+\widetilde{\kappa}_6}{B} M_0 C_m \le \frac{\max(P+\kappa_{\mathrm{feat}},\,2(P-1)+\widetilde{\kappa}_6)}{B} M_0 (\mathbb{E}[H_{B_t}^2])_{mm}$ for every m, so $|\mathrm{BS}^\star(\Sigma_0^\star)| \le \big(3 + \frac{\max(P+\kappa_{\mathrm{feat}},\,2(P-1)+\widetilde{\kappa}_6)}{B}\big) M_0$ and $|\mathrm{BS}^\star(\Sigma_0^\star)|/(2(1-\beta)/\eta^\star) \le O(\eta^\star) = O(\sqrt{\lambda_k}) \to 0$ by Theorem 8(3). □

Proof of Theorem 14. (1) For $y \ge y^\star$, $\eta \ge \eta^\star(y)$ and $\rho(\mathcal{T}_y) \ge 1$: by Theorem 9(1)–(2) when $\eta < \eta_{\mathrm{det}}(y)$, and when $\eta \ge \eta_{\mathrm{det}}(y)$ because $\mathcal{T}_y \succeq \mathcal{T}_{0,y}$ on the cone and $\rho(\mathcal{T}_{0,y}) = \rho(A_{0,y})^2 \ge 1$ (proof of Theorem 9(3)); so Corollary 4 gives $\mathbb{E}\|w_t\|^2 + \mathbb{E}\|w_{t-1}\|^2 \ge c(y)t$ for $t \ge 1$; summing over $t = 1, \dots, K$ gives $\frac{1}{K}\sum_{t=1}^K \mathbb{E}\|w_t\|^2 \ge \frac{1}{4}c(y)(K+1) - \frac{\mathbb{E}\|w_0\|^2}{2K}$ and (a) follows from $\mathrm{tr}(\Phi(y)\mathbb{E}[w_t w_t^\top]) \ge \lambda_{\min}(\Phi(y))\mathbb{E}\|w_t\|^2$. For $y \in [y_{\min}, y^\star)$, $\rho(\mathcal{T}_y) < 1$ gives $\mathbb{E}[w_t w_t^\top] \to \Sigma_0(y)$. By perturbation theory of simple isolated eigenvalues of the C^1 self-adjoint family $\mathcal{L}_0(y)^{-1/2}\mathcal{B}_\beta(y)\mathcal{L}_0(y)^{-1/2}$, $u(y) \triangleq 1 - \eta/\eta^\star(y)$, $\Sigma_0^\star(y)$ and the reduced resolvent $\widetilde{R}_\perp(y) = \sigma_\xi^{-2} R_\perp(y)$ of (10) are C^1 in a two-sided neighbourhood of $y^\star$, with $u(y^\star) = 0$ and $u'(y^\star) = g_0 < 0$. Taking the trace of (10) against $\frac{1}{2}\Phi(y)$ gives (14). Since $G_{\mathrm{slow}}(y_{\min}) < 0$ and $\lim_{y\uparrow y^\star} G_{\mathrm{slow}}(y) = +\infty$, a root $y_{\mathrm{eq}} \in (y_{\min}, y^\star)$ exists. With $\varepsilon \triangleq \sigma_\xi^2/v_{\mathrm{PS}}$, multiplying $G_{\mathrm{slow}}(y) = 0$ by $u(y)/v_{\mathrm{PS}}$ gives $\mathcal{F}(u,\varepsilon) \triangleq u - \varepsilon\big[\bar{c}(y(u)) + \frac{u}{2}\mathrm{tr}(\Phi(y(u))\widetilde{R}_\perp(y(u)))\big] = 0$, which is C^1 in a two-sided neighbourhood of $(0,0)$ with $\mathcal{F}(0,0) = 0$, $\partial_u \mathcal{F}(0,0) = 1$ and $\bar{c}(y^\star) = \bar{c}_\star > 0$; the implicit function theorem gives the unique root $u_{\mathrm{eq}} = \bar{c}_\star \varepsilon + O(\varepsilon^2)$, at which $G'_{\mathrm{slow}}(y_{\mathrm{eq}}) = \frac{|g_0|\bar{c}_\star \sigma_\xi^2}{u_{\mathrm{eq}}^2}(1 + O(u_{\mathrm{eq}})) > 0$, and $\frac{\eta^\star - \eta}{\eta} = \frac{u}{1-u}$ together with $1 - r(y_{\mathrm{eq}}) = d_\star u_{\mathrm{eq}} + O(u_{\mathrm{eq}}^2)$ gives (15). For (c), when $w_0 = w_{-1} = 0$ and $\Phi \succ 0$, $\mathbb{E}[w_t w_t^\top] \preceq \Sigma_0(y_{\min})$ for all $t \ge 0$ at $y_{\min}$ (by positivity of $\mathcal{T}_{y_{\min}}^s$), so $G_{\mathrm{slow},K}(y_{\min}) \le G_{\mathrm{slow}}(y_{\min}) < 0$, while $G_{\mathrm{slow},K}(y) > 0$ on $[y^\star, y_{\max}]$ by (a); continuity of $G_{\mathrm{slow},K}$ on $[y_{\min}, y_{\max}]$ implies that every root of $G_{\mathrm{slow},K}$ lies in $(y_{\min}, y^\star)$. Since $\sup_{y\in[y_-,y_+]} \rho(\mathcal{T}_y) < 1$, $(G_{\mathrm{slow},K}, G'_{\mathrm{slow},K}) \to (G_{\mathrm{slow}}, G'_{\mathrm{slow}})$ uniformly on $[y_-, y_+]$ as $K \to \infty$, giving a unique root $y_{\mathrm{eq},K} \in [y_-, y_+]$ for large K with $y_{\mathrm{eq},K} \to y_{\mathrm{eq}}$. By Theorem 23(1), for any $\mathcal{F}_{k-1}$-measurable initial state $Q_{k,0}$ at epoch k, $\mathbb{E}[\frac{1}{K}\sum_{t=1}^K g_{y,t}(w_{k,t}, y_k) \mid \mathcal{F}_{k-1}] = G_{\mathrm{slow},K}(y_k) + b_{k,K}$ with $|b_{k,K}| = \big|\frac{1}{2K}\sum_{t=1}^K \mathrm{tr}(\Phi(y_k)(\mathcal{T}_{y_k}^t(\mathbb{E}[Q_{k,0} \mid \mathcal{F}_{k-1}]))_{11})\big| \le \frac{C_+ R_0^2}{K}$. By the mean-value theorem on $[y_-, y_+]$, $G_{\mathrm{slow},K}(y_k) = \bar{G}'_k(y_k - y_{\mathrm{eq},K})$ with $\bar{G}'_k \in [m_0, L_K]$, so for $0 < \eta_y \le 1/L_K$ one has $|a_k| \triangleq |y_k - y_{\mathrm{eq},K} - \eta_y G_{\mathrm{slow},K}(y_k)| \le (1 - \eta_y m_0)|y_k - y_{\mathrm{eq},K}|$; applying $(a_k - \eta_y b_{k,K})^2 \le (1 + \eta_y m_0)a_k^2 + (1 + \frac{1}{\eta_y m_0})\eta_y^2 b_{k,K}^2 \le (1 - \eta_y m_0)(y_k - y_{\mathrm{eq},K})^2 + \frac{2\eta_y}{m_0} b_{k,K}^2$ and non-expansiveness of $\Pi_{[y_-,y_+]}$ yields $\mathbb{E}[(y_{k+1} - y_{\mathrm{eq},K})^2 \mid \mathcal{F}_{k-1}] \le (1 - \eta_y m_0)(y_k - y_{\mathrm{eq},K})^2 + \eta_y^2 s_K^2 + \frac{2\eta_y C_+^2 R_0^4}{m_0 K^2}$ and the stated bound.

(2) Applying (14) with $\Phi = 2I$ and $v_{\mathrm{PS}} = \gamma$ to the secant family on $(\gamma^\star, \gamma_{\max}]$ gives $F(\gamma) \triangleq \mathrm{tr}\,\Sigma_0^{\mathrm{sec}}(\gamma) - \gamma = \frac{\eta\,\mathrm{tr}(\Sigma_0^\star(\gamma)\Sigma_\xi)}{2(1-\beta)\,\mathrm{tr}(H_{\mathrm{sec}}(\gamma)\Sigma_0^\star(\gamma)^2)(1 - \eta/\eta_{\mathrm{sec}}^\star(\gamma))} + \mathrm{tr}\,R_\perp^{\mathrm{sec}}(\gamma) - \gamma$, which tends to $+\infty$ as $\gamma \downarrow \gamma^\star$ and is ≤ 0 at $\gamma_{\max}$; solving $(1 - \eta/\eta_{\mathrm{sec}}^\star(\gamma))F(\gamma) = 0$ in $u = 1 - \eta/\eta_{\mathrm{sec}}^\star(\gamma)$ by the implicit function theorem as in (1) gives (16). □

B Further properties of the second-moment operator

B.1 Momentum diagonalizes the mean-square threshold

We keep the setting of Theorem 9: $H = \sum_k \lambda_k u_k u_k^\top \succ 0$ with $\lambda_1 \ge \cdots \ge \lambda_P > 0$, per-sample Hessians H_i sampled with replacement in batches of size B, $\Delta H_i = H_i - H$, $\mathcal{V}(X) = \frac{1}{B}\mathbb{E}[\Delta H_i X \Delta H_i]$, $\kappa_\beta = \frac{1+\beta}{1-\beta}$, $\eta_{\mathrm{det}} = \frac{2(1+\beta)}{\lambda_1}$ and $f = \eta/\eta_{\mathrm{det}} \in (0,1)$. All matrices are written in the eigenbasis of H. Let $\mu_1 > \cdots > \mu_R$ be the distinct eigenvalues of H, P_r the orthogonal projector on the eigenspace

of μ_r, and

$$\Delta_{\min} \triangleq \min_{r\neq s} |\mu_r - \mu_s| \qquad (\Delta_{\min} = +\infty \text{ if } R = 1),$$

the smallest gap between *distinct* eigenvalues. On $\mathcal{H} = (\mathbb{S}^P, \langle X, Y\rangle_F)$ we use the orthonormal basis $\{E_{kk}\}_k \cup \{F_{jk} = (E_{jk} + E_{kj})/\sqrt{2}\}_{j<k}$.

The commutant splitting. Let

$$\mathcal{D} \triangleq \{X \in \mathbb{S}^P : [H, X] = 0\} = \{X : X_{jk} = 0 \text{ if } \lambda_j \neq \lambda_k\}, \qquad \mathcal{O} \triangleq \mathcal{D}^{\perp} = \{X : X_{jk} = 0 \text{ if } \lambda_j = \lambda_k\}.$$

The orthogonal projection of $\mathcal{H}$ onto $\mathcal{D}$ is the pinching $\mathcal{P}(X) = \sum_r P_r X P_r$, a positive map. When the spectrum is simple, $\mathcal{D}$ is the set of diagonal matrices. Since $\mathcal{L}_{\eta,\beta}$ multiplies entry (j,k) by $\ell_{jk}(\eta,\beta)$, both $\mathcal{L}_{\eta,\beta}$ and $\mathcal{N}_{\eta,\beta} = \eta\mathcal{L}_{\eta,\beta}^{-1}$ leave $\mathcal{D}$ and $\mathcal{O}$ invariant. On $\mathcal{D}$ the double commutator vanishes and the weight is constant on each eigenspace block,

$$\ell_{jk} = \ell_r \triangleq 2\mu_r\Big(1 - \frac{\eta\mu_r}{2(1+\beta)}\Big) \in \big[2(1-f)\mu_r,\, 2\mu_r\big] \qquad (\lambda_j = \lambda_k = \mu_r), \tag{17}$$

while on $\mathcal{O}$

$$\ell_{jk} = \Big[\lambda_j + \lambda_k - \frac{\eta\lambda_j\lambda_k}{1+\beta}\Big] + \frac{\eta\beta(\lambda_j - \lambda_k)^2}{(1-\beta)^2(1+\beta)} \;\geq\; (1-f)(\lambda_j + \lambda_k) + \frac{\eta\beta\Delta_{\min}^2}{(1-\beta)^2(1+\beta)}, \tag{18}$$

because $\frac{\eta\lambda_j\lambda_k}{1+\beta} = \frac{2f\lambda_j\lambda_k}{\lambda_1} \leq f(\lambda_j + \lambda_k)$ (as $\lambda_j\lambda_k \leq \lambda_1\lambda_k$ and $\lambda_j\lambda_k \leq \lambda_1\lambda_j$). Write the blocks of $\mathcal{V}$ (self-adjoint on $\mathcal{H}$) as $\mathcal{V}_{DD} = \mathcal{P}\mathcal{V}|_{\mathcal{D}}$, $\mathcal{V}_{OD} = (I-\mathcal{P})\mathcal{V}|_{\mathcal{D}}$, $\mathcal{V}_{DO} = \mathcal{V}_{OD}^*$, $\mathcal{V}_{OO} = (I-\mathcal{P})\mathcal{V}|_{\mathcal{O}}$; norms of blocks are operator norms on $\mathcal{H}$ and do not depend on (η,β). For a simple spectrum $\mathcal{V}_{DD}$ is the $P \times P$ matrix

$$\Gamma_{kl} \triangleq \langle E_{kk}, \mathcal{V}(E_{ll})\rangle_F = (\mathcal{V}(E_{ll}))_{kk} = \frac{1}{B}\mathbb{E}\big[(u_k^{\top}\Delta H_i u_l)^2\big] \;\geq 0, \tag{19}$$

symmetric and entrywise non-negative. The *diagonal part* of $\mathcal{K}_{\eta,\beta}$ is

$$\mathcal{K}^D_{\eta,\beta} \triangleq \frac{\eta}{1-\beta}\mathcal{L}_{\eta,\beta}^{-1}\mathcal{P}\mathcal{V}\Big|_{\mathcal{D}} \quad \text{on } \mathcal{D}, \qquad \text{for a simple spectrum the matrix } \frac{\eta}{1-\beta}\operatorname{diag}(1/\ell_{kk})\,\Gamma.$$

By Theorem 9(1), $\mathcal{K}_{\eta,\beta}$ is similar to $\widetilde{\mathcal{K}} = \frac{1}{1-\beta}\mathcal{N}^{1/2}\mathcal{V}\mathcal{N}^{1/2}$, and in a basis adapted to $\mathcal{D}\oplus\mathcal{O}$

$$\widetilde{\mathcal{K}} = \begin{pmatrix} \mathsf{A} & \mathsf{B} \\ \mathsf{B}^* & \mathsf{C} \end{pmatrix}, \quad \mathsf{A} = \tfrac{1}{1-\beta}\mathcal{N}_D^{1/2}\mathcal{V}_{DD}\mathcal{N}_D^{1/2}, \;\; \mathsf{B} = \tfrac{1}{1-\beta}\mathcal{N}_D^{1/2}\mathcal{V}_{DO}\mathcal{N}_O^{1/2}, \;\; \mathsf{C} = \tfrac{1}{1-\beta}\mathcal{N}_O^{1/2}\mathcal{V}_{OO}\mathcal{N}_O^{1/2}, \tag{20}$$

where $\mathcal{N}_D, \mathcal{N}_O$ are the restrictions of $\mathcal{N}_{\eta,\beta}$; A is similar to $\mathcal{K}^D_{\eta,\beta}$.

Commutator-energy coordinates. On $\mathcal{O}$ let D_Δ multiply entry (j,k) by $|\lambda_j - \lambda_k| > 0$, and on $\mathcal{D}$ let $\Lambda_D^{-1/2}(X) \triangleq H^{-1/4}XH^{-1/4}$ (the matrix $\Lambda^{-1/2}$ for a simple spectrum). For $X \in \mathcal{O}$ put $z = D_\Delta X$, i.e. $z_{jk} = |\lambda_j - \lambda_k|X_{jk}$, so that $\|z\|_F^2 = \|[H,X]\|_F^2$. In the Rayleigh denominator of Theorem 9(2) the double-commutator part of the off-diagonal weight is then isotropic:

$$\sum_{\lambda_j\neq\lambda_k} \frac{\ell_{jk}(\eta,\beta)}{\eta} X_{jk}^2 = \sum_{\lambda_j\neq\lambda_k} \Big(\frac{\lambda_j+\lambda_k}{\eta} - \frac{\lambda_j\lambda_k}{1+\beta}\Big)X_{jk}^2 + \frac{\beta}{(1-\beta)^2(1+\beta)}\|z\|_F^2$$

(the factor η sits in ℓ_{jk} itself: $\ell_{jk} \supset \frac{\eta\beta}{(1-\beta)^2(1+\beta)}(\lambda_j - \lambda_k)^2$). Pairs inside one eigenspace stay in $\mathcal{D}$. The constants below are written in these coordinates:

$$\kappa_\Delta \triangleq \big\|D_\Delta^{-1}\mathcal{V}_{OD}\Lambda_D^{-1/2}\big\|, \qquad v_\Delta \triangleq \lambda_{\max}\big(D_\Delta^{-1}\mathcal{V}_{OO}D_\Delta^{-1}\big), \qquad v'_\Delta \triangleq \big\|D_\Delta^{-1}\mathcal{V}_{OO}D_\Delta^{-1}\big\|, \tag{21}$$

and, at a given η, $b_\Delta \triangleq \|\widehat{D}^{1/2}\mathcal{V}_{DO}D_\Delta^{-1}\|$ with $\widehat{D} = \mathcal{L}_{\eta,\beta}^{-1}|_{\mathcal{D}}$. Each pair (j,k) is charged with its own gap; neither $\Delta_{\min}$ nor λ_P is pulled out of the off-diagonal block.

Lemma 17 (Eigenvalue bounds for a symmetric 2×2 block operator)**.** *Let* $M = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}$ *be self-adjoint on* $\mathcal{D}\oplus\mathcal{O}$, $a = \lambda_{\max}(A)$, $c = \lambda_{\max}(C)$, $b = \|B\|$. *Then*

$$a \;\le\; \lambda_{\max}(M) \;\le\; \frac{a+c}{2} + \sqrt{\Big(\frac{a-c}{2}\Big)^2 + b^2},$$

and if $a > c$ *then* $\lambda_{\max}(M) \le a + \frac{b^2}{a-c}$; *in particular* $\lambda_{\max}(M) \le a + b^2/(a - \|C\|)$ *whenever* $a > \|C\|$.

Proof. The lower bound is the Rayleigh quotient of M at $(x,0)$ with x a top eigenvector of A (Cauchy interlacing). Let $\mu = \lambda_{\max}(M)$ with eigenvector (x,y). If $\mu \le c$, the upper bound holds because its right side is at least $\max(a,c)$. If $\mu > c$, then $(\mu - C)$ is invertible, $y = (\mu - C)^{-1}B^*x$, and $Ax + By = \mu x$ gives $\big(A + B(\mu - C)^{-1}B^*\big)x = \mu x$ with $x \neq 0$ (otherwise $y = 0$). Taking the Rayleigh quotient at x and using $(\mu - C)^{-1} \preceq (\mu - c)^{-1}I$, $\mu \le a + \frac{\|B^*x\|^2}{(\mu-c)\|x\|^2} \le a + \frac{b^2}{\mu - c}$, that is $(\mu - a)(\mu - c) \le b^2$, whose largest solution is the stated bound. If $a > c$, then $\mu \ge a > c$ and $\mu - a \le b^2/(\mu - c) \le b^2/(a - c)$. Finally $c \le \|C\|$. □

Theorem 18 (Two-sided bound on $\rho(\mathcal{K}_{\eta,\beta})$ by its diagonal part)**.** *Let* $\beta \in (0,1)$ *and* $\eta \in (0, \eta_{\det})$. *Write* $\widehat{D} = \mathcal{L}_{\eta,\beta}^{-1}|_{\mathcal{D}}$, $r_D \triangleq \lambda_{\max}(\widehat{D}^{1/2}\mathcal{V}_{DD}\widehat{D}^{1/2})$, $\widehat{\omega} \triangleq \max_{\lambda_j \neq \lambda_k} 1/\ell_{jk}(\eta,\beta)$ *and*

$$\varepsilon_{\eta,\beta} \triangleq \frac{(1-\beta)^2(1+\beta)}{\eta\beta\Delta_{\min}^2}.$$

(1) $\mathcal{K}_{\eta,\beta}^D$ *preserves* $\mathcal{D}\cap\mathbb{S}_+^P$, *and* $\rho(\mathcal{K}_{\eta,\beta}^D) = \lambda_{\max}(\mathsf{A}) = \frac{\eta}{1-\beta}r_D$. *With* $a = \rho(\mathcal{K}_{\eta,\beta}^D)$, $b = \|\mathsf{B}\|$ *and* $c = \lambda_{\max}(\mathsf{C}) \le \|\mathsf{C}\|$,

$$\rho(\mathcal{K}_{\eta,\beta}^D) \le \rho(\mathcal{K}_{\eta,\beta}) \le \frac{a+c}{2} + \sqrt{\Big(\frac{a-c}{2}\Big)^2 + b^2}, \qquad \rho(\mathcal{K}_{\eta,\beta}) \le \rho(\mathcal{K}_{\eta,\beta}^D) + \frac{\|\mathsf{B}\|^2}{\rho(\mathcal{K}_{\eta,\beta}^D) - \|\mathsf{C}\|} \;\; \text{if } \rho(\mathcal{K}_{\eta,\beta}^D) > \|\mathsf{C}\|.$$

(2) $\widehat{\omega} \le \varepsilon_{\eta,\beta}$, $\|\mathsf{B}\|^2 \le \big(\frac{\eta}{1-\beta}\big)^2\widehat{\omega}\,\|\widehat{D}^{1/2}\mathcal{V}_{DO}\|^2$ *and* $\|\mathsf{C}\| \le \frac{\eta}{1-\beta}\widehat{\omega}\,\|\mathcal{V}_{OO}\|$. *(3) If* $r_D > \widehat{\omega}\|\mathcal{V}_{OO}\|$, *the prefactors* $\eta/(1-\beta)$ *cancel in the relative error:*

$$0 \le \frac{\rho(\mathcal{K}_{\eta,\beta}) - \rho(\mathcal{K}_{\eta,\beta}^D)}{\rho(\mathcal{K}_{\eta,\beta}^D)} \le \frac{\widehat{\omega}\,\|\widehat{D}^{1/2}\mathcal{V}_{DO}\|^2}{r_D\big(r_D - \widehat{\omega}\|\mathcal{V}_{OO}\|\big)}. \tag{22}$$

With $\rho_V \triangleq \lambda_{\max}(\mathcal{V}_{DD}) \ge \frac{\mathbb{E}\|\Delta H_i\|_F^2}{BP} > 0$, $r_D \ge \rho_V/(2\lambda_1)$ *and* $\|\widehat{D}^{1/2}\mathcal{V}_{DO}\|^2 \le \|\mathcal{V}_{DO}\|^2/(2(1-f)\lambda_P)$, *so that whenever* $2\lambda_1\varepsilon_{\eta,\beta}\|\mathcal{V}_{OO}\| < \rho_V$,

$$0 \le \frac{\rho(\mathcal{K}_{\eta,\beta}) - \rho(\mathcal{K}_{\eta,\beta}^D)}{\rho(\mathcal{K}_{\eta,\beta}^D)} \le \frac{2\lambda_1^2\,\varepsilon_{\eta,\beta}\,\|\mathcal{V}_{DO}\|^2}{(1-f)\lambda_P\,\rho_V\big(\rho_V - 2\lambda_1\varepsilon_{\eta,\beta}\|\mathcal{V}_{OO}\|\big)}.$$

(3′) (Commutator-energy form.) If $\eta\beta r_D > (1-\beta)^2(1+\beta)v'_\Delta$, *then*

$$0 \le \frac{\rho(\mathcal{K}_{\eta,\beta}) - \rho(\mathcal{K}_{\eta,\beta}^D)}{\rho(\mathcal{K}_{\eta,\beta}^D)} \le \frac{(1+\beta)(1-\beta)^2\,b_\Delta^2}{\beta\,\eta\,r_D\big(r_D - (1-\beta)^2(1+\beta)v'_\Delta/(\beta\eta)\big)}.$$

(4) At fixed f, $\eta = 2f(1+\beta)/\lambda_1$ and $\varepsilon_{\eta,\beta} = \frac{(1-\beta)^2\lambda_1}{2f\beta\Delta_{\min}^2}$, so as $\beta \to 1$

$$\rho(\mathcal{K}^D_{\eta,\beta}) \geq \frac{f\kappa_\beta\rho_V}{\lambda_1^2} \to \infty, \qquad \frac{\rho(\mathcal{K}_{\eta,\beta}) - \rho(\mathcal{K}^D_{\eta,\beta})}{\rho(\mathcal{K}^D_{\eta,\beta})} = O\Big(\frac{(1-\beta)^2}{\beta\Delta_{\min}^2}\Big), \qquad \rho(\mathcal{K}_{\eta,\beta}) - \rho(\mathcal{K}^D_{\eta,\beta}) = O\Big(\frac{1-\beta}{\beta\Delta_{\min}^2}\Big),$$

with the constants of (3) or (3′); with (3′) the relative error is at most $\frac{(1-\beta)^2\lambda_1 b_\Delta^2}{2f\beta\, r_D\,(r_D - (1-\beta)^2\lambda_1 v'_\Delta/(2f\beta))}$, where b_Δ and r_D stay bounded at fixed f and no $\Delta_{\min}$ appears. (5) If $\mathcal{V}_{DO} = 0$ ($\mathcal{V}$ maps $\mathcal{D}$ into $\mathcal{D}$), then $\rho(\mathcal{K}_{\eta,\beta}) = \max\big(\rho(\mathcal{K}^D_{\eta,\beta}), \lambda_{\max}(\mathsf{C})\big)$ exactly, for every $\beta \in [0,1)$.

Proof. (1) On $\mathcal{D}$, $\mathcal{L}^{-1}$ multiplies the block $P_r X P_r$ by $1/\ell_r > 0$ by (17), so $\mathcal{K}^D$ is a composition of the positive maps $\mathcal{V}$, $\mathcal{P}$ and a positive blockwise scaling, and preserves $\mathcal{D} \cap \mathbb{S}^P_+$, a proper cone in $\mathcal{D}$. It is similar to the self-adjoint A (through $\mathcal{N}_D^{1/2}$), so its spectrum is real, and by the Krein–Rutman theorem in finite dimension its spectral radius is an eigenvalue, hence $\rho(\mathcal{K}^D) = \lambda_{\max}(\mathsf{A})$; $\mathsf{A} = \frac{\eta}{1-\beta}\widehat{D}^{1/2}\mathcal{V}_{DD}\widehat{D}^{1/2}$ gives the last expression. By Theorem 9(1), $\rho(\mathcal{K}) = \lambda_{\max}(\widetilde{\mathcal{K}})$, and Lemma 17 applied to (20) gives the bounds. (2) The first claim is (18). $\mathcal{N}_O = \eta\,\mathcal{L}^{-1}|_{\mathcal{O}}$ is diagonal in the basis $\{F_{jk}\}_{\lambda_j \neq \lambda_k}$ with entries at most $\eta\widehat{\omega}$, so $\|\mathcal{N}_O^{1/2}\| \leq (\eta\widehat{\omega})^{1/2}$ and $\|\mathcal{N}_O\| \leq \eta\widehat{\omega}$; with $\mathcal{N}_D^{1/2} = \eta^{1/2}\widehat{D}^{1/2}$ the two bounds follow from submultiplicativity. (3) Insert (2) and $a = \frac{\eta}{1-\beta}r_D$ in the second bound of (1) and divide by a. For the explicit form: $I \in \mathcal{D}$ and $\langle I, \mathcal{V}(I)\rangle_F = \frac{1}{B}\mathbb{E}\|\Delta H_i\|_F^2$, so $\rho_V \geq \frac{1}{B}\mathbb{E}\|\Delta H_i\|_F^2/\|I\|_F^2$. If y is a unit top eigenvector of $\mathcal{V}_{DD}$ and $x = \widehat{D}^{-1/2}y$, the Rayleigh quotient of $\widehat{D}^{1/2}\mathcal{V}_{DD}\widehat{D}^{1/2}$ at x is $\rho_V/\langle y, \widehat{D}^{-1}y\rangle \geq \rho_V \min\widehat{D}$, and $\min\widehat{D} \geq 1/(2\lambda_1)$, $\max\widehat{D} \leq 1/(2(1-f)\lambda_P)$ by (17); the right side of (22) is increasing in $\widehat{\omega}$ and $\widehat{\omega} \leq \varepsilon_{\eta,\beta}$. (3′) By (18), $\mathcal{N}_O$ multiplies entry (j,k) by $\eta/\ell_{jk} \leq q\,(\lambda_j - \lambda_k)^{-2}$ with $q = \frac{(1-\beta)^2(1+\beta)}{\beta}$. For diagonal $0 \leq \mathsf{a} \leq \mathsf{b}$ entrywise and any M, $\|M\mathsf{a}\| = \|M\mathsf{b}(\mathsf{a}/\mathsf{b})\| \leq \|M\mathsf{b}\|$ and $\|\mathsf{a}M\mathsf{a}\| \leq \|\mathsf{b}M\mathsf{b}\|$, so $\|\mathsf{B}\|^2 \leq \frac{\eta q}{(1-\beta)^2}b_\Delta^2 = \frac{\eta(1+\beta)}{\beta}b_\Delta^2$ and $\|\mathsf{C}\| \leq \frac{q}{1-\beta}v'_\Delta = \frac{1-\beta^2}{\beta}v'_\Delta$. Insert in the second bound of (1) with $a = \frac{\eta}{1-\beta}r_D$ and divide by a. (4) Substitute $\eta = 2f(1+\beta)/\lambda_1$ in (3) or (3′); $\rho(\mathcal{K}^D) = \frac{\eta}{1-\beta}r_D \geq \frac{2f(1+\beta)}{\lambda_1(1-\beta)}\frac{\rho_V}{2\lambda_1}$. The absolute error is the relative one times $\rho(\mathcal{K}^D) \leq \frac{\eta}{1-\beta}\frac{\|\mathcal{V}_{DD}\|}{2(1-f)\lambda_P} = O(\kappa_\beta)$. (5) With $\mathsf{B} = 0$, $\widetilde{\mathcal{K}}$ is block diagonal. □

At fixed f the stationary regime is long lost as $\beta \to 1$ ($\rho(\mathcal{K}) \to \infty$), so (4) is a statement about the operator, not about the threshold. The threshold itself sits at $\eta = \Theta(1-\beta)$, where $\varepsilon_{\eta,\beta} = \Theta(1-\beta)$ and the relative error loses one power of $1-\beta$. The cleanest way to see it is the η-free form of Theorem 9(3): with $\mathcal{L}_0(X) = HX + XH$ and $\mathcal{B}_\beta$ as there, let

$$\mathcal{G}_\beta \triangleq \mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}, \qquad \widetilde{\mathcal{V}}(X) \triangleq \mathcal{P}\big(H^{-1/4}\mathcal{V}(H^{-1/4}XH^{-1/4})H^{-1/4}\big) \text{ on } \mathcal{D}, \qquad \mathcal{H}_D(X) \triangleq HX \text{ on } \mathcal{D}.$$

For a simple spectrum $\widetilde{\mathcal{V}}$ is the matrix $\widetilde{\Gamma} = \Lambda^{-1/2}\Gamma\Lambda^{-1/2}$, $\widetilde{\Gamma}_{kl} = \frac{\mathbb{E}[(u_k^\top \Delta H_i u_l)^2]}{B\sqrt{\lambda_k\lambda_l}}$, and $\mathcal{H}_D = \Lambda$. In the splitting $\mathcal{D} \oplus \mathcal{O}$,

$$\mathcal{G}_\beta = \begin{pmatrix} \frac{1}{2(1+\beta)}\big(\mathcal{H}_D + \kappa_\beta\widetilde{\mathcal{V}}\big) & \frac{1}{1-\beta}\mathcal{L}_{0,D}^{-1/2}\mathcal{V}_{DO}\mathcal{L}_{0,O}^{-1/2} \\ \frac{1}{1-\beta}\mathcal{L}_{0,O}^{-1/2}\mathcal{V}_{OD}\mathcal{L}_{0,D}^{-1/2} & \mathcal{S}_O + \frac{1}{1-\beta}\mathcal{L}_{0,O}^{-1/2}\mathcal{V}_{OO}\mathcal{L}_{0,O}^{-1/2} \end{pmatrix}, \tag{23}$$

where $\mathcal{S}_O$ multiplies entry (j,k), $\lambda_j \neq \lambda_k$, by $\big[\frac{\lambda_j\lambda_k}{1+\beta} - \frac{\beta(\lambda_j-\lambda_k)^2}{(1-\beta)^2(1+\beta)}\big]/(\lambda_j+\lambda_k)$. (On $\mathcal{D}$, $\mathcal{L}_0$ multiplies block r by $2\mu_r$, so $\mathcal{L}_{0,D}^{-1/2} = 2^{-1/2}H^{-1/4}\cdot H^{-1/4}$ and $\mathcal{L}_{0,D}^{-1/2}\big(\frac{H\cdot H}{1+\beta} + \frac{\mathcal{P}\mathcal{V}}{1-\beta}\big)\mathcal{L}_{0,D}^{-1/2} = \frac{1}{2(1+\beta)}(\mathcal{H}_D + \kappa_\beta\widetilde{\mathcal{V}})$.) Write $\mathcal{G}_{DD}, \mathcal{G}_{DO}, \mathcal{G}_{OO}$ for the blocks, $g_D \triangleq \lambda_{\max}(\mathcal{G}_{DD})$, and define the *diagonal threshold*

$$\eta_D(\beta) \triangleq \frac{1}{g_D} = \frac{2(1+\beta)}{\lambda_{\max}\big(\mathcal{H}_D + \kappa_\beta\widetilde{\mathcal{V}}\big)} \qquad \Big(= \frac{2(1+\beta)}{\lambda_{\max}(\Lambda + \kappa_\beta\widetilde{\Gamma})} \text{ for a simple spectrum}\Big). \tag{24}$$

Proposition 19 (The diagonal threshold as an M-matrix condition; reduction to Theorems 5 and 8)**.** *Let $\beta \in [0,1)$. (1) $\eta_D \le \eta_{\det}$ and, for $\eta \in (0,\eta_{\det})$, $\rho(\mathcal{K}^D_{\eta,\beta}) < 1 \iff \eta < \eta_D$. For a simple spectrum the following are equivalent: $\rho(\mathcal{K}^D_{\eta,\beta}) < 1$; $\operatorname{diag}(\ell_{kk}) - \frac{\eta}{1-\beta}\Gamma$ is a non-singular M-matrix; $I - \frac{\eta}{2(1+\beta)}\big(\Lambda + \kappa_\beta \widetilde{\Gamma}\big)$ is a non-singular M-matrix; $\eta < \eta_D$. In terms of the normalized step variances $c_k = \frac{2\lambda_k}{\eta(1+\beta)}\gamma_{0,k}$, the diagonal equations read $\big[I - \frac{\eta}{2(1+\beta)}(\Lambda + \kappa_\beta \Gamma \Lambda^{-1})\big]c = \frac{\sigma_\xi^2}{1-\beta^2} +$ (terms in the off-diagonal of Σ_0), the last vanishing when $\mathcal{V}_{DO} = 0$. (2) If $\widetilde{\Gamma} = \operatorname{diag}(\gamma_k) + gg^\top$ with $g_k > 0$, then $\eta_D = 2(1+\beta)/\mu_\star$, where $\mu_\star$ is the unique $\mu > \max\big(0, \max_k(\lambda_k + \kappa_\beta\gamma_k)\big)$ at which (25) holds with $\eta = 2(1+\beta)/\mu$:*

$$\frac{\eta}{2(1-\beta)} \sum_{k=1}^{P} \frac{g_k^2}{1 - \frac{\eta\lambda_k}{2(1+\beta)} - \frac{\eta\gamma_k}{2(1-\beta)}} = 1. \tag{25}$$

For the random features of Theorem 5, $\Gamma_{kl} = \frac{1}{B}\big(\lambda_k\lambda_l + (1+\kappa_{\text{feat}})\lambda_k^2\delta_{kl}\big)$, so $g_k^2 = \lambda_k/B$, $\gamma_k = (1+\kappa_{\text{feat}})\lambda_k/B$, and (25) is (4); for the features of Theorem 8, $\Gamma_{kl} = \frac{1}{B}\big(s_k s_l + (1+\kappa_{\text{feat}})s_k^2\delta_{kl}\big)$, $g_k^2 = s_k^2/(B\lambda_k)$, $\gamma_k = (1+\kappa_{\text{feat}})s_k^2/(B\lambda_k)$, and (25) is the equation of Theorem 8(2). In both cases $\mathcal{V}_{DO} = 0$.

Proof. (1) $\lambda_{\max}(\mathcal{H}_D + \kappa_\beta\widetilde{\mathcal{V}}) \ge \langle E_{11}, (\mathcal{H}_D + \kappa_\beta\widetilde{\mathcal{V}})E_{11}\rangle_F \ge \lambda_1$ because $\widetilde{\mathcal{V}}$ is a positive map, so $\eta_D \le \eta_{\det}$. As in Theorem 9(1), $\rho(\mathcal{K}^D) < 1 \iff \lambda_{\max}(\mathsf{A}) < 1 \iff \mathcal{L}_D - \frac{\eta}{1-\beta}\mathcal{V}_{DD} \succ 0$ on $\mathcal{D}$; on $\mathcal{D}$, $\mathcal{L}_D - \frac{\eta}{1-\beta}\mathcal{V}_{DD} = \mathcal{L}_{0,D}^{1/2}\big(I - \eta\mathcal{G}_{DD}\big)\mathcal{L}_{0,D}^{1/2}$ (the commutator is absent on $\mathcal{D}$), which is positive definite iff $\eta g_D < 1$. For a simple spectrum, $\mathcal{K}^D = \frac{\eta}{1-\beta}\operatorname{diag}(1/\ell_{kk})\Gamma \ge 0$ entrywise and $\operatorname{diag}(\ell_{kk}) - \frac{\eta}{1-\beta}\Gamma = \operatorname{diag}(\ell_{kk})(I - \mathcal{K}^D)$ is a Z-matrix that is a non-singular M-matrix iff $\rho(\mathcal{K}^D) < 1$ [Berman and Plemmons, 1994, Ch. 6]; the identity

$$\operatorname{diag}(\ell_{kk}) - \frac{\eta}{1-\beta}\Gamma = 2\Lambda^{1/2}\Big[I - \frac{\eta}{2(1+\beta)}\big(\Lambda + \kappa_\beta\widetilde{\Gamma}\big)\Big]\Lambda^{1/2}$$

(use $\ell_{kk} = 2\lambda_k - \eta\lambda_k^2/(1+\beta)$ and $\frac{\eta\kappa_\beta}{1+\beta} = \frac{\eta}{1-\beta}$) transfers the property to the symmetric Z-matrix in brackets, which is a non-singular M-matrix iff it is positive definite. The equation for c is the (k,k) entry of Theorem 2(2) divided by $\eta(1+\beta)$, as in the proof of Theorem 5; $(\mathcal{V}(\Sigma_0))_{kk} = \sum_l \Gamma_{kl}\gamma_{0,l} + \langle E_{kk}, \mathcal{V}_{DO}(\Sigma_0)_{\mathcal{O}}\rangle_F$. (2) By the rank-one determinant lemma, $\det(\mu I - \Lambda - \kappa_\beta \operatorname{diag}(\gamma) - \kappa_\beta gg^\top) = \prod_k(\mu - \lambda_k - \kappa_\beta\gamma_k)\big(1 - \kappa_\beta\sum_k g_k^2/(\mu - \lambda_k - \kappa_\beta\gamma_k)\big)$; the sum decreases from $+\infty$ to 0 on $(\max_k(\lambda_k + \kappa_\beta\gamma_k), \infty)$ ($g_k > 0$ for every k), and by interlacing the top eigenvalue lies there, so it is the unique root of the bracket on that interval. At $\mu = 2(1+\beta)/\eta$, multiplying numerator and denominator by $\eta/(2(1+\beta))$ and using $\frac{\eta\kappa_\beta}{2(1+\beta)} = \frac{\eta}{2(1-\beta)}$ gives (25). The entries of Γ are $(\mathcal{V}(E_{ll}))_{kk}$ from the formulas for $\mathcal{V}$ in the proofs of Theorems 5 and 8; there $(\mathcal{V}(X))_{jk} = \frac{s_j s_k}{B}X_{jk}$ for $j \ne k$ (with $s = \lambda$ in Theorem 5), which vanishes for diagonal X, so $\mathcal{V}_{DO} = 0$. With $g_k^2 = \lambda_k/B$ and $\gamma_k = (1+\kappa_{\text{feat}})\lambda_k/B$, the denominator of (25) is $1 - \frac{\eta\lambda_k}{2(1+\beta)}\alpha_{\text{diag}}$. □

Theorem 20 (Momentum diagonalizes the threshold)**.** *Let $\beta \in (0,1)$ and $\eta^\star = \eta_{\text{crit}}^{\text{EoSS},P}(\beta, B)$ be the mean-square ceiling of Theorem 9. Then $\eta^\star = 1/\lambda_{\max}(\mathcal{G}_\beta)$, $\eta^\star \le \eta_D$, and the gap $1/\eta^\star - 1/\eta_D \ge 0$ has the following upper bounds, listed from the tightest; each holds whenever its condition holds. (1) (Eigenvector split; the tightest of the bounds.) Let u be a unit top eigenvector of $\mathcal{G}_{DD}$, $\gamma_D \triangleq g_D - \lambda_2(\mathcal{G}_{DD})$ its spectral gap ($\gamma_D = +\infty$ if $\dim\mathcal{D} = 1$), $P_\perp = I - u\langle u, \cdot\rangle$ on $\mathcal{D}$, and $\mathsf{E} \in \{\mathsf{E}_\Delta, \bar{\mathsf{E}}\}$ one of the majorants of (2) (defined when $\theta_\beta < 1$, resp. $\bar{\theta} < 1$). If*

$$\gamma_D + \langle u, \mathsf{E}u\rangle > \|P_\perp \mathsf{E} P_\perp\|, \tag{26}$$

then

$$0 \le \frac{1}{\eta^\star} - \frac{1}{\eta_D} \le c''_\beta \triangleq \langle u, \mathsf{E}u\rangle + \frac{\|P_\perp \mathsf{E}u\|^2}{\gamma_D + \langle u, \mathsf{E}u\rangle - \|P_\perp \mathsf{E} P_\perp\|}.$$

If $\lambda_{\max}(\widetilde{\mathcal{V}})$ *is simple with gap* $\operatorname{gap}(\widetilde{\mathcal{V}}) > 0$*, then* $\gamma_D \ge \frac{\operatorname{gap}(\widetilde{\mathcal{V}})}{2(1-\beta)} - \frac{\lambda_1}{1+\beta}$*, so* (26) *holds for all* β *close enough to* 1*, and with* $\mathsf{E} = \mathsf{E}_\Delta$*,* $c''_\beta = c_\infty + O(1-\beta)$*: the leading term is exact and the bound is asymptotically sharp. The flat and weakly coupled directions of* A *enter only through* $\|P_\perp \mathsf{E}u\|^2/\gamma_D = O(1-\beta)$*. (2) (Loewner majorants of the Schur term; commutator-energy form.) If* $\theta_\beta \triangleq (1-\beta^2)v_\Delta/\beta < 1$*,*

$$0 \le \frac{1}{\eta^\star} - \frac{1}{\eta_D} \le c'_\beta \triangleq \frac{1+\beta}{2\beta}\,\frac{\kappa_\Delta^2}{1-\theta_\beta} \xrightarrow[\beta\to 1]{} \kappa_\Delta^2 = \big\|D_\Delta^{-1}\mathcal{V}_{OD}\Lambda_D^{-1/2}\big\|^2.$$

A weighted form, defined whenever the Schur step is: with $s_{jk} = \lambda_j + \lambda_k$*,* $\bar{c}_{jk} \triangleq \frac{\beta(\lambda_j-\lambda_k)^2}{(1-\beta)^2(1+\beta)s_{jk}} + g_D - \frac{\lambda_j\lambda_k}{(1+\beta)s_{jk}}$*,* $\bar{\mathcal{C}} = \operatorname{diag}(\bar{c}_{jk})$ *on* $\mathcal{O}$*,* $\mathcal{E}_{OO} = \frac{1}{1-\beta}\mathcal{L}_{0,O}^{-1/2}\mathcal{V}_{OO}\mathcal{L}_{0,O}^{-1/2}$ *and* $\bar{\theta} = \lambda_{\max}(\bar{\mathcal{C}}^{-1/2}\mathcal{E}_{OO}\bar{\mathcal{C}}^{-1/2}) < 1$*,* $\frac{1}{\eta^\star} - \frac{1}{\eta_D} \le \frac{\|\bar{\mathcal{C}}^{-1/2}\mathcal{G}_{OD}\|^2}{1-\bar{\theta}}$*. In operator form:* $\mathsf{E}_\Delta \triangleq \alpha_\beta A^* A$ *with* $A = D_\Delta^{-1}\mathcal{V}_{OD}\Lambda_D^{-1/2}$ *and* $\alpha_\beta = \frac{1+\beta}{2\beta(1-\theta_\beta)}$*, and* $\bar{\mathsf{E}} \triangleq \frac{1}{1-\bar{\theta}}\mathcal{G}_{DO}\bar{\mathcal{C}}^{-1}\mathcal{G}_{OD}$*, both positive semidefinite on* $\mathcal{D}$*, satisfy* $0 \preceq \mathcal{G}_{DO}(\mu_\eta^\star - \mathcal{G}_{OO})^{-1}\mathcal{G}_{OD} \preceq \mathsf{E}$ *at* $\mu_\eta^\star = 1/\eta^\star$*, and the two scalar bounds are* $c'_\beta = \|\mathsf{E}_\Delta\|$ *and* $\|\bar{\mathsf{E}}\|$*. Two further majorants: let* $T = \bar{\mathcal{C}}^{-1/2}\mathcal{E}_{OO}\bar{\mathcal{C}}^{-1/2}$ *and* T_+ *its positive part (*$\mathcal{V}_{OO}$*, hence* T*, need not be positive semidefinite, and* T *cannot replace* T_+*). Since* $\frac{1}{1-t} \le 1 + \frac{t_+}{1-\theta}$ *for* $t \le \bar{\theta} < 1$*, and* $1 + \frac{t_+}{1-\theta} \le \frac{1}{1-\bar{\theta}}$ *when moreover* $\bar{\theta} \ge 0$*,* $\bar{\mathsf{E}}_1 \triangleq \mathcal{G}_{DO}\bar{\mathcal{C}}^{-1}\mathcal{G}_{OD} + \frac{1}{1-\bar{\theta}}\mathcal{G}_{DO}\bar{\mathcal{C}}^{-1/2}T_+\bar{\mathcal{C}}^{-1/2}\mathcal{G}_{OD}$ *satisfies* $\mathcal{G}_{DO}(\mu_\eta^\star - \mathcal{G}_{OO})^{-1}\mathcal{G}_{OD} \preceq \bar{\mathsf{E}}_1$*, with* $\bar{\mathsf{E}}_1 \preceq \bar{\mathsf{E}}$ *when* $\bar{\theta} \ge 0$ *(for* $\bar{\theta} < 0$ *the order can fail); and whenever* $g_D > \lambda_{\max}(\mathcal{G}_{OO})$*,* $\mu_\eta^\star \ge g_D$ *gives* $\mathcal{G}_{DO}(\mu_\eta^\star - \mathcal{G}_{OO})^{-1}\mathcal{G}_{OD} \preceq \mathsf{E}_{\mathrm{op}} \triangleq \mathcal{G}_{DO}(g_D - \mathcal{G}_{OO})^{-1}\mathcal{G}_{OD}$*. Both may be used as* E *in (1); over the 42 cells of* `paper/checks/diagonalization-burnin.py` *(section S8) the ratio of (1) to the true gap is* 1–17.3 *with* $\bar{\mathsf{E}}_1$ *and* 1–18.2 *with* E_{op}*, against* 1–17.5 *with* $\bar{\mathsf{E}}$*; the largest ratios come from a small* γ_D*, not from the scalar factor. (3) (Operator form.) Whenever* $g_D > \lambda_{\max}(\mathcal{G}_{OO})$*,*

$$0 \le \frac{1}{\eta^\star} - \frac{1}{\eta_D} \le \frac{\|\mathcal{G}_{DO}\|^2}{g_D - \lambda_{\max}(\mathcal{G}_{OO})}.$$

(4) ($\Delta_{\min}$*,* λ_P *form.)* $\|\mathcal{G}_{DO}\| \le \frac{\|\mathcal{V}_{DO}\|}{2\lambda_P(1-\beta)}$ *and* $g_D - \lambda_{\max}(\mathcal{G}_{OO}) \ge \frac{\beta\Delta_{\min}^2}{2\lambda_1(1-\beta)^2(1+\beta)} - \frac{\|\mathcal{V}_{OO}\|}{2\lambda_P(1-\beta)}$*. Hence, if* $\beta\Delta_{\min}^2 > (1-\beta^2)\lambda_1\|\mathcal{V}_{OO}\|/\lambda_P$*,*

$$0 \le \frac{1}{\eta^\star} - \frac{1}{\eta_D} \le c_\beta \triangleq \frac{\lambda_1(1+\beta)\|\mathcal{V}_{DO}\|^2}{2\lambda_P^2\big(\beta\Delta_{\min}^2 - (1-\beta^2)\lambda_1\|\mathcal{V}_{OO}\|/\lambda_P\big)},$$

a quantity that stays bounded as $\beta \to 1$*. (5)* $\frac{1}{\eta_D} \in \Big[\frac{\lambda_{\max}(\widetilde{\mathcal{V}})}{2(1-\beta)},\ \frac{\lambda_{\max}(\widetilde{\mathcal{V}})}{2(1-\beta)} + \frac{\lambda_1}{2(1+\beta)}\Big]$ *with* $\lambda_{\max}(\widetilde{\mathcal{V}}) \ge \frac{\mathbb{E}\|\Delta H_i\|_F^2}{B\,\mathrm{tr}\,H} > 0$*. Consequently*

$$1 \le \frac{\eta_D}{\eta^\star} \le 1 + \frac{2(1-\beta)c_\beta^{\mathrm{best}}}{\lambda_{\max}(\widetilde{\mathcal{V}})}, \qquad \eta^\star = \frac{2(1-\beta)}{\lambda_{\max}(\widetilde{\mathcal{V}})}\big(1 + O(1-\beta)\big) \quad (\beta \to 1),$$

with c_β^{best} *the least of the defined bounds (1)–(4), so the relative error of the diagonal threshold is* $\frac{2(1-\beta)c_\infty}{\lambda_{\max}(\widetilde{\mathcal{V}})}(1 + o(1))$ *by Proposition 21, and (1) attains this to first order (at most* $O\big((1-\beta)\kappa_\Delta^2/\lambda_{\max}(\widetilde{\mathcal{V}})\big)$ *by (2)) (and* $O\big((1-\beta)/(\beta\Delta_{\min}^2)\big)$ *in the cruder form), one power of* $1-\beta$ *less*

than the operator statement of Theorem 18(4), because $\eta^\star = \Theta(1-\beta)$. *(6) If* $\mathcal{V}_{DO} = 0$, *then* $1/\eta^\star = \max\big(g_D, \lambda_{\max}(\mathcal{G}_{OO})\big)$, *and* $\eta^\star = \eta_D$ *as soon as* $\lambda_{\max}(\mathcal{G}_{OO}) \le g_D$, *in particular for all* β *close enough to* 1 *when* $R \ge 2$ *(and in the settings of Theorems 5 and 8 for every* β, *by those theorems).*

Proof. For $\eta \in (0, \eta_{\det})$, Theorem 9 gives $\rho(\mathcal{T}) < 1 \iff \rho(\mathcal{K}) < 1 \iff \mathcal{L}_0 - \eta\mathcal{B}_\beta \succ 0 \iff \eta\lambda_{\max}(\mathcal{G}_\beta) < 1$ (note $\lambda_{\max}(\mathcal{G}_\beta) \ge g_D > 0$), and $\rho(\mathcal{T}) \ge 1$ for $\eta \ge \eta_{\det}$ (there $\rho(\mathcal{T}) \ge \rho(\mathcal{T}_0) = \rho(\bar{F})^2 \ge 1$, because $\mathcal{T} - \mathcal{T}_0$ is positive, see Section B.2, and the spectral radius is monotone on positive maps of a proper cone). Since $\lambda_{\max}(\mathcal{G}_\beta) \ge g_D \ge 1/\eta_{\det}$ (Proposition 19(1)), the stable set is exactly $(0, 1/\lambda_{\max}(\mathcal{G}_\beta))$. The lower bound $1/\eta^\star \ge g_D$ is Lemma 17. (3) is Lemma 17 applied to (23). (4) $\mathcal{L}_{0,D} \succeq 2\lambda_P I$ and $\mathcal{L}_{0,O} \succeq 2\lambda_P I$, which gives the first bound. On $\mathcal{O}$, $\frac{\lambda_j\lambda_k}{\lambda_j+\lambda_k} \le \frac{\lambda_1}{2}$ and $\lambda_j+\lambda_k \le 2\lambda_1$, so $\mathcal{S}_O \preceq \big(\frac{\lambda_1}{2(1+\beta)} - \frac{\beta\Delta_{\min}^2}{2\lambda_1(1-\beta)^2(1+\beta)}\big)I$, and Weyl's inequality gives $\lambda_{\max}(\mathcal{G}_{OO}) \le \frac{\lambda_1}{2(1+\beta)} - \frac{\beta\Delta_{\min}^2}{2\lambda_1(1-\beta)^2(1+\beta)} + \frac{\|\mathcal{V}_{OO}\|}{2\lambda_P(1-\beta)}$, while $g_D \ge \frac{\lambda_1}{2(1+\beta)}$ by Proposition 19(1). Dividing $\|\mathcal{G}_{DO}\|^2$ by the difference gives c_β. (2) Let $\mu = \lambda_{\max}(\mathcal{G}_\beta)$. Since $\mu \ge g_D \ge \frac{\lambda_1}{2(1+\beta)} \ge \frac{\lambda_j\lambda_k}{(1+\beta)s_{jk}}$, $\mu - \mathcal{S}_O = \bar{\mathcal{C}} + (\mu - g_D)I \succeq \bar{\mathcal{C}} \succeq \mathcal{C}$, where $\mathcal{C}$ multiplies entry (j,k) by $c_{jk} = \frac{\beta(\lambda_j-\lambda_k)^2}{(1-\beta)^2(1+\beta)s_{jk}}$. If $\mathcal{E}_{OO} \preceq \theta\,\mathcal{C}'$ for $\mathcal{C}' \in \{\mathcal{C}, \bar{\mathcal{C}}\}$ with $\theta < 1$, then $\mu - \mathcal{G}_{OO} \succeq \mathcal{C}' - \mathcal{E}_{OO} \succeq (1-\theta)\mathcal{C}' \succ 0$, so $\mu > \lambda_{\max}(\mathcal{G}_{OO})$ and, as in the proof of Lemma 17, μ is an eigenvalue of $\mathcal{G}_{DD} + \mathcal{G}_{DO}(\mu - \mathcal{G}_{OO})^{-1}\mathcal{G}_{OD} \preceq \mathcal{G}_{DD} + \frac{1}{1-\theta}\mathcal{G}_{DO}\mathcal{C}'^{-1}\mathcal{G}_{OD}$, whence $\mu \le g_D + \frac{\|\mathcal{C}'^{-1/2}\mathcal{G}_{OD}\|^2}{1-\theta}$. For $\mathcal{C}' = \bar{\mathcal{C}}$ this is the weighted form. For $\mathcal{C}' = \mathcal{C}$: $\mathcal{C}^{-1/2}\mathcal{L}_{0,O}^{-1/2} = \big(\frac{(1-\beta)^2(1+\beta)}{\beta}\big)^{1/2} D_\Delta^{-1}$, so the least admissible θ is $\frac{1}{1-\beta}\frac{(1-\beta)^2(1+\beta)}{\beta} v_\Delta = \theta_\beta$, and $\mathcal{C}^{-1/2}\mathcal{G}_{OD} = \big(\frac{1+\beta}{\beta}\big)^{1/2} D_\Delta^{-1}\mathcal{V}_{OD}\mathcal{L}_{0,D}^{-1/2}$ with $\mathcal{L}_{0,D}^{-1/2} = 2^{-1/2}\Lambda_D^{-1/2}$, so $\|\mathcal{C}^{-1/2}\mathcal{G}_{OD}\|^2 = \frac{1+\beta}{2\beta}\kappa_\Delta^2$ and $\frac{1}{1-\theta_\beta}\mathcal{G}_{DO}\mathcal{C}^{-1}\mathcal{G}_{OD} = \mathsf{E}_\Delta$. Both majorants are positive semidefinite, as is the Schur term itself ($\mu - \mathcal{G}_{OO} \succ 0$). (1) By the proof of (2), μ is an eigenvalue of $\mathcal{G}_{DD} + \mathsf{S}$ with $0 \preceq \mathsf{S} \preceq \mathsf{E}$, hence $\mu \le \lambda_{\max}(\mathcal{G}_{DD} + \mathsf{E})$. Write $M = \mathcal{G}_{DD} + \mathsf{E}$ in the splitting $\mathcal{D} = \operatorname{span}(u) \oplus u^\perp$. Since $\mathcal{G}_{DD}u = g_D u$, the (u,u) entry is $a = g_D + \langle u, \mathsf{E}u\rangle$, the off-diagonal block is $P_\perp \mathsf{E}u$ (as $P_\perp\mathcal{G}_{DD}u = 0$), and the $u^\perp$ block is $P_\perp(\mathcal{G}_{DD} + \mathsf{E})P_\perp$, whose top eigenvalue is at most $\lambda_2(\mathcal{G}_{DD}) + \|P_\perp\mathsf{E}P_\perp\| = g_D - \gamma_D + \|P_\perp\mathsf{E}P_\perp\| \triangleq \bar{c}$ by Weyl's inequality. Under (26), $a > \bar{c}$, and the proof of Lemma 17 ($\mu - a \le b^2/(\mu - c) \le b^2/(a - \bar{c})$ for $c \le \bar{c} < a \le \mu$) gives $\lambda_{\max}(M) \le a + \|P_\perp\mathsf{E}u\|^2/(a - \bar{c})$, which is the bound. For the asymptotics: $\mathcal{G}_{DD} = \frac{1}{2(1-\beta)}\widetilde{\mathcal{V}} + \frac{1}{2(1+\beta)}\mathcal{H}_D$ with $0 \preceq \mathcal{H}_D \preceq \lambda_1 I$, so by Weyl $\gamma_D \ge \frac{\operatorname{gap}(\widetilde{\mathcal{V}})}{2(1-\beta)} - \frac{\lambda_1}{1+\beta}$, and by the Davis–Kahan theorem $u = X_\infty + O(1-\beta)$ (up to sign). Since $\theta_\beta \to 0$ and $\alpha_\beta \to 1$, $\langle u, \mathsf{E}_\Delta u\rangle = \alpha_\beta\|Au\|^2 = \|AX_\infty\|^2 + O(1-\beta) = c_\infty + O(1-\beta)$ (Proposition 21), $\|\mathsf{E}_\Delta\| = O(1)$, and the second term is $O(1-\beta)$. (5) $0 \preceq \mathcal{H}_D \preceq \lambda_1 I$ on $\mathcal{D}$ and Weyl's inequality give the bracket for $g_D = 1/\eta_D$ (with $\frac{\kappa_\beta}{2(1+\beta)} = \frac{1}{2(1-\beta)}$). $\widetilde{\mathcal{V}}$ is a positive self-adjoint map on $\mathcal{D}$; its Rayleigh quotient at $H^{1/2} \in \mathcal{D}$ is $\operatorname{tr}(\mathcal{V}(I))/\operatorname{tr} H = \frac{\mathbb{E}\|\Delta H_i\|_F^2}{B \operatorname{tr} H}$. Then $\frac{\eta_D}{\eta^\star} - 1 = \eta_D\big(\frac{1}{\eta^\star} - \frac{1}{\eta_D}\big) \le \frac{2(1-\beta)}{\lambda_{\max}(\widetilde{\mathcal{V}})} c_\beta^{\text{best}}$, and $\frac{1}{\eta^\star} = \frac{\lambda_{\max}(\widetilde{\mathcal{V}})}{2(1-\beta)} + O(1)$. (6) With $\mathcal{G}_{DO} = 0$, $\mathcal{G}_\beta$ is block diagonal; $\lambda_{\max}(\mathcal{G}_{OO}) \to -\infty$ like $-(1-\beta)^{-2}$ when $\mathcal{O} \ne \{0\}$, while $g_D = \Theta((1-\beta)^{-1})$. □

Proposition 21 (First-order correction of the threshold as $\beta \to 1$)**.** *Assume* $\lambda_{\max}(\widetilde{\mathcal{V}})$ *is a simple eigenvalue of* $\widetilde{\mathcal{V}}$ *on* $\mathcal{D}$, *with unit eigenvector* $X_\infty \in \mathcal{D}$, *and set* $Y_\infty = H^{-1/4}X_\infty H^{-1/4}$. *Then, as* $\beta \to 1$,

$$\frac{1}{\eta^\star} = \frac{1}{\eta_D} + c_\infty + O(1-\beta), \qquad c_\infty \triangleq \sum_{\substack{j\ne k \\ \lambda_j \ne \lambda_k}} \frac{(\mathcal{V}(Y_\infty))_{jk}^2}{(\lambda_j - \lambda_k)^2}, \qquad \frac{\eta_D}{\eta^\star} - 1 = \frac{2(1-\beta)c_\infty}{\lambda_{\max}(\widetilde{\mathcal{V}})} + O\big((1-\beta)^2\big).$$

In commutator-energy coordinates $c_\infty = \big\|D_\Delta^{-1}\mathcal{V}_{OD}\Lambda_D^{-1/2}X_\infty\big\|^2 \le \kappa_\Delta^2 = \lim_{\beta\to 1} c'_\beta$: *the limit of*

the scalar bound of Theorem 20(2) is the same quadratic form evaluated in its worst direction instead of at the dominant diagonal mode X_∞. In the scale of the paper, $\mu = 2(1+\beta)/\eta$: $\mu^\star = \mu_D + 2(1+\beta)c_\infty + O(1-\beta)$ with $\mu_D = 2(1+\beta)/\eta_D = \lambda_{\max}(\mathcal{H}_D + \kappa_\beta \widetilde{\mathcal{V}}) = \Theta((1-\beta)^{-1})$. In particular the relative error of the diagonal threshold is of exact order $1-\beta$ iff $\mathcal{V}(Y_\infty) \notin \mathcal{D}$, i.e. iff the curvature noise maps the dominant diagonal mode off the commutant of H.

Proof. Let $\epsilon = 1-\beta$, $\mu = \lambda_{\max}(\mathcal{G}_\beta)$ and $\mathsf{G}_1 = \mathcal{L}_{0,D}^{-1/2}\mathcal{V}_{DO}\mathcal{L}_{0,O}^{-1/2}$, which does not depend on β; $\mathcal{G}_{DO} = \mathsf{G}_1/\epsilon$. By (23), $\epsilon\mathcal{G}_{DD} = \frac{1}{2}\widetilde{\mathcal{V}} + O(\epsilon)$ and $\epsilon^2\mathcal{G}_{OO} = -\frac{\beta}{1+\beta}\mathcal{Q}_\Delta + O(\epsilon)$ with $\mathcal{Q}_\Delta$ multiplying entry (j,k) of $\mathcal{O}$ by $(\lambda_j-\lambda_k)^2/(\lambda_j+\lambda_k) \geq \Delta_{\min}^2/(2\lambda_1)$. By Theorem 20, $\mu = g_D + O(1) = O(\epsilon^{-1})$, so $\epsilon^2(\mu - \mathcal{G}_{OO}) = \frac{\beta}{1+\beta}\mathcal{Q}_\Delta + O(\epsilon)$ is invertible for small ϵ and, as in the proof of Lemma 17, μ is an eigenvalue of $\mathcal{G}_{DD} + \mathsf{S}$ with $\mathsf{S} = \mathcal{G}_{DO}(\mu - \mathcal{G}_{OO})^{-1}\mathcal{G}_{OD} = \mathsf{G}_1\big(\frac{\beta}{1+\beta}\mathcal{Q}_\Delta + O(\epsilon)\big)^{-1}\mathsf{G}_1^* = \mathsf{S}_\infty + O(\epsilon)$, $\mathsf{S}_\infty = 2\mathsf{G}_1\mathcal{Q}_\Delta^{-1}\mathsf{G}_1^*$. The top eigenvalue of $\mathcal{G}_{DD}$ is separated from the rest of its spectrum by $\frac{\mathrm{gap}(\widetilde{\mathcal{V}})}{2\epsilon} + O(1)$, with unit eigenvector $x_\beta = X_\infty + O(\epsilon)$. Lemma 17, applied to $\mathcal{G}_{DD} + \mathsf{S}$ in the basis $(x_\beta, x_\beta^\perp)$, gives $\lambda_{\max}(\mathcal{G}_{DD} + \mathsf{S}) = g_D + \langle x_\beta, \mathsf{S}x_\beta\rangle + O(\|\mathsf{S}\|^2\epsilon)$ and shows that all other eigenvalues of $\mathcal{G}_{DD} + \mathsf{S}$ are below $g_D - \frac{\mathrm{gap}}{2\epsilon} + O(1) < g_D \leq \mu$, so μ is that top eigenvalue. Hence $\mu = g_D + \langle X_\infty, \mathsf{S}_\infty X_\infty\rangle + O(\epsilon)$. Finally $\langle X_\infty, \mathsf{S}_\infty X_\infty\rangle = 2\sum_{o\in\mathcal{O}}\langle F_o, \mathcal{V}(\mathcal{L}_{0,D}^{-1/2}X_\infty)\rangle_F^2/(\lambda_j-\lambda_k)^2$, and $\mathcal{L}_{0,D}^{-1/2}X_\infty = Y_\infty/\sqrt{2}$, $\langle F_{jk}, Z\rangle_F = \sqrt{2}Z_{jk}$, which gives c_∞ (the sum over $j<k$ counted twice); equivalently $\langle X_\infty, \mathsf{S}_\infty X_\infty\rangle = 2\|D_\Delta^{-1}\mathcal{V}_{OD}\mathcal{L}_{0,D}^{-1/2}X_\infty\|^2 = \|D_\Delta^{-1}\mathcal{V}_{OD}\Lambda_D^{-1/2}X_\infty\|^2$. The last display uses $\eta_D = \frac{2(1-\beta)}{\lambda_{\max}(\widetilde{\mathcal{V}})}(1 + O(1-\beta))$. □

The two regimes. (a) *Fixed* $f = \eta/\eta_{\det}$ (Theorem 18(4)): $\rho(\mathcal{K})$ and $\rho(\mathcal{K}^D)$ both grow like κ_β, their difference is $O(1-\beta)$, and the relative error is $O((1-\beta)^2)$. This regime lies beyond the ceiling for β near 1. (b) *At the threshold* (Theorem 20, Proposition 21): $\eta^\star = \Theta(1-\beta)$, $\mu_D = 2(1+\beta)/\eta_D = \Theta((1-\beta)^{-1})$, and $\mu^\star = \mu_D + 2(1+\beta)c_\infty + O(1-\beta)$, so the relative shift $\mu^\star/\mu_D - 1 = \eta_D/\eta^\star - 1$ is $\Theta(1-\beta)$ when $c_\infty > 0$. The loss of one power is the factor $1/\eta$ in $\varepsilon_{\eta,\beta}$ evaluated at $\eta = \eta^\star$.

Remark 22 (Scope: the rank-one limit, degenerate, small-gap and flat modes). (i) The rank-one secular equation of Theorem 5 is not the $\beta \to 1$ limit for arbitrary non-commuting per-sample Hessians, in two ways. The $\beta \to 1$ limit is the diagonal M-matrix condition of Proposition 19 with the full non-negative matrix Γ of (19); it is a rank-one secular equation only when $\widetilde{\Gamma}$ is diagonal plus rank one, which is what independent feature coordinates produce (Theorems 5, 8). And it is not exact at any $\beta < 1$ when $c_\infty > 0$: by Proposition 21, $\eta_D/\eta^\star - 1 = \Theta(1-\beta)$ whenever $c_\infty > 0$. What momentum does is suppress the off-commutant entries of Σ_0 through the weight $\frac{\eta\beta(\lambda_j-\lambda_k)^2}{(1-\beta)^2(1+\beta)}$, which grows faster than the $\frac{\eta}{1-\beta}$ amplification of the curvature noise. As $\beta \to 1$, $\frac{2(1-\beta)}{\eta^\star} \to \lambda_{\max}(\widetilde{\mathcal{V}})$, which for random features is $\lambda_{\max}\big(\frac{1}{B}(\sqrt{\lambda}\sqrt{\lambda}^\top + (1+\kappa_{\mathrm{feat}})\Lambda)\big)$, the batch trace-sharpness plateau $\operatorname{tr} H/B$ of Corollary 6 up to a term of size $|1+\kappa_{\mathrm{feat}}|\lambda_1/B$. (ii) Degenerate eigenvalues. Entries (j,k) inside one eigenspace carry no double-commutator weight, so they belong to $\mathcal{D}$; the limit is block diagonal, with $\widetilde{\mathcal{V}}$ acting on block-diagonal matrices through the pinching $\mathcal{P}$. It is not the diagonal of Γ alone: within an eigenspace of multiplicity m, $\widetilde{\mathcal{V}}$ acts on $\mathbb{S}^m$. (iii) Small gaps. Every bound scales with $1/\Delta_{\min}^2$, and the explicit form (4) of Theorem 20 requires $\beta\Delta_{\min}^2 > (1-\beta^2)\lambda_1\|\mathcal{V}_{OO}\|/\lambda_P$. For a cluster of eigenvalues closer than $(1-\beta)$ times the scale of the curvature noise, put the cluster into $\mathcal{D}$: Lemma 17 and Theorem 20(3) hold verbatim for any orthogonal splitting of $\mathcal{H}$ into $\mathcal{L}$-invariant coordinate subspaces, with $\Delta_{\min}$ the gap between clusters, but inside a cluster the small double-commutator weights remain and η_D loses the closed form (24). (iv) Flat modes. λ_P enters

through $\|\widehat{D}^{1/2}\mathcal{V}_{DO}\|^2 \le \|\mathcal{V}_{DO}\|^2/(2(1-f)\lambda_P)$, $\|\mathcal{G}_{DO}\| \le \|\mathcal{V}_{DO}\|/(2\lambda_P(1-\beta))$ and the condition in (iii); a mode with $\lambda_P \to 0$ and curvature noise coupling it to the rest (the collapse of Theorem 8) makes these constants grow like λ_P^{-1} to λ_P^{-2}, so the diagonal reduction needs β closer to 1. A tether $\mu > 0$ replaces λ_P by $\lambda_P + \mu$. The $\Delta_{\min}$, λ_P constants are crude because they charge every pair, including the top couplings $(1,k)$, with the smallest gap and the flattest mode: in the checks c_β exceeds the operator form of Theorem 20(3) by factors 10 to 5×10^4 and is defined in 23 of 42 cells. The commutator-energy constant c'_β charges each pair with its own gap; it lies between 0.16 and 11 times the operator form and is defined in 35 of 42 cells, and the weighted form between 0.16 and 1.1 times, defined in all 42. In the near-flat cell $c'_\beta \approx 262\, c_\infty$ at $\beta = 0.9999$: κ_Δ^2 takes the worst direction of A, the flat diagonal mode ($\Lambda_D^{-1/2} \approx 7$ there), while c_∞ weighs that mode by its small component in X_∞. The eigenvector split of Theorem 20(1) removes this: its first term is $\langle u, \mathsf{E}u\rangle \to c_\infty$, and the flat direction enters only the second-order term, divided by $\gamma_D = \Theta((1-\beta)^{-1})$; in the near-flat cell it is 5.1, 1.6, 1.2, 1.04, 1.004, 1.0005 times the true value at $\beta = 0.8, \dots, 0.9999$.

Numerical check (`paper/checks/diagonalization-burnin.py`). $\mathcal{T}$ is built in Kraus form on $\mathbb{S}^{2P}$ by enumerating all N^B batches of non-commuting positive semidefinite per-sample Hessians ($P \in \{3,4,5\}$, $N \in \{4,5,6,8\}$, $B \in \{1,2\}$; Gram matrices of random Gaussian factors, centered perturbations of a H with a double and a triple eigenvalue, a near-flat mode, and Rademacher features), and $\rho(\mathcal{T})$ is computed as $\lim \|\mathcal{T}^{2^K}\|^{1/2^K}$ by 46 normalized squarings. In all 14 cells (7 models, $\beta \in \{0.5, 0.9\}$), $\rho(\mathcal{K}) = 1$ at $\eta^\star = 1/\lambda_{\max}(\mathcal{G}_\beta)$ to 10^{-12}, $|\rho(\mathcal{T}(\eta^\star)) - 1| \le 1.1\times10^{-13}$, and $\rho(\mathcal{T})$ crosses 1 between $0.999\,\eta^\star$ and $1.001\,\eta^\star$. Both inequalities of Theorem 18(1) and the bound (22) hold in all 30 cells at $f = \frac{1}{2}$, $\beta \in \{0.5, 0.8, 0.9, 0.95, 0.99\}$; the log-slope of the relative error in $1-\beta$ between $\beta = 0.95$ and 0.99 is 2.015–2.024 in the six models, including the degenerate ones. At the threshold, $(\eta_D/\eta^\star - 1)$ has log-slope 0.98–1.00 between $\beta = 0.999$ and 0.9999 and $1/\eta^\star - 1/\eta_D$ agrees with c_∞ of Proposition 21 to 4 digits at $\beta = 0.9999$ (e.g. 7.051 against 7.050, 0.9065 against 0.9065). For Rademacher features $\mathcal{V}_{DO} = 0$ and $\eta^\star$, η_D and the root of (4) agree to 12 significant digits. Scalar constants against the operator forms (ratio range; cells where the constant is defined): at the threshold, the eigenvector split of Theorem 20(1) is 0.0038–10.3 times the operator form (3) with E_Δ (35/42) and 0.0038–1.75 times with $\bar{\mathsf{E}}$ (42/42), and 1.000–17.7 (resp. 1.000–17.5) times the true $1/\eta^\star - 1/\eta_D$, the ratio tending to 1 as $\beta \to 1$ in every model (≤ 1.001 at $\beta = 0.9999$); condition (26) held in every cell where the majorant was defined, so its only failures are the 7 small-β cells with $\theta_\beta \ge 1$, where $\bar{\mathsf{E}}$ is used. The other forms: $c_\beta/\mathrm{op} \in [10, 5.1\times10^4]$ (23/42), $c'_\beta/\mathrm{op} \in [0.16, 11]$ (35/42), weighted form $\in [0.16, 1.1]$ (42/42); against the true $1/\eta^\star - 1/\eta_D$ the ranges are $[41, 1.3\times10^7]$, $[2.0, 588]$ and $[1.8, 262]$. At fixed f, the $\Delta_{\min}$ form of Theorem 18(3) is 21–1.3×10^3 times (22) (21/30) and the form (3′) is 0.016–3.9 times it (25/30); against the true relative error, 53–5.3×10^4 and 1.5–40. Every scalar bound holds in every cell where it is defined; $c_\infty = \|D_\Delta^{-1}\mathcal{V}_{OD}\Lambda_D^{-1/2}X_\infty\|^2$ matches the sum form to 10^{-10} and $\kappa_\Delta^2/c_\infty \in \{2.0, 2.2, 2.1, 3.4, 3.8, 262\}$ in the six models.

B.2 Burn-in in Loewner order and the spectral gap of the second-moment operator

Let $z_t = (w_t, w_{t-1})$, $Q_t = \mathbb{E}[z_t z_t^\top]$, $\mathcal{T}(Q) = \mathbb{E}[F_t Q F_t^\top]$ on $\mathbb{S}^{2P}$ as in Lemma 3, and $G = \mathrm{blkdiag}(\eta^2\Sigma_\xi, 0)$. Assume **(H0)** $\mathbb{E}[z_0] = 0$, or $\mathbb{E}[(H_{B_t} - H)a\,\xi_t^\top] = 0$ for every $a \in \mathbb{R}^P$. Then $Q_{t+1} = \mathcal{T}(Q_t) + G$ for every t: conditioning on (F_t, ξ_t), independent of z_t, the quadratic term is $\mathcal{T}(Q_t)$, and the cross term $-\eta\mathbb{E}[F_t z_t \xi_t^\top] = -\eta\mathbb{E}[F_t\,\mathbb{E}[z_t]\,\xi_t^\top]$ vanishes under (H0) because $\mathbb{E}[z_t] = \bar{F}^t\mathbb{E}[z_0]$ with $\bar{F} = \mathbb{E}[F_t]$ and $\mathbb{E}[F_t a\xi_t^\top] = -\eta\,\mathrm{blkdiag}(\mathbb{E}[(H_{B_t} - H)a\xi_t^\top], 0)$ for the first block row.

Theorem 23 (Burn-in in Loewner order)**.** *Assume (H0) and $\rho(\mathcal{T}) < 1$, and let $Q_\infty = (I - \mathcal{T})^{-1}(G) = \sum_{k\ge 0} \mathcal{T}^k(G) \succeq 0$. (1) $Q_\infty - Q_t = \mathcal{T}^t(Q_\infty - Q_0)$ for every $t \ge 0$. (2) For every $\theta \in (\rho(\mathcal{T}), 1)$, $\theta > 0$, the series $Q_\theta \triangleq \sum_{k\ge 0} \theta^{-k}\mathcal{T}^k(I)$ converges, $Q_\theta \succeq I$ and $\mathcal{T}(Q_\theta) = \theta(Q_\theta - I) \preceq \theta Q_\theta$. With $c_\pm \triangleq \min\{s \ge 0 : \pm(Q_\infty - Q_0) \preceq sQ_\theta\} \le \|(Q_\infty - Q_0)_\pm\|$,*

$$-c_-\theta^t Q_\theta \;\preceq\; Q_\infty - Q_t \;\preceq\; c_+\theta^t Q_\theta \qquad (t \ge 0).$$

If $Q_0 \preceq Q_\infty$ (for instance $Q_0 = 0$, the start at the minimizer), then $c_- = 0$ and $0 \preceq Q_\infty - Q_t \preceq c_+\theta^t Q_\theta$; in particular $0 \le \mathbb{E}\|w_\infty\|^2 - \mathbb{E}\|w_t\|^2 \le c_+\theta^t \operatorname{tr}\big((Q_\theta)_{11}\big)$. (3) If $\beta \in (0,1)$ and the transitions are irreducible, i.e. no subspace $\{0\} \ne \mathcal{W} \ne \mathbb{R}^{2P}$ satisfies $F(H_B)\mathcal{W} \subseteq \mathcal{W}$ for almost every batch, then $\mathcal{T}$ has an eigenmatrix $Q_* \succ 0$ with $\mathcal{T}(Q_*) = \rho(\mathcal{T})Q_*$, and (2) holds with $\theta = \rho(\mathcal{T})$ and Q_θ replaced by Q_*:

$$-c_-^*\rho(\mathcal{T})^t Q_* \preceq Q_\infty - Q_t \preceq c_+^*\rho(\mathcal{T})^t Q_*, \qquad c_\pm^* = \lambda_{\max}\big(\pm Q_*^{-1/2}(Q_\infty - Q_0)Q_*^{-1/2}\big)_+.$$

Proof. (1) Subtract $Q_{t+1} = \mathcal{T}(Q_t) + G$ from $Q_\infty = \mathcal{T}(Q_\infty) + G$ and iterate. (2) $\rho(\mathcal{T}/\theta) < 1$ gives convergence; $Q_\theta \succeq I$ because every term is positive semidefinite (positivity of $\mathcal{T}^k$) and the $k = 0$ term is I; $\mathcal{T}(Q_\theta) = \sum_{k\ge 0}\theta^{-k}\mathcal{T}^{k+1}(I) = \theta\sum_{k\ge 1}\theta^{-k}\mathcal{T}^k(I) = \theta(Q_\theta - I)$. Since $\mathcal{T}$ is linear and positive, $X \preceq Y$ implies $\mathcal{T}(X) \preceq \mathcal{T}(Y)$, so $\mathcal{T}^t(Q_\theta) \preceq \theta^t Q_\theta$ by induction. Apply $\mathcal{T}^t$ to $-c_-Q_\theta \preceq Q_\infty - Q_0 \preceq c_+Q_\theta$ and use (1). The bound on $c_\pm$ follows from $Q_\theta \succeq I$; the trace statement takes the trace of the $(1,1)$ block. (3) $\mathcal{T}$ maps the proper cone $\mathbb{S}_+^{2P}$ into itself, so by the finite-dimensional Krein–Rutman (Perron–Frobenius) theorem [Berman and Plemmons, 1994, Thm. 1.3.2] $\rho(\mathcal{T})$ is an eigenvalue with an eigenvector $Q_* \in \mathbb{S}_+^{2P} \setminus \{0\}$. $\rho(\mathcal{T}) > 0$: by Jensen, $\mathcal{T}(I) = \mathbb{E}[F_tF_t^\top] \succeq \bar{F}\bar{F}^\top \succ 0$ ($\det\bar{F} = \beta^P \ne 0$), so $\mathcal{T}(I) \succeq \sigma I$ for some $\sigma > 0$ and $\mathcal{T}^n(I) \succeq \sigma^n I$. If $x \perp \operatorname{range}(Q_*)$, then $0 = \rho(\mathcal{T})\, x^\top Q_* x = x^\top\mathcal{T}(Q_*)x = \mathbb{E}\|Q_*^{1/2}F_t^\top x\|^2$, so $F_t^\top x \in \ker Q_* = \operatorname{range}(Q_*)^\perp$ almost surely: $F_t^\top$ preserves $\operatorname{range}(Q_*)^\perp$, hence F_t preserves $\operatorname{range}(Q_*)$, almost surely. Irreducibility and $Q_* \ne 0$ give $\operatorname{range}(Q_*) = \mathbb{R}^{2P}$. The rest is (2) with $\mathcal{T}^t(Q_*) = \rho(\mathcal{T})^t Q_*$. □

Irreducibility can fail: if every H_B commutes with H, each $\operatorname{span}\{(u_k, 0), (0, u_k)\}$ is invariant and Q_* may be singular; (2) needs no such condition. The matrix Q_ε of the proof of Theorem 5, which satisfies $\mathcal{T}(Q_\varepsilon) \preceq \theta Q_\varepsilon$ for some $\theta < 1$, can be used in (2) in place of Q_θ.

The regular splitting and the spectral gap. Write $\mathcal{T} = \mathcal{T}_0 + \Delta$ with $\mathcal{T}_0(Q) = \bar{F}Q\bar{F}^\top$ (the noise-free part) and $\Delta(Q) = \mathbb{E}[(F_t - \bar{F})Q(F_t - \bar{F})^\top] = \operatorname{blkdiag}(\eta^2\mathcal{V}(Q_{11}), 0)$; both are positive, and $\rho(\mathcal{T}_0) = \rho(\bar{F})^2 < 1$ for $\eta \in (0, \eta_{\det})$. Let $\mathcal{R} = (I - \mathcal{T}_0)^{-1}\Delta$, $\Pi(Q) = Q_{11}$ and $\mathcal{J}(X) = (I - \mathcal{T}_0)^{-1}\big(\operatorname{blkdiag}(\eta^2\mathcal{V}(X), 0)\big)$ for $X \in \mathbb{S}^P$.

Theorem 24 (Regular splitting: $\rho(\mathcal{R}) = \rho(\mathcal{K})$ and a two-sided spectral gap)**.** *Let $\beta \in [0, 1)$ and $\eta \in (0, \eta_{\det})$, with $\beta > 0$ in parts (3)–(6). (1) $\mathcal{R} = \mathcal{J} \circ \Pi$ and $\Pi \circ \mathcal{J} = \mathcal{K}_{\eta,\beta}$. Hence $\mathcal{R}$ and $\mathcal{K}_{\eta,\beta}$ have the same non-zero eigenvalues with the same algebraic multiplicities, and $r \triangleq \rho(\mathcal{R}) = \rho(\mathcal{K}_{\eta,\beta})$ exactly. (2) Let $X_* \in \mathbb{S}_+^P \setminus \{0\}$ be the Krein–Rutman eigenmatrix of $\mathcal{K}_{\eta,\beta}$ (Theorem 9(1)), $Y_* = \operatorname{blkdiag}(\eta^2\mathcal{V}(X_*), 0) \succeq 0$ and $W = \mathcal{J}(X_*) = \sum_{k\ge 0}\mathcal{T}_0^k(Y_*) \succeq 0$. Then $\mathcal{R}(W) = rW$, $\Pi(W) = rX_*$, $(I - \mathcal{T}_0)(W) = Y_*$, and*

$$\mathcal{T}(W) = rW + (1 - r)\mathcal{T}_0(W) = W - (1 - r)Y_*. \tag{27}$$

(3) $W \succ 0$ iff no eigenvector of H is orthogonal to $\operatorname{range}\mathcal{V}(X_*)$; in particular if $\mathcal{V}(X_*) \succ 0$. (4) Lower bound. With $s_{\text{low}} = \max\{s \ge 0 : \mathcal{T}_0(W) \succeq sW\}$, $\rho(\mathcal{T}) \ge r + (1-r)s_{\text{low}}$ if $r \le 1$, and $\rho(\mathcal{T}) \ge 1$ if $r \ge 1$. If $W \succ 0$, $s_{\text{low}} = \lambda_{\min}(W^{-1/2}\mathcal{T}_0(W)W^{-1/2}) > 0$. (5) The one-step upper bound is empty.

If $r \le 1$, $W \succ 0$ and $\mathcal{T}_0(W) \preceq sW$, then $\rho(\mathcal{T}) \le r + (1-r)s$; but for this W the least such s is exactly 1, so the bound reads $\rho(\mathcal{T}) \le 1$. (6) Two-step upper bound. If $r \le 1$ and $\mathcal{V}(X_*) \succ 0$, then $Z \triangleq Y_* + \mathcal{T}(Y_*) \succ 0$, $W \succ 0$, $\mathcal{T}^2(W) = W - (1-r)Z$, and with $m_2 \triangleq \lambda_{\min}(W^{-1/2} Z W^{-1/2}) > 0$, $\rho(\mathcal{T})^2 \le 1 - (1-r)m_2$. Therefore, for $r < 1$,

$$\frac{(1-r)\, m_2}{2} \;\le\; 1 - \rho(\mathcal{T}) \;\le\; (1-r)(1 - s_{\text{low}}), \qquad r = \rho(\mathcal{K}_{\eta,\beta}).$$

More generally, if $Z_m \triangleq \sum_{k<m} \mathcal{T}^k(Y_*) \succ 0$ and $W \succ 0$, then $\rho(\mathcal{T})^m \le 1 - (1-r)\lambda_{\min}(W^{-1/2} Z_m W^{-1/2})$.

Proof. (1) $\Delta = \operatorname{blkdiag}(\eta^2 \mathcal{V}(\Pi\,\cdot), 0)$ gives $\mathcal{R} = \mathcal{J} \circ \Pi$. For $Y \in \mathbb{S}^P_+$, $Q = (I - \mathcal{T}_0)^{-1} \operatorname{blkdiag}(\eta^2 Y, 0)$ is the stationary second moment of the noise-free recursion (2) with $\mathcal{V} = 0$ and $\Sigma_\xi = Y$ (driven by centered Gaussian noise, so $\mathbb{E}[w_t] = 0$), and Theorem 2(2) gives $\mathcal{L}_{\eta,\beta}(Q_{11}) = \frac{\eta}{1-\beta} Y$, i.e. $\Pi(I - \mathcal{T}_0)^{-1} \operatorname{blkdiag}(\eta^2 Y, 0) = \frac{\eta}{1-\beta} \mathcal{L}^{-1}_{\eta,\beta}(Y)$ ($\mathcal{L}_{\eta,\beta}$ is invertible by Theorem 9(1)); both sides are linear in Y and $\mathbb{S}^P_+$ spans $\mathbb{S}^P$, so it holds on $\mathbb{S}^P$. With $Y = \mathcal{V}(X)$, $\Pi\mathcal{J} = \frac{\eta}{1-\beta} \mathcal{L}^{-1} \mathcal{V} = \mathcal{K}$. For linear maps $\mathcal{J} : \mathbb{S}^P \to \mathbb{S}^{2P}$ and $\Pi : \mathbb{S}^{2P} \to \mathbb{S}^P$, $\mathcal{J}\Pi$ and $\Pi\mathcal{J}$ have the same non-zero eigenvalues with multiplicities ($\det(zI - \mathcal{J}\Pi) = z^{\dim \mathbb{S}^{2P} - \dim \mathbb{S}^P} \det(zI - \Pi\mathcal{J})$). The phrase "compression to the $(1,1)$ block" in the proof of Theorem 9 is made precise by this factorization; a compression alone would not fix the spectral radius. (2) $\mathcal{R}(W) = \mathcal{J}(\Pi\mathcal{J} X_*) = \mathcal{J}(\mathcal{K} X_*) = r\mathcal{J}(X_*) = rW$ and $\Pi W = \mathcal{K} X_* = r X_*$. $(I - \mathcal{T}_0)W = Y_*$ by definition of $\mathcal{J}$, and $\Delta(W) = \operatorname{blkdiag}(\eta^2 \mathcal{V}(r X_*), 0) = r Y_*$. Hence $\mathcal{T}(W) = \mathcal{T}_0(W) + r Y_* = W - Y_* + r Y_* = W - (1-r)Y_*$, and $rW + (1-r)\mathcal{T}_0(W) = rW + (1-r)(W - Y_*)$ is the same matrix. $W \succeq 0$ because $Y_* \succeq 0$ and $\mathcal{T}_0$ is positive. (3) $x \in \ker W$ iff $Y_*^{1/2} (\bar{F}^\top)^k x = 0$ for all k, iff x is orthogonal to the reachable subspace of the pair $(\bar{F}, \operatorname{range} Y_*)$. By the Popov–Belevitch–Hautus test, that subspace is $\mathbb{R}^{2P}$ iff no left eigenvector $z = (p, q)$ of $\bar{F}$ is orthogonal to $\operatorname{range} Y_* = \operatorname{range} \mathcal{V}(X_*) \oplus 0$. From $z^\top \bar{F} = \nu z^\top$ with $\bar{F} = \begin{pmatrix} A & -\beta I \\ I & 0 \end{pmatrix}$, $A = (1+\beta)I - \eta H$: $\nu \neq 0$ ($\det \bar{F} = \beta^P$), $q = -\beta p/\nu$ and $Ap = (\nu + \beta/\nu)p$, so p lies in an eigenspace of H; conversely every real eigenvector p of H, $Hp = \lambda p$, gives left eigenvectors with $\nu^2 - ((1+\beta) - \eta\lambda)\nu + \beta = 0$. Orthogonality of z to $\operatorname{range} Y_*$ is $p \perp \operatorname{range} \mathcal{V}(X_*)$ (for complex p in an eigenspace, its real or imaginary part is a non-zero real eigenvector with the same property). (4) By (27), $\mathcal{T}(W) = rW + (1-r)\mathcal{T}_0(W) \succeq (r + (1-r)s_{\text{low}})W$ when $r \le 1$, and $\mathcal{T}(W) = W + (r-1)Y_* \succeq W$ when $r \ge 1$. If $\mathcal{T}(W) \succeq cW$ with $c \ge 0$ and $W \in \mathbb{S}^{2P}_+ \setminus \{0\}$, then $\mathcal{T}^n(W) \succeq c^n W$, so $\|\mathcal{T}^n\| \ge c^n \|W\|/\|W\|$ and $\rho(\mathcal{T}) \ge c$ by Gelfand's formula. $\mathcal{T}_0(W) = \bar{F} W \bar{F}^\top \succ 0$ when $W \succ 0$. (5) If $W \succ 0$ and $\mathcal{T}(W) \preceq \theta W$ then every $X \in \mathbb{S}^{2P}_+$ satisfies $X \preceq \|X\| \|W^{-1}\| W$, so $\mathcal{T}^n(X) \preceq \theta^n \|X\| \|W^{-1}\| W$ and $\rho(\mathcal{T}) \le \theta$ by Lemma 3; with $\theta = r + (1-r)s$ this is the stated bound. Now $W - \mathcal{T}_0(W) = Y_* \succeq 0$ gives $s \le 1$, and for $x = (0, y)$, $y \neq 0$, $x^\top \mathcal{T}_0(W) x = x^\top W x - x^\top Y_* x = x^\top W x > 0$ because the $(2,2)$ block of Y_* is zero, so $s \ge 1$. This is why a one-step contraction of W under $\mathcal{T}_0$ cannot be used here. (6) Applying $\mathcal{T}$ to (27): $\mathcal{T}^2(W) = \mathcal{T}(W) - (1-r)\mathcal{T}(Y_*) = W - (1-r)(Y_* + \mathcal{T}(Y_*))$. With $a = \eta^2 \mathcal{V}(X_*) \succ 0$, the quadratic form of Z at (x, y) is $x^\top a x + \mathbb{E}\big[(A_t x + y)^\top a (A_t x + y)\big]$, $A_t = (1+\beta)I - \eta H_{B_t}$ (because $F_t^\top(x, y) = (A_t x + y, -\beta x)$), which vanishes only if $x = 0$ and then $y = 0$; so $Z \succ 0$. $W \succ 0$ by (3). Then $\mathcal{T}^2(W) \preceq (1 - (1-r)m_2)W$ and, as in (5) applied to $\mathcal{T}^2$, $\rho(\mathcal{T})^2 \le 1 - (1-r)m_2$. For $r < 1$, $\rho(\mathcal{T}) \le 1$ and $1 - \rho(\mathcal{T}) \ge \frac{1 - \rho(\mathcal{T})^2}{2} \ge \frac{(1-r)m_2}{2}$; the upper bound on $1 - \rho(\mathcal{T})$ is (4). The m-step statement is the same computation with $\mathcal{T}^m(W) = W - (1-r)Z_m$. □

So the spectral gap of the full second-moment operator is the gap $1 - \rho(\mathcal{K}_{\eta,\beta})$ of the reduced operator times a factor between $m_2/2$ and $1 - s_{\text{low}}$, both read off the single matrix $W = \mathcal{J}(X_*)$; with Theorem 23, $t \ge \log(c_+/\delta)/\log(1/\rho(\mathcal{T}))$ steps bring $Q_\infty - Q_t$ below δQ_*, and $\log(1/\rho(\mathcal{T})) \ge 1 - \rho(\mathcal{T}) \ge (1 - \rho(\mathcal{K}))m_2/2$. The lower bound is loose in the checks: m_2 is small because W is ill-conditioned.

Numerical check (`paper/checks/diagonalization-burnin.py`). In five cells ($P \in \{3,4,5\}$, a double eigenvalue, $\beta \in \{0.5, 0.8, 0.9, 0.95\}$, $\eta \in \{0.7, 0.8, 0.9\}\eta^\star$): $|\rho(\mathcal{R}) - \rho(\mathcal{K})| \le 2 \times 10^{-14}$; $\|\mathcal{R}W - rW\|/\|W\| \le 4 \times 10^{-14}$ and the residual of (27) is $\le 6 \times 10^{-16}$; $X_* \succ 0$, $W \succ 0$, $Q_* \succ 0$ in every cell; the one-step constant is $s = 1.000000000000$ in every cell, as Theorem 24(5) predicts; the two-sided gap bound holds, e.g. $1.6 \times 10^{-4} \le 1 - \rho(\mathcal{T}) = 1.1 \times 10^{-2} \le 2.5 \times 10^{-2}$ at $P = 3$, $\beta = 0.9$, $\eta = 0.9\eta^\star$. The Loewner envelopes of Theorem 23 hold for $t \le 300$ from $Q_0 = 0$ and from a Q_0 with $Q_\infty - Q_0$ indefinite, with both Q_* and Q_θ, $\theta = (1 + \rho(\mathcal{T}))/2$ (smallest normalized eigenvalue of the envelope gap $\ge -4 \times 10^{-14}$, round-off).